FedSA-Drift: Structure- and Drift-Aware Federated Vision Transformers for Multimodal Skin Lesion Classification

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ABSTRACT

Federated learning allows healthcare institutions to train shared machine learning models on patient records without exchanging raw data, supporting compliance with privacy regulations such as GDPR and HIPAA. Heterogeneity across institutions, however, induces client drift, unstable convergence and unfair per-client performance, and these effects intensify when structured clinical metadata is fused with high-capacity Vision Transformers. This paper proposes FedSA-Drift (Structure- and Drift-Aware Federated Learning) for multimodal skin lesion classification. A SwinV2-Base backbone fused with a metadata branch is separated into three parameter groups—early vision layers, deeper Transformer layers, and client-specific multimodal heads—each aggregated by a different rule: standard FedAvg, drift-aware inverse-distance weighting, and no aggregation, respectively. All correction occurs at the server during aggregation: no penalty term is added to any local objective and no regularisation hyperparameter is introduced. Our central result is obtained on the held-out ISIC 2019 test collection under severe heterogeneity ($\alpha = 0.1$, $K = 5$), where one client holds no samples for four of the eight classes. There, standard FedAvg attains the highest mean per-client accuracy (0.4702) yet abandons four of the eight diagnostic classes outright, recording *zero* melanoma recall for every client—including the institution holding 3815 of the federation's 3844 melanoma cases (0/1327, 95% Wilson interval [0.000, 0.003])—because size-weighted aggregation is dominated by a client that holds none. Structure-aware decomposition restores all eight classes at a cost of 15.9 percentage points of mean accuracy, and drift-aware weighting recovers 11.5 of those points while raising the worst-client floor by 21.5 points and reducing the inter-client standard deviation of accuracy by 67%. At the melanoma-holding client, recall rises from 0.000 to 0.573 with the decomposition and to 0.727 with drift weighting, across three disjoint confidence intervals. Balanced accuracy separates these very different clinical profiles by roughly one point (0.3778 against 0.3681): aggregate metrics do not reveal complete class abandonment. At the milder settings, worst-client accuracy rises from 0.4800 to 0.5705 at $\alpha = 0.5$ with a 39% reduction in inter-client standard deviation. These proof-of-concept results, spanning three federated configurations on a single dataset, suggest improved class coverage, stability and fairness.

Keywords: client drift, federated learning, heterogeneous data, medical image analysis, multimodal learning, non-IID data, skin lesion classification, Vision Transformer

1 INTRODUCTION

Melanoma is responsible for a disproportionate number of deaths from skin cancer. It is one of the most common cancers in the world (Arnold et al., 2022; Hay et al., 2014). To improve patient outcomes, early and accurate diagnosis is a must. Automated classification of skin lesions using dermoscopic images has greatly improved recently. Much of this improvement has been accomplished through deep learning and large-scale vision models such as Vision Transformers (Aguirre et al., 2024).

However, most highly effective models currently rely upon centralized environments where the model is created by pooling all of the data into one single repository. This assumption rarely applies to clinical environments, since due to extremely strict privacy laws and institutional policies, medical data is complex to share across hospitals. The CCPA and the GDPR both limit the direct ability to share the raw data collected by hospitals. Therefore, the laws that restrict the sending of data make multi-institutional collaborations for various medical purposes legally and ethically complex (Voigt and Von dem Bussche, 2017; Law, 2023).

Federated learning (FL) is an alternative to this issue. Institutions can share model parameters with each other, but not their raw data. This allows multiple institutions to collaborate without sacrificing patient privacy (McMahan et al., 2017; Rieke et al., 2020). Nevertheless, FL has one major challenge in the medical realm: there is heterogeneity in the data from different hospitals. Medical data in the real world is not independently and identically distributed (that is, non-IID). All of the differences in patient demographics, the number of patients with a certain disease, the imaging devices and device settings used to image patients (metadata) all contribute to statistical heterogeneity in the datasets (Zhao et al., 2018; Li et al., 2020).

The complexity of multimodal learning tasks compounds the challenge. Analysis of skin lesions, for example, requires both imaging as well as clinical metadata which includes age, gender and anatomical location (Nunnari et al., 2020; Gessert et al., 2020). When done in a centralized setting, patient metadata typically increases diagnostic performance. Patient metadata contains information about the clinical context that cannot be derived from visual features only (Gessert et al., 2020; Nunnari et al., 2020).

The aggregation of multimodal data creates additional challenge due to heterogeneity when utilizing federated systems. Image distributions alone are already heterogeneous among various healthcare institutions due to many differences between imaging devices and patient characteristics. When considering the distribution of metadata, it is much more heterogeneous than image distributions. For example, within a paediatric clinic, there may be significant representation of a constrained age range. In a specialist centre or referral clinic, there may be substantial representation of only one or very few anatomical sites. The manner in which sex and body location are recorded may also vary across electronic health record systems. Thus, fusing two heterogeneous modalities and distributing them across clients through a shared model only serves to further exacerbate instability. Therefore, client drift is exacerbated in multimodal federated systems compared with those only containing imaging (Zhu et al., 2025).

The aggregation of all model parameters uniformly is not appropriate for multimodal architectures. The visual backbone of a model and the metadata fusion component perform different tasks within a model, and each modality may be subject to varying distributions among each of the clients. If these two different modalities are aggregated in the same manner, the visual representation will be corrupted. Likewise, the metadata components will not be allowed to acclimate or adapt to local institutional characteristics (Zhu et al., 2025). Therefore, in order to build a stable and equitable multimodal federated system, client drift must be addressed and the structural differences between model components must be respected. Figure 1 illustrates this federated setup.

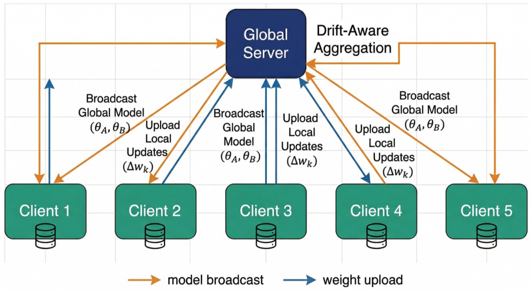


Figure 1. Overview of the federated learning setup with 5 clients. Each client trains locally on private data and uploads model updates to the global server. The server aggregates updates via drift-aware inverse-distance weighting and broadcasts the updated global model parameters back to all clients.

1.1 Existing Solutions and Limitations

Federated Averaging (FedAvg) is the de facto standard for federated optimization algorithms (McMahan et al., 2017). It averages the models from all clients weighted according to the size of the local dataset for each client. The FedAvg algorithm works well when local data distributions are similar to one another. However, when data is non-IID, the local models begin to drift; this means that the local models will converge toward different optimal points, resulting in a biased or unstable aggregate model (Karimireddy et al., 2020; Li et al., 2020).

Different methods have been proposed to address this problem. There are proximal regularization methods that introduce a penalty to any local update that deviates from the global model (Li et al., 2020). Alternatively, there are also methods that use control variates or partial personalization strategies (Karimireddy et al., 2020; Collins et al., 2021). However, using these additional methods typically requires the introduction of additional hyperparameters or a modification of the objectives that are normally employed to train local models.

For multimodal federated learning, the majority of approaches consider the model a single set of parameters. That is, all layers of all models are aggregated in the same way, regardless of the role of each layer. This does not take into account that the early feature extractors and the deep semantic layers will behave very differently under any change in distribution. As a result, uniform aggregation produces instability in shared representations and limits the ability of institutions to specialize (Zhu et al., 2025).

In addition to this, many researchers and practitioners tend to focus on global metrics such as overall accuracy or AUC. For clinical applications, however, global performance alone is not adequate. For example, a model that performs well on average, but fails at one institution, will create inequities and diminish trust (Darzi et al., 2024; Li et al., 2021).

1.2 Research Gap

There are still critical gaps that exist in federated medical learning despite past progress.

Sensitivity to layer-wise distributional heterogeneity: Compared to the earlier feature extraction layers, the performance of later deep Transformer model layers is more adversely impacted by distributional heterogeneity. Research on federated Vision Transformers indicates that if the multi-head attention remains misaligned across multiple client devices, the performance of the other clients will suffer due to feature collapse (Darzi et al., 2024). Studies evaluating partially personalized models show that uniformly aggregating shared model parameters negatively impacts the device drift experienced by clients (Zhu et al., 2025). Most current methods uniformly aggregate all shared model parameters while failing to take into account the different functional behaviours exhibited by the various layers of the model.

Multimodal personalization: Structured clinical metadata frequently varies greatly across hospitals (e.g., age, gender and body structure) (Nunnari et al., 2020). When using multimodal federated models, if the shared visual and the personalized components of the metadata are not kept separate from each other, the divergence of each will increase, thereby decreasing stability (Zhu et al., 2025). Most current methods do not appropriately enforce a separation between these two components.

Client-level stability under severe non-IID conditions: Individual clients may experience substantial drops in performance under conditions of severe non-IID distributional heterogeneity, even though the global metric remains satisfactory. Recent research on federated Vision Transformers shows that there is a critical requirement for the robustness of the worst client in order to stabilise client performance (Darzi et al., 2024; Zhao et al., 2025). The majority of current methods are primarily directed toward achieving a global objective and do not provide a mechanism for optimising worst-case client performance.

Moreover, many of the technical methods used to stabilise client performance impose extensive computation costs on client devices, while it is very important that efficiency be considered when developing practical applications for resource-limited environments (Wu et al., 2024). There is therefore a clear need for a federated method that stabilizes critical representation layers and improves fairness without complex local regularization.

1.3 Objectives

The objective of this research is to develop a federated multimodal skin lesion classification framework that incorporates the concept of data structure upon which it is designed. Additionally, the framework will be designed with the express intent of being resilient to the different forms of data distribution (heterogeneous data distributions).

This research will accomplish its objectives by developing:

- A method to decompose parameters that separates global visual representations from client-specific multimodal heads.
- An aggregation mechanism to address client divergence using available data from clients on the server-side.
- An examination of how the model behaves under IID versus non-IID conditions, as simulated using a Dirichlet distribution.
- A comparison between the performance of federated learning versus the maximum achievable performance of centralized learning.
- An evaluation of fairness in terms of worst-case client accuracy and the inter-client standard deviation of accuracy for all models.

1.4 Major Contributions

This paper makes four algorithmic and design contributions and three empirical ones. We separate them because two of the strongest results below are findings about evaluation rather than new machinery. The algorithmic contributions are:

- **Structure-Aware Parameter Decomposition:** Model parameters are decomposed in a manner that separates early backbone layers, deep Transformer layers, and client-specific multimodal heads. With this design, the server can selectively aggregate contributions from clients based on the functional role each parameter group plays.
- **Drift-Aware Inverse-Distance Aggregation:** The aggregation mechanism designed for the server-side reduces the overall contribution of outlier clients based on how far they have drifted from prior models. No changes to local training objectives need to be made, nor are any additional regularization hyperparameters required.
- **Fairness at the Client Level:** The results indicated that worst-performing clients benefited from drift-aware aggregation versus decomposition alone, while the overall variance between clients was significantly lower.
- **Low Computational Overhead:** The server-side cost of drift-aware aggregation is negligible relative to local training (Section 4.7.1), and clients are unmodified, storing no auxiliary state. We measured computational and communication cost only; we did not evaluate deployment, which would additionally involve network conditions, edge hardware and institutional integration.

The empirical contributions are:

- **Class Abandonment Under Severe Skew:** We show that standard FedAvg can drive recall on a clinically critical class to exactly zero under severe label skew while retaining competitive aggregate accuracy, and that this failure is invisible to overall and balanced accuracy alike. This argues for reporting per-class coverage as a first-class metric in federated medical imaging.
- **Component Decomposition Ablation:** A three-way comparison at $\alpha = 0.1$—no decomposition, decomposition only, and full FedSA-Drift—separates the roles of the two mechanisms. The parameter decomposition is responsible for preserving class coverage, while drift-aware weighting is responsible for recovering utility and reducing client dispersion. Neither component is redundant and neither is sufficient alone.
- **Proof-of-Concept Heterogeneity Study:** The performance of FedSA-Drift was evaluated from near-IID through severely heterogeneous data distributions using three Dirichlet concentration parameters ($\alpha \in \{0.1, 0.5, 1.0\}$) and two client counts ($K \in \{3, 5\}$) on the ISIC 2019 dataset, with the severe setting additionally validated on a held-out test split. This constitutes an initial empirical characterization; evaluation at larger client counts is left to future work.

2 LITERATURE SURVEY

2.1 Survey of Existing Methods

2.1.1 Transformer-Based Vision Models

Automated classification of skin lesions has moved from CNN methods to Vision Transformers (ViTs) that leverage self-attention to capture long-range dependencies. ViTs outperform traditional CNN methods, especially when the class distribution is imbalanced (Aguirre et al., 2024). Hybrid models are now the most common state-of-the-art methods due to their ability to combine local and global features. Strong attention-based fusion models combining ConvNeXt and ViT have produced state-of-the-art performance on ISIC datasets (Naeem et al., 2025). Multi-stream architectures using transformers can extract multi-scale patterns in dermatology (Alshdaifat et al., 2025). Other approaches include segmentation/classification fusion (Li et al., 2025), ensemble models that take into account metadata (Hasan and Rifat, 2024), and transformer ensembles that utilize uncertainty as part of the fusion process (Zoravar et al., 2024).

2.1.2 Skin Lesion Datasets and Benchmarking

ISIC 2019 and ISIC 2020 are recognized as the standard benchmarks for skin lesion classification because they provide a number of challenges, such as class imbalance, heterogeneous quality of the images, and the presence of out-of-distribution samples. The top-performing methods typically utilize multi-resolution architectures, extensive augmentation of the training data, and a loss function that balances the classes (Gessert et al., 2020). Additionally, incorporating structured demographic information into the classification process increases diagnostic performance (Nunnari et al., 2020). However, the presence of artifacts such as occlusions caused by hair or measurement markers makes the preprocessing of the data more complex (Wegley et al., 2026). Finally, the presence of demographic imbalances in the datasets implies that there may be issues with classification fairness or generalizability (Alipour et al., 2024).

2.1.3 Federated Learning in Medical Imaging

Federated Learning (FL) allows for private, cooperative training that meets the GDPR and HIPAA laws (Guan et al., 2024). Initial implementations used small convolutional neural networks that ran Federated Averaging to learn from multiple institutions (Subramanian et al., 2025). By utilizing communication-efficient techniques, such as asynchronous updates at the layer level, the amount of bandwidth required was significantly reduced (Yaqoob et al., 2023). However, there are still issues with data being non-independent and identically distributed (non-IID) between the various participating institutions, and class imbalance exists as well. Dynamic Adaptive Focal Loss is one method of correcting for class imbalance, but it also increases the computational burden and sensitivity to hyperparameters involved (Zhao et al., 2025).

2.1.4 Federated Vision Transformers

Federated settings with Vision Transformers create additional computation and communication challenges compared to Federated Convolutional Neural Networks, due in part to their architecture. Strategies to increase efficiency include masking patches during training (Wu et al., 2024) and pruning layers in the ViT architecture (Paxton et al., 2024). In addition to being adversely affected by non-IID data, feature collapse is also an issue with Federated ViT learning. Some alignment-based methods have been developed to ensure consistent representation of local and global attention (Darzi et al., 2024). Lastly, personalization and continual learning strategies are being investigated to improve the adaptability of the Federated ViT learning process (Zuo et al., 2024; Yao et al., 2024).

2.1.5 Explainability, Security and System Architectures

Concerns regarding the robustness, privacy, and interpretability of federated systems have been documented. One way to improve the stability of representations in a federated system is through self-ensembling (Almalik et al., 2022). Techniques for mitigating the effects of adversarial attacks include diffusion-based defenses (Imam et al., 2023). Privacy-preserving techniques, such as random projection, have been suggested as a way to improve communication security between federated clients and providers (Narimani et al., 2024). In multimodal systems, partial personalization leads to better adaptation at the local client; however, it increases the potential for client drift, and thus optimization-based corrections are required (Zhu et al., 2025).

2.2 Comparative Analysis

Standalone architectures have consistently been outperformed by hybrid CNN–Transformer models when used in centralized cases (Naeem et al., 2025; Alshdaifat et al., 2025). In federated environments, a trade-off exists between model consistency and computational resource efficiency. Alignment-based approaches result in improved model stability but come with a very high overhead cost. Efficiency-based methods reduce the amount of computational resources required to run the models, but at the expense of representational fidelity.

2.3 Relation to Personalized and Drift-Corrected Federated Learning

FedSA-Drift combines partial parameter personalization with adaptive server-side aggregation. Neither ingredient is new, and we do not claim otherwise. What follows states precisely what is and is not different from the established alternatives, since the combination is the contribution rather than either component.

Partial personalization. FedPer (Collins et al., 2021) and FedRep (Collins et al., 2021) keep a classification head local and share a representation; FedBABU trains the body with the head frozen and personalizes only at the end; FedBN keeps normalisation statistics local. FedSA-Drift's Group C is the same idea as FedPer's private head, extended to a multimodal head that also carries the metadata branch. The difference from all four is not the existence of a private head but that the *shared* portion is itself split, with early and deep layers aggregated by different rules—FedPer, FedRep, FedBABU and FedBN all apply one aggregation rule to everything they share.

Drift correction. FedProx adds a proximal term to the local objective; SCAFFOLD maintains per-client control variates that correct the local update direction; FedDyn adds a dynamic regulariser; FedNova normalises for unequal local step counts. All four act on the client. FedSA-Drift leaves the local objective and optimiser untouched and acts only on the aggregate. This is the practical distinction we emphasise: no additional hyperparameter is tuned, no auxiliary state of model size is stored per client, and clients require no modification. It is also a weaker form of correction, since the server cannot see the trajectory that produced a client's parameters.

What is genuinely new here is narrow. Applying depth-dependent aggregation rules within a Vision Transformer backbone—plain averaging for early layers, drift-weighted averaging for deep layers—together with a private multimodal head, is a configuration we have not found in the prior work. Our empirical finding about class abandonment is independent of that configuration and would apply to any comparison of aggregation schemes.

We have not measured against these methods. None of FedProx, SCAFFOLD, FedBN, FedRep, FedPer, FedBABU, FedNova or FedDyn was re-implemented under our conditions. Our comparisons are to FedAvg, to the decomposition without drift weighting, and to a no-decomposition variant. We therefore cannot state whether the gains reported here would persist against a strong personalized baseline such as FedRep or FedBN, and we do not claim they would. Section 4.5.1 isolates the contribution of each of our own components, which is a different and narrower question than superiority over the personalized federated learning literature. Establishing the latter requires the comparison set above and is the primary item of future work.

2.4 Limitations in Prior Work

1. **Client Drift in Heterogeneous Settings:** The client drift phenomenon refers to a non-IID data distribution present in heterogeneous settings, which may cause instability in convergence and thus reduce global performance (Zhu et al., 2025).
2. **Destructive Efficiency Techniques:** Aggressive masking and/or pruning can cause disruption of the representation of hierarchical aspects of transformer models (Wu et al., 2024).
3. **Complex Loss Engineering:** Because of the introduction of adaptive loss functions, the number of hyperparameters has increased, which adds computational overhead in the form of extra computation (Zhao et al., 2025).
4. **Feature Collapse under Uniform Aggregation:** Due to the standard aggregation process used in federated learning, the diversity of hierarchical features will not be maintained under uniform aggregation (Darzi et al., 2024).

2.5 Mathematical and Computational Complexity Analysis

2.5.1 Formal Computational Complexity

The use of large Vision Transformer models in federated learning can require a lot of extra computational resources due to their size, especially when the device is an edge device (Wu et al., 2024). For a model with parameter size $|w|$, local training complexity scales as:

$$\mathcal{O}(E \cdot |w|), \tag{1}$$

where E is the number of local epochs. Optimization-based approaches further increase this cost due to additional penalty terms and dual-variable updates. Alignment-based methods also introduce overhead proportional to the attention dimensions, typically $\mathcal{O}(hd^2)$ (Darzi et al., 2024).

2.5.2 Communication Efficiency Considerations

Standard federated learning models transmit all the parameters of the model for every round of training, which results in a very high communication cost. Therefore, due to the full parameter transmission of large transformer models, communication costs are high as well. Compression and selective updates can help reduce bandwidth usage, but typically reduce the performance and representational capacity of the model.

Table 1. Comparison of hybrid and federated skin lesion classification methods and their limitations.

Author(s)	Model / Framework	Key Innovation	Limitation
Naeem et al. (2025)	MedFusionNet	Attention-based fusion of ConvNeXt and ViT	Centralized architecture; does not address federated privacy constraints
Gessert et al. (2020)	Multi-Resolution EfficientNets	Ensemble with metadata fusion (ISIC 2019 winner)	Requires centralized metadata aggregation
Zhao et al. (2025)	FL with DAFL	Dynamic loss adaptation for class imbalance	Hyperparameter-sensitive; increased computational overhead
Wu et al. (2024)	EFTViT	Patch masking for efficient federated ViT training	Disrupts spatial and hierarchical representations
Darzi et al. (2024)	FedMHA	Attention alignment to mitigate feature collapse	Computationally expensive regularization
Zhu et al. (2025)	FedAPM	ADMM-based personalization to reduce drift	High optimization complexity
Yaqoob et al. (2023)	Async-FL-CNN	Asynchronous updates for communication efficiency	Limited to CNNs
Paxton et al. (2024)	Skewness-Guided Pruning	Model compression for edge deployment	Reduces model capacity
Narimani et al. (2024)	FedRP	Random projection for privacy preservation	Does not address data heterogeneity

3 METHODOLOGY

3.1 Methodology Overview

This part discusses how to create the FedSA-Drift training workflow step by step.

1. Set up the federated multimodal problem and notation.
2. Create heterogeneous client datasets with Dirichlet partitions.
3. Create a centralized upper bound baseline.
4. Decompose the parameters of the model into groups that preserve the structure of the data.
5. Train the model using local computation and then aggregate the results globally using drift-aware weighting.

Figure 2 shows the structure-aware federated aggregation architecture.

3.2 Problem Setup and System View

The goal of FedSA-Drift is to train a shared model across K clients without exchanging any raw data between clients. Each client holds their own local dataset:

$$\mathcal{D}_k = \{(x_i^k, m_i^k, y_i^k)\}_{i=1}^{n_k} \tag{2}$$

where x_i^k represents the i-th dermoscopic image, m_i^k represents the clinical metadata associated with that image in structured format, and $y_i^k \in \{1, \ldots, C\}$ represents the label or class of the lesion in the image. The total number of samples across K clients is $N = \sum_{k=1}^{K} n_k$.

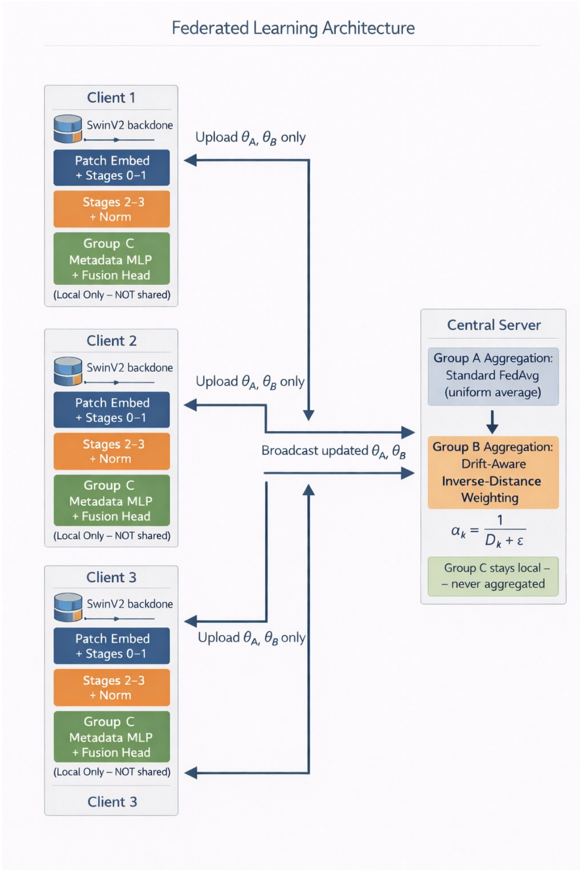


Figure 2. Structure-aware federated aggregation architecture.

The common goal is to develop a multimodal classifier defined as:

$$f_\theta : (x, m) \mapsto \hat{y} \tag{3}$$

where θ is the parameter vector of the model and $\hat{y}$ is the predicted classification label.

3.3 Step 1: Dirichlet Non-IID Data Partitioning

Each client's data is sampled from a Dirichlet prior to simulate the true heterogeneity that exists across multiple clients:

$$\pi_k \sim \mathrm{Dir}(\alpha) \tag{4}$$

where $\pi_k \in \Delta^{C-1}$ is the class-proportion vector for client k and $\alpha > 0$ is the concentration parameter. A smaller value for α gives more skewed class distributions across clients, and a larger value provides more uniform ones. We evaluate $\alpha \in \{1.0, 0.5, 0.1\}$. At $\alpha = 0.1$ the draw is degenerate in a way that matters for the results: with $K = 5$ one client receives no samples whatsoever for four of the eight classes, two more are missing three and two classes respectively, and that same largest client receives 53% of the federation's data. A partition is accepted only if every client holds at least two classes and at least 100 samples; this admits the $\alpha = 0.1$ draw used here, whose per-client class counts are listed in Section 4.3.3.

The fraction of class-c samples at client k is:

$$p_{kc} = \frac{n_{kc}}{n_k} \tag{5}$$

The variance of these proportions across multiple clients is given by:

$$\mathrm{Var}(p_{kc}) = \frac{\alpha(C\alpha - \alpha)}{(C\alpha)^2(C\alpha + 1)} \tag{6}$$

As $\alpha \to 0$, the variance increases, which represents the degree to which distributions across multiple clients will diverge based on the value assigned to the alpha parameter, leading to client gradients diverging:

$$\nabla F_k(\theta) \neq \nabla F_j(\theta), \quad k \neq j \tag{7}$$

This gradient misalignment is the core challenge in the federated learning scenario.

3.4 Step 2: Centralized Training Baseline

The data of all clients must be pooled together in order to create a reference upper bound:

$$\mathcal{D} = \bigcup_{k=1}^{K} \mathcal{D}_k \tag{8}$$

Empirical risk minimization (ERM) is used as the central target:

$$F(\theta) = \frac{1}{N} \sum_{k=1}^{K} \sum_{i=1}^{n_k} \ell(f_\theta(x_i^k, m_i^k), y_i^k) \tag{9}$$

where $\ell(\cdot, \cdot)$ is the per-sample cross-entropy loss. Written schematically as gradient descent, the update dynamics are:

$$\theta^{t+1} = \theta^t - \eta \nabla F(\theta^t) \tag{10}$$

All reported experiments use AdamW rather than plain gradient descent; Equation (10) is written in this form only to define the objective being descended.

where $\nabla F(\theta^t) = \frac{1}{N} \sum_k \sum_i \nabla \ell(f_\theta(x_i^k, m_i^k), y_i^k)$.

The model is trained for E epochs using AdamW with a cosine learning rate schedule:

$$\theta^E = \mathrm{AdamW}(\theta^0, \mathcal{D}) \tag{11}$$

In a centralized context, there is no client drift and therefore no communication overhead. As such, the performance of this model acts as a reference upper bound for all subsequent federated experiments.

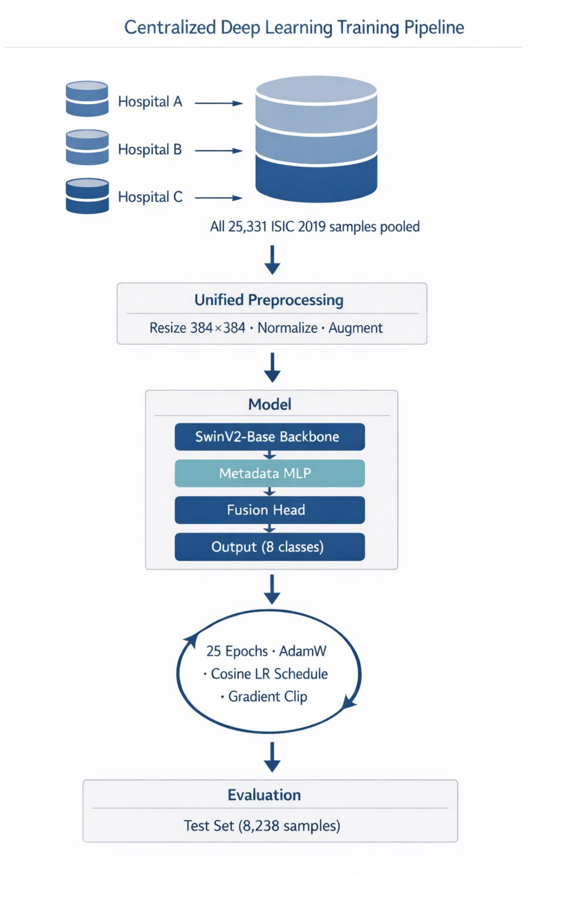


Figure 3. Centralized training pipeline using pooled multimodal data.

3.5 Step 3: Federated Optimization Objective

In federated learning, the global goal is the weighted average of all of the local objectives:

$$F(\theta) = \sum_{k=1}^{K} \frac{n_k}{N} F_k(\theta) \tag{12}$$

where $F_k(\theta)$ is the risk value for each client and is defined as:

$$F_k(\theta) = \frac{1}{n_k} \sum_{i=1}^{n_k} \ell(f_\theta(x_i^k, m_i^k), y_i^k) \tag{13}$$

The clients' local objectives combine into the global optimization objective:

$$\nabla F(\theta) = \sum_{k=1}^{K} \frac{n_k}{N} \nabla F_k(\theta) \tag{14}$$

When there are differences between the distributions of clients, this approximation becomes biased due to client drift.

3.6 Step 4: Structure-Aware Parameter Decomposition

To mitigate drift, FedSA-Drift assigns each parameter to one of three functional groups:

$$\theta = \{\theta_A, \theta_B, \theta_{C,k}\} \tag{15}$$

- **Group A** (θ_A) contains the patch embedding and early SwinV2 stages (0–1). These capture low-level visual features that generalize across institutions, and are aggregated by standard FedAvg.
- **Group B** (θ_B) contains the late SwinV2 stages (2–3) and layer normalization. These encode high-level semantics that are most sensitive to distribution shifts, and are aggregated with drift-aware weighting.
- **Group C** ($\theta_{C,k}$) contains the client-specific metadata MLP and fusion head. These are never transmitted to the server.

The predicted output for sample i at client k is:

$$\hat{y}_i^k = g_{\theta_{C,k}}\Big(h_{\theta_A,\theta_B}(x_i^k),\, m_i^k\Big) \tag{16}$$

where h_{θ_A,θ_B} is the shared SwinV2 backbone and $g_{\theta_{C,k}}$ is the local fusion head. As long as Group C is maintained locally, this enables the models to be personalized for each institution while still securing the confidentiality of the metadata.

3.7 Step 5: Local Client Optimization

Clients apply local training of the models via AdamW, where the backbone is trained at a 0.1η learning rate and the metadata MLP and fusion head use the η learning rate. In addition, a cosine schedule with linear warmup is used for all parameters in both groups. The update rule is:

$$\theta_k^{t+1} = \text{AdamW}(\theta^t, \nabla F_k(\theta^t), \eta) \tag{17}$$

For the structured (grouped) parameters:

$$\theta_{A,k}^{t+1} = \text{AdamW}\big(\theta_A^t,\, \nabla_{\theta_A} F_k,\, 0.1\eta\big) \tag{18}$$

$$\theta_{B,k}^{t+1} = \text{AdamW}\big(\theta_B^t,\, \nabla_{\theta_B} F_k,\, 0.1\eta\big) \tag{19}$$

where $\nabla_{\theta_A} F_k$ and $\nabla_{\theta_B} F_k$ are the partial gradients of local risk at the clients. These per-client updates diverge across institutions under non-IID conditions. It is for this reason that drift-aware aggregation is needed, as described in Section 3.8.

3.8 Step 6: Server-Side Drift-Aware Aggregation

Once local training is complete, the parameters for Groups A and B are uploaded to the server, where standard FedAvg is used to aggregate Group A and drift-aware weighting is used to de-emphasise updates from clients in Group B that deviate from the consensus. Furthermore, there are no changes to the local training objectives or the addition of hyperparameters.

3.8.1 Distance Computation

The server computes the mean across clients for every parameter tensor $l \in \theta_B$:

$$\bar{w}_l = \frac{1}{K} \sum_{k=1}^{K} w_{k,l} \tag{20}$$

where $w_{k,l}$ is the tensor uploaded by client k. To calculate the client drift from the mean across clients, the Euclidean distance between the client update and the mean is computed:

$$D_{k,l} = \|w_{k,l} - \bar{w}_l\|_2 \tag{21}$$

The L_2 norm is used here because model parameters exist within a Euclidean space. It is computationally efficient and gives an accurate measurement of how far every client update deviates from the mean across clients.

3.8.2 Inverse-Distance Weighting

Each client receives an unnormalised weight calculated using its drift:

$$\alpha_{k,l} = \frac{1}{D_{k,l} + \epsilon} \tag{22}$$

where $\epsilon = 10^{-6}$ avoids division by zero. This therefore gives more weight to clients that are near consensus and proportionally less weight to clients that are far from consensus.

The inverse approach is preferred to an exponential, e.g., $\exp(-\beta D_{k,l})$, as it has the following advantages:

- It does not require any additional sensitivity parameter β.
- It will not cause a complete collapse of weights toward 0 in the case of large values of $D_{k,l}$.
- A smooth heavy-tailed profile of decay in the weights is achieved.

Weights are normalized so that they sum to 1:

$$\hat{\alpha}_{k,l} = \frac{\alpha_{k,l}}{\sum_{j=1}^{K} \alpha_{j,l}} \tag{23}$$

The total combined Group B tensor is:

$$\theta_{B,l}^{t+1} = \sum_{k=1}^{K} \hat{\alpha}_{k,l} \cdot w_{k,l} \tag{24}$$

3.8.3 Convergence Analysis (Informal Argument)

Assumptions. The following informal argument assumes that each local objective F_k is L-smooth (i.e., $\|\nabla F_k(\mathbf{u}) - \nabla F_k(\mathbf{v})\| \leq L\|\mathbf{u} - \mathbf{v}\|$ for all $\mathbf{u}, \mathbf{v}$) and that stochastic gradient noise has bounded second moment. This analysis is not a formal convergence proof and does not derive a rate; its purpose is to make precise the intuitive stability benefit of drift-aware attenuation under standard conditions.

Let $\Delta_k^t = \mathbf{w}_k^t - \mathbf{w}^t$ denote the local update of client k at round t. Standard FedAvg combines updates as:

$$\mathbf{w}^{t+1} = \mathbf{w}^t + \sum_{k=1}^{K} p_k \Delta_k^t \tag{25}$$

where $p_k = n_k/N$. In FedSA-Drift, each Group B update is attenuated by a scalar factor $\gamma_k^t \in (0, 1]$ derived from the inverse-distance weight $\hat{\alpha}_{k,l}$, giving an effective update $\tilde{\Delta}_k^t = \gamma_k^t \Delta_k^t$:

$$\mathbf{w}^{t+1} = \mathbf{w}^t + \sum_{k=1}^{K} p_k \tilde{\Delta}_k^t \tag{26}$$

Since $\gamma_k^t \leq 1$ for all k, the ℓ_2 norm of the aggregate step is bounded:

$$\left\| \sum_{k=1}^{K} p_k \tilde{\Delta}_k^t \right\|_2 \leq \left\| \sum_{k=1}^{K} p_k \Delta_k^t \right\|_2 \tag{27}$$

Clients with the largest drift (large $D_{k,l}$, hence small γ_k^t) receive the smallest weight, pulling the aggregate toward the cross-client consensus.

We must be explicit about the gap between this expression and Algorithm 2. Equation (27) holds for an aggregate of the form $\sum_k p_k \gamma_k \Delta_k$ with size weights $p_k = n_k/N$ retained and $\gamma_k \leq 1$ applied on top. The implemented Group B rule does not do this: it *replaces* the size weights with normalised inverse-distance weights $\hat{\alpha}_{k,l}$ that sum to one, so a client close to consensus can receive a weight substantially larger than its p_k. The contraction in Equation (27) therefore does not follow for the algorithm as implemented, and we do not claim it does. A second gap is that the aggregation rules of Equation (24) and its Group A counterpart combine parameters $\mathbf{w}_{k,l}$ whereas Equations (27) onward reason about updates Δ_k; these coincide only when every client starts the round from the same point, which holds for Groups A and B but not for the personalized Group C.

What follows below should accordingly be read as motivation for why down-weighting divergent clients is expected to reduce round-to-round oscillation, not as a proof about the deployed procedure. A convergence analysis of normalised inverse-distance aggregation is left to future work.

3.8.4 Properties and Failure Modes of the Weighting

The weight assigned to client k for parameter tensor l is $\hat{\alpha}_{k,l} = (D_{k,l} + \epsilon)^{-1} / \sum_j (D_{j,l} + \epsilon)^{-1}$ with $\epsilon = 10^{-6}$. Three properties follow directly and should be stated, since two of them are potential failure modes.

The scheme can be dominated by a single client. Writing $D_{\min}$ and $D_{\max}$ for the smallest and largest drift among the K clients,

$$\hat{\alpha}_{\max} \;\le\; \left[1 + (K-1)\frac{D_{\min}+\epsilon}{D_{\max}+\epsilon}\right]^{-1}. \tag{28}$$

As $D_{\min} \to 0$ with $\epsilon \ll D_{\max}$, this bound approaches 1: a client whose Group B parameters coincide with the cross-client mean absorbs essentially the entire aggregate. Because $\bar{\mathbf{w}}_l$ is the unweighted mean, $\sum_k(\mathbf{w}_{k,l} - \bar{\mathbf{w}}_l) = 0$, so $D_{\min}$ cannot vanish unless all clients are identical; but it can become small, and the guard against this is ϵ alone. We did not observe divergence at any setting tested, including α=0.1, but we have not characterised the regime in which it would occur, and a bounded or temperature-controlled variant would be the natural remedy.

Dataset size is discarded, and the resulting bias must be stated. Unlike FedAvg, the Group B rule does not weight by n_k: a client holding 1,015 samples and one holding 11,339 are treated identically if their drift is equal. This is deliberate, but it changes the objective being optimised and warrants a formal statement.

Federated averaging with $p_k = n_k/N$ is the natural estimator for the pooled empirical risk $F(\mathbf{w}) = \sum_k p_k F_k(\mathbf{w})$, in the sense that the aggregate is an unbiased combination of per-client quantities under that weighting. Replacing p_k with $\hat{\alpha}_{k,l}$ introduces a per-tensor bias

$$\mathbf{b}_l \;=\; \sum_{k=1}^{K} \big(\hat{\alpha}_{k,l} - p_k\big)\,\mathbf{w}_{k,l}, \tag{29}$$

so the aggregate no longer estimates the minimiser of F. The magnitude of $\mathbf{b}_l$ is governed by the relationship between drift and dataset size. Where the two are positively associated—large clients producing more stable, near-consensus updates because they hold more data—the reweighting is mild and $\hat{\alpha}_{k,l} \approx p_k$. Where they are negatively associated, the reweighting is large and deliberate.

The α=0.1 partition is the second case, and it is the case the method is built for. Client 2 holds 52.7% of the federation's data yet contains only four of the eight classes and no melanoma at all. Under size weighting it determines the aggregate, which is precisely the mechanism producing the zero melanoma recall documented in Section 4.5.1; under inverse-distance weighting its influence is reduced to the extent that its parameters diverge from consensus. The bias in Equation (29) is therefore not incidental—it is the effect being sought. What FedSA-Drift optimises is closer to a robust consensus among clients than to the pooled empirical risk, and we should describe it that way rather than as an unbiased federated estimator.

Clinically, this is a defensible trade in a specific circumstance and a poor one in others. A federation containing a large referral centre with a skewed case mix is the motivating case: size weighting hands that centre the model, and the resulting model can fail silently on categories it rarely sees. Where instead the largest site is also the most representative—a plausible and common situation—discarding n_k discards genuine statistical information, and FedSA-Drift would be expected to underperform FedAvg. We have not tested that regime. Combined with the concentration bound above, the failure mode to be aware of is a small site whose parameters sit near the mean acquiring influence disproportionate to the evidence it contributes.

An obvious middle ground, $\hat{\alpha}_{k,l} \propto n_k/(D_{k,l} + \epsilon)$, retains size information while still suppressing divergence and would interpolate between the two objectives. We did not evaluate it, and we do not claim the size-free form is optimal.

On the choice of Euclidean distance. The reviewer question of whether cosine similarity would serve better admits a partly analytical answer, because the two are not interchangeable in this formulation.

Every client begins the round from the same broadcast point, so $\mathbf{w}_{k,l} = \mathbf{w}_l^t + \Delta_{k,l}$ where $\Delta_{k,l}$ is client k's local update. The cross-client mean is $\bar{\mathbf{w}}_l = \mathbf{w}_l^t + \bar{\Delta}_l$, and therefore

$$D_{k,l} = \left\|\mathbf{w}_{k,l} - \bar{\mathbf{w}}_l\right\|_2 = \left\|\Delta_{k,l} - \bar{\Delta}_l\right\|_2 . \tag{30}$$

The shared component cancels exactly. Although the rule is written over parameters, it measures the deviation of each client's *update* from the mean update, which is the quantity of interest.

Cosine similarity has no such property. Since $\|\Delta_{k,l}\| \ll \|\mathbf{w}_l^t\|$ after a single local epoch, $\cos(\mathbf{w}_{k,l}, \bar{\mathbf{w}}_l) = 1 - \mathcal{O}\big(\|\Delta\|^2/\|\mathbf{w}^t\|^2\big)$ for every client: the measure is dominated by the shared broadcast point and is near-degenerate. In a numerical check with $\|\Delta\|/\|\mathbf{w}^t\| = 0.04$, representative of one local epoch at our learning rates, the spread of cosine similarities across five clients was 1.4×10^{-5}, against a spread of 1.7×10^{-2} for the same quantity computed on the updates. Substituting cosine for Euclidean in Equation (30) without also re-centring would therefore produce weights that are essentially uniform, and the method would degenerate to plain averaging over Group B.

The meaningful alternative is cosine similarity on the updates, $\cos(\Delta_{k,l}, \bar{\Delta}_l)$, which discards magnitude and penalises only directional disagreement. This is a substantively different criterion: it would treat a client that moved far along the consensus direction as undrifted, whereas the Euclidean rule penalises it. Which is preferable is an empirical question we have not settled, and we make no claim of optimality for the Euclidean form. We note only that it is not arbitrary—by Equation (30) it is the natural metric on updates—and that the naive substitution the question suggests would not work. Softmax, exponential and inverse-square weightings likewise remain untested.

Applying the standard descent lemma for L-smooth objectives and taking expectations over the stochastic gradient noise:

$$\mathbb{E}[F(\mathbf{w}^{t+1})] \;\le\; F(\mathbf{w}^t) - \eta\|\nabla F(\mathbf{w}^t)\|^2 + \frac{\eta^2 L}{2}\,\mathbb{E}\left\|\sum_k p_k \tilde{\Delta}_k^t\right\|^2 \tag{31}$$

The third term—the noise amplification from the aggregate step—is bounded above by the corresponding FedAvg quantity via Eq. (27). Intuitively, by suppressing drifted updates, FedSA-Drift reduces the round-to-round oscillation of the global model without altering local training objectives. Deriving tight convergence rates under non-IID data and formally bounding gradient divergence under this aggregation scheme are left to future theoretical analysis.

3.9 Step 7: End-to-End Federated Training Process

Algorithm 1 shows the centralized baseline. Algorithm 2 shows the full FedSA-Drift federated procedure across T communication rounds.

Compared to the centralized system (Algorithm 1), the federated implementation (Algorithm 2) is structurally different. In centralized training, optimization runs directly on the pooled data, while in federated training, clients perform on-site training and the server enforces a global consensus over the shared backbone. The Group C parameters that clients create using their on-site data are never sent to the

Algorithm 1 Centralized Training Baseline

Require: Dataset $\mathcal{D} = \bigcup_{k=1}^{K} \mathcal{D}_k$, pretrained backbone weights θ_A^0, θ_B^0, total epochs E, learning rate η
Ensure: Trained model parameters θ^E

Initialize model: $\theta^0 \leftarrow \{\theta_A^0,\ \theta_B^0,\ \theta_C^0\}$
for epoch $e = 1$ to E **do**
  for each mini-batch $(x_i, m_i, y_i) \in \mathcal{D}$ **do**
    Compute logits: $\hat{y}_i = f_\theta(x_i, m_i)$
    Compute class-weighted CE loss with label smoothing:

$$\mathcal{L} = \mathcal{L}_{CE}^{\text{weighted}}(\hat{y}_i,\ y_i)$$

    Update via AdamW with cosine LR schedule:

$$\theta \leftarrow \theta - \eta\, \nabla_\theta\, \mathcal{L}$$

  end for
  Evaluate on test set; record Balanced Accuracy, F1, ROC-AUC
end for
return θ^E

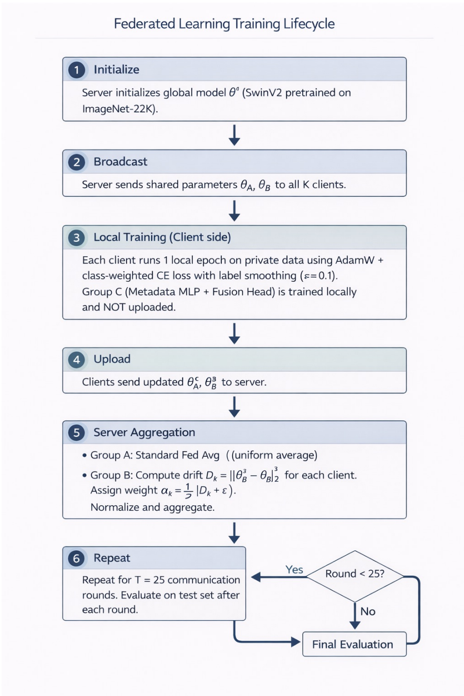


Figure 4. Federated training lifecycle across communication rounds.

server. This structure preserves the privacy of institutional metadata while still providing the capability for institutional specialisation.

3.9.1 Training Flow Diagram

Figure 4 illustrates one communication round of the training process.

Algorithm 2 FedSA-Drift: Structure-Aware Federated Training with Drift-Aware Aggregation

Require: K clients with local datasets $\{\mathcal{D}_k\}_{k=1}^{K}$, rounds T, local epochs E, learning rate η, $\epsilon = 10^{-6}$
Ensure: Global shared parameters $\theta_A^T,\ \theta_B^T$; local parameters $\{\theta_{C,k}^T\}$

Server: Initialize $\theta_A^0,\ \theta_B^0$ from pretrained SwinV2-Base
Each client k initializes local head $\theta_{C,k}^0$ independently
for round $t = 1$ to T **do**
  Server broadcasts $\theta_A^t,\ \theta_B^t$ to all K clients
  for each client $k = 1$ to K **(in parallel) do**
    Load received backbone: $\theta_{A,k} \leftarrow \theta_A^t,\ \ \theta_{B,k} \leftarrow \theta_B^t$
    for local epoch $e = 1$ to E **do**
      for each mini-batch $(x_i^k, m_i^k, y_i^k) \in \mathcal{D}_k$ **do**
        $\hat{y}_i^k = g_{\theta_{C,k}}\Big(h_{\theta_{A,k},\theta_{B,k}}(x_i^k),\ m_i^k\Big)$
        $\mathcal{L}_k = \mathcal{L}_{CE}^{\text{weighted}}(\hat{y}_i^k,\ y_i^k)$
        Update all local params via AdamW (diff. LR): $\theta_{A,k},\ \theta_{B,k},\ \theta_{C,k} \leftarrow \text{AdamW}(\mathcal{L}_k,\ \eta)$
      end for
    end for
    Upload $\theta_{A,k}^{t+1},\ \theta_{B,k}^{t+1}$ to server *(Group C stays local)*
  end for
  Server: Group A aggregation (standard FedAvg):

$$\theta_A^{t+1} = \sum_{k=1}^{K} \frac{n_k}{N}\, \theta_{A,k}^{t+1}$$

  Server: Group B drift-aware aggregation:
  for each parameter tensor $l \in \theta_B$ **do**
    Compute cross-client mean: $\bar{w}_l = \frac{1}{K}\sum_{k=1}^{K} w_{k,l}$
    Compute per-client drift: $D_{k,l} = \|w_{k,l} - \bar{w}_l\|_2$
    Compute inverse-drift weights: $\alpha_{k,l} = \frac{1}{D_{k,l}+\epsilon}$
    Normalize: $\hat{\alpha}_{k,l} = \alpha_{k,l} \big/ \sum_{j=1}^{K} \alpha_{j,l}$
    Aggregate: $\theta_{B,l}^{t+1} = \sum_{k=1}^{K} \hat{\alpha}_{k,l}\, w_{k,l}$
  end for
  Evaluate global model $(\theta_A^{t+1}, \theta_B^{t+1})$ on test set
end for
return $\theta_A^T,\ \theta_B^T,\ \{\theta_{C,k}^T\}_{k=1}^{K}$

3.10 Dataset

FedSA-Drift is evaluated on the ISIC 2019 dataset, which consists of 25,331 dermoscopic images that encompass eight categories: Melanoma (MEL), Melanocytic Nevus (NV), Basal Cell Carcinoma (BCC), Actinic Keratosis (AK), Benign Keratosis (BKL), Dermatofibroma (DF), Vascular Lesion (VASC) and Squamous Cell Carcinoma (SCC). The unknown `UNK` class was removed before training. A separate test set of 8,238 images is available for final evaluation.

3.11 Data Accounting

Because several reported quantities depend on exactly which images are used where, we state the split arithmetic in full.

The ISIC 2019 training archive contains 25,331 labelled dermoscopic images across the eight classes. We hold out a stratified 15% validation split of 3,800 images; the remaining **21,531** images form the training partition that is distributed across clients by the Dirichlet procedure. Summing the per-client class counts reported in Section 4.3.3 reproduces this figure exactly ($1015 + 2687 + 11339 + 4945 + 1545 = 21{,}531$), and the per-class totals reconcile with the archive: for example melanoma contributes 3,844 images to the training partition and 678 to the validation split, together the 4,522 melanoma images the archive contains. No image appears in more than one split, although sibling images of the same lesion do; this is quantified in the Limitations section and bounds how the validation figures should be read.

The ISIC 2019 *test* collection is a separate set of 8,238 images and is not drawn from the 25,331. Its released ground truth assigns 2,047 of those images to an `UNK` (outlier) category that lies outside our eight-class label space; since the model has no `UNK` output, these are excluded and all test-split results are computed on the remaining **6,191** labelled images. Melanoma accounts for 1,327 of them. These 1,327 test melanomas are distinct from the 4,522 in the training archive; the two collections are disjoint, and the melanoma counts should not be summed against the archive total.

One consequence is worth stating plainly. The centralized balanced accuracy of 0.8867 is measured on the internal validation split, which is drawn from the same archive as the training data. It is therefore not comparable with balanced accuracies reported on the ISIC 2019 test collection by challenge entries, and we do not present it as such. Our own test-split figures are substantially lower than our validation figures across every configuration—for example 0.3681 against 0.4382 for FedSA-Drift at α=0.1—which is the expected consequence of evaluating on a genuinely held-out collection.

The images are resized to 384×384 pixels and normalised using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). The augmentation applied to the training images varies and includes random resized crops, random horizontal and vertical flips, rotations up to 45°, color jittering, RandAugment (2 ops, magnitude 9) and random erasing ($p = 0.3$). All augmentations are applied during the training process only.

3.12 Federated Data Partitioning

The 21,531 samples of the training partition (Section 3.11) are distributed across K clients in accordance with a Dirichlet distribution centered at the concentration parameter α. For instance, $\alpha = 1.0$ simulates near-IID conditions, $\alpha = 0.5$ simulates moderately heterogeneous class distributions, and $\alpha = 0.1$ simulates severe skew in which individual clients may hold no samples at all for several classes. All three values are evaluated in this work. Each client is allocated a minimum of 2 class labels and at least 100 samples. If the partition does not pass these criteria, a new random seed is attempted, for a maximum of 50 times.

3.13 Model Architecture

The model architecture consists of a SwinV2-Base (`swinv2_base_window12to24_192to384`, pre-trained on ImageNet-22K) backbone combined with a metadata branch. The SwinV2-Base implements shifted-window self-attention using 4 hierarchical levels to produce a 1024-dimensional feature representation from images. A two-layer metadata MLP ($13 \rightarrow 128 \rightarrow 128$ with GELU activation and dropout) provides a 128-dimensional feature representation of the age, sex, and anatomical site of each patient. This gives a total of 1152 dimensions as a concatenation of 1024 dimensions from the images and 128 dimensions from the metadata. The combined 1152-dimensional representation is then input to a fusion head that reduces the representation from $1152 \rightarrow 512 \rightarrow 8$ to classify each lesion into one of eight categories.

3.14 Training Protocol

The centralized baseline is trained on the same 21,531-image training partition with no client split, for 25 epochs, and is evaluated on the same 3,800-image validation split. It therefore serves as an upper bound under identical data conditions, not as a comparison against externally reported numbers.

Using federated training, FedAvg is used as the baseline aggregation method. The process for each communication round is as follows:

1. The global model is broadcast from the server to all K clients.
2. Each client performs local training on its own data.
3. Each client uploads the parameters to the server for both Group A and Group B.
4. The server combines the updates and sends out a new global model.

A total of 25 communication rounds are completed. The local training loss is computed using class-weighted cross-entropy with label smoothing ($\epsilon_s = 0.1$):

$$\mathcal{L}_k = \mathcal{L}_{CE}^{\text{weighted}}(f_\theta(x^k, m^k),\, y^k) \tag{32}$$

Both drift correction and aggregation updates are completed only at the server; therefore, all local training objectives remain unchanged.

3.15 Optimization Details

The AdamW optimizer is configured via hyperparameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}) with differential learning rates. The SwinV2 backbone uses 0.1η, while $\eta = 4 \times 10^{-4}$ is used for the metadata MLP and the fusion head. The learning rate schedule consists of a cosine decay following a linear warmup of 2 epochs. Gradient clipping is applied so that gradients remain at most 1.0.

Each class is weighted by the inverse of its class frequency. At every round, the `WeightedRandomSampler` oversamples the low-frequency classes at each client. A single batch size of 32 (16 per optimizer step with 2 gradient accumulation steps (effective batch 32 on one device)) is used during centralized training. The batch size is set to 16 with 1 local epoch per round at each federated client.

All experiments were completed using FP16 mixed precision on a single NVIDIA A40 GPU with 48 GB of RAM. All federated experiments are simulated rather than deployed: clients are trained sequentially within a single process, so no network communication takes place. The $\alpha \in \{0.5, 1.0\}$ configurations were each run on one NVIDIA A40; the three α=0.1 configurations were run concurrently on three A40s, one configuration per device, which affects wall-clock time only.

3.16 Evaluation Metrics

Two evaluation sets are used and are kept distinct throughout. Round-by-round metrics during training, and every figure reporting communication rounds, are computed on the 3,800-image *validation* split. Results labelled *test* are computed on the separate ISIC 2019 test collection, restricted to its 6,191 labelled images as described in Section 3.11. The following metrics are reported:

- Top-1 Accuracy and Balanced Accuracy
- Macro F1-Score and Macro ROC-AUC
- Macro Average Precision
- Per-class classification report and confusion matrix

In addition to the above metrics, fairness evaluation is performed using:

- Worst-case client accuracy (minimum across all clients)
- Inter-client standard deviation of accuracy

Every dispersion percentage reported in this paper is a reduction in this *standard deviation*, not in its square. For reference, a 39% reduction in standard deviation corresponds to a 62.4% reduction in variance; we quote the former throughout.

4 RESULTS AND DISCUSSION

The performance of FedSA-Drift is assessed along three dimensions: global classification accuracy, worst-client fairness and computational efficiency. Numerical comparisons with recent state-of-the-art methods are also provided.

4.1 Centralized Upper Bound Performance

FedSA-Drift was trained in a centralized fashion on the 21,531-image training partition for 25 epochs, reaching a peak Balanced Accuracy of **0.8867** at Epoch 21 on the internal validation split. This figure serves only as a reference ceiling obtained under data conditions identical to the federated runs. It is not a generalization estimate: the validation split shares lesions with the training partition (see Limitations), and it is not comparable with balanced accuracies reported on the official ISIC 2019 test collection. Top-1 Accuracy at this point was 0.8566, Macro ROC-AUC was 0.9607 and Macro F1-score was 0.8237. Epoch 21 is the maximum rather than the onset of a plateau: balanced accuracy is lower at Epochs 22–25. We also note that this epoch is selected on the same validation split on which the figure is reported, so 0.8867 is an optimistically selected value and is used here only as a reference ceiling under matched data conditions. Training curves can be found in Figure 5. The terminal output is also available in Figure 6.

The centralized baseline is reported here only as an internal reference ceiling; we make no comparison against externally reported figures, for the reasons given in Section 3.11 and the Limitations.

4.2 Consolidated Federated Performance

The results from each experimental configuration have been summarized in Table 2. These summary results consist of the centralized upper bound together with the three levels of Dirichlet heterogeneity ($\alpha \in \{1.0, 0.5, 0.1\}$) and the aggregation schemes evaluated at each, including the no-decomposition FedAvg baseline added at α=0.1. All comparisons made after this step reference Table 2.

4.3 Mitigating Exacerbated Drift and Ensuring Client Fairness

In multimodal federated networks that keep metadata heads local but share a common backbone, client drift can be exacerbated as the visual backbone is driven toward different local optima (Zhu et al., 2025). This paper confirms this through experiments with FedSA-Drift and describes it in more detail below.

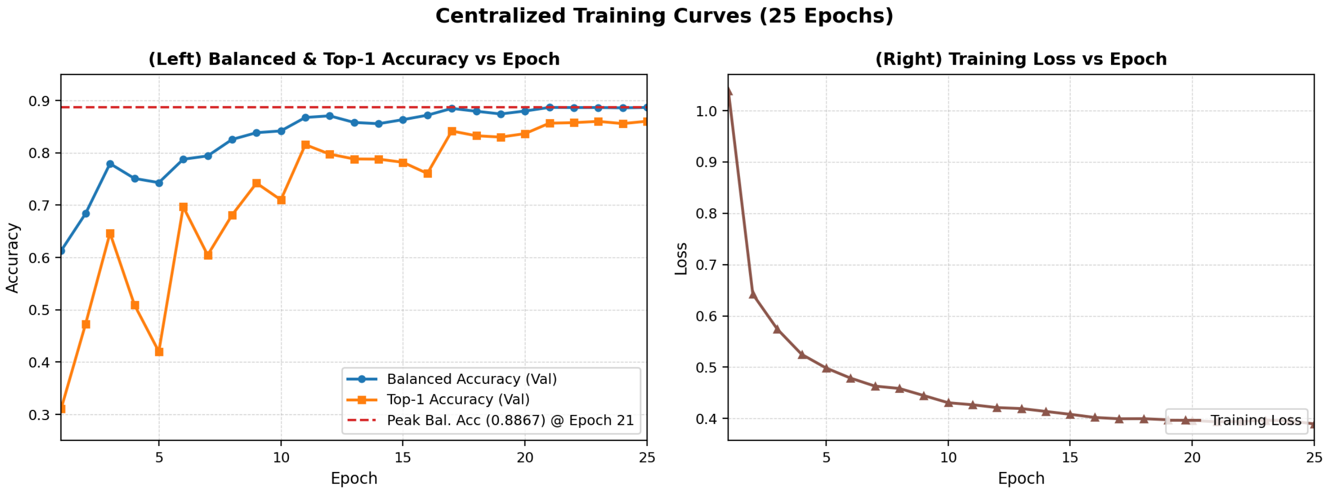


Figure 5. Centralized training curves over 25 epochs: (Left) Balanced Accuracy and Top-1 Accuracy; (Right) Training Loss. Peak Balanced Accuracy of 0.8867 is reached at Epoch 21.

Table 2. Consolidated Performance of FedSA-Drift Across All Experimental Configurations

Experiment	Config (α, K)	Drift Strategy	Global Bal. Acc	Worst Client Acc	Std Dev (Acc)	Avg Round Time
Centralized	N/A	N/A	0.8867	N/A	N/A	1066 s
Fed-2 (Baseline)	α=1.0, K=3	Uniform (Off)	**0.8314**	0.6479	0.0182	1297 s
Fed-1 (Ours)	α=1.0, K=3	Adaptive (On)	0.8209	**0.7850**	**0.0076**	1301 s
Fed-4 (Baseline)	α=0.5, K=5	Uniform (Off)	**0.7622**	0.4800	0.0833	1439 s
Fed-3 (Ours)	α=0.5, K=5	Adaptive (On)	0.7066	**0.5705**	**0.0511**	1450 s
Fed-7 (FedAvg)†	α=0.1, K=5	No decomposition	0.4460	—	—	1229 s
Fed-6 (Baseline)	α=0.1, K=5	Uniform (Off)	0.4477	0.1953	0.1294	1398 s
Fed-5 (Ours)	α=0.1, K=5	Adaptive (On)	**0.5196**	**0.4297**	**0.0551**	1397 s

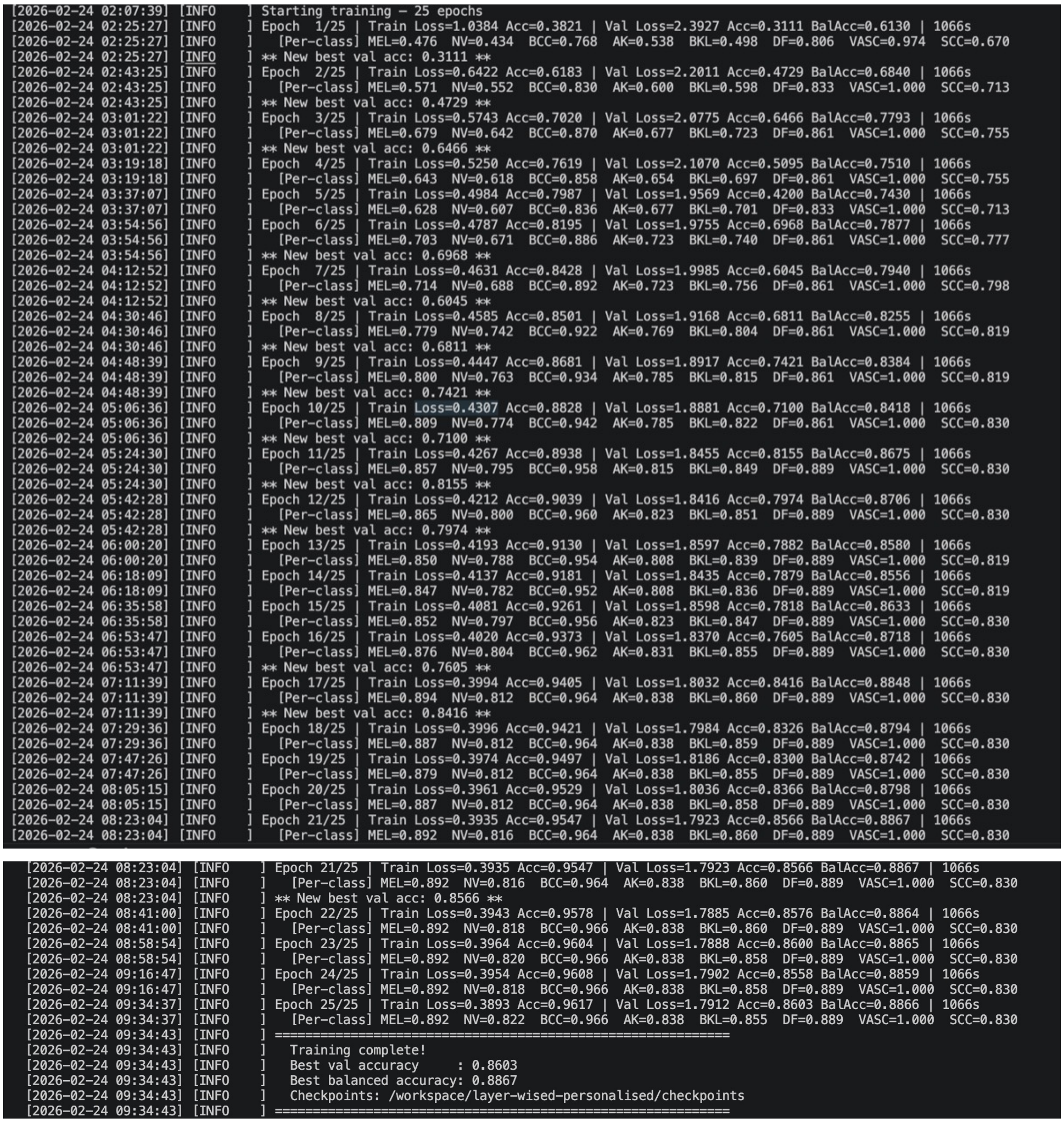
†Fed-7 removes the A/B/C decomposition entirely, so no client-specific head exists and every client evaluates an identical model; worst-client accuracy and inter-client dispersion are undefined for this configuration rather than favourable. Its Global Bal. Acc is also the only entry in this column produced by a fully aggregated network: for the decomposed arms the shared model carries the common initial Group C head rather than an aggregated one (see Limitations). The column is comparable within the decomposed family but not across it; Table 4 gives the like-for-like per-client comparison.

4.3.1 Near-IID Setting ($\alpha = 1.0$, $K = 3$)

With K=3 clients and α=1.0, the class distributions are nearly uniform across clients. Two different aggregation methodologies are compared: 1) standard FedAvg, and 2) FedSA-Drift that uses drift-aware weighting on Group B.

Under these mild levels of client heterogeneity, FedAvg achieves a balanced global accuracy of 0.8314; however, the worst-performing client drops to 0.6479, which would not be apparent from the global average. Using FedSA-Drift, the worst-client accuracy increases by **+13.71 points** to 0.7850, and the inter-client standard deviation decreases by **58%** from 0.0182 to 0.0076 (observed in a single experimental trial). Measured on the shared global model, FedSA-Drift reaches 92.6% and the uniform-aggregation baseline 93.8% of the centralized balanced accuracy, both on the same internal validation split.

The round-by-round training curves and terminal outputs (Figures 7 and 8) clearly demonstrate that the two methodologies converge toward the centralized upper bound, and that FedSA-Drift achieves a tighter client accuracy band and a higher worst-client floor compared to FedAvg.



```
[2026-02-24 02:07:39] [INFO    ] Starting training — 25 epochs
[2026-02-24 02:25:27] [INFO    ] Epoch  1/25 | Train Loss=1.0384 Acc=0.3821 | Val Loss=2.3927 Acc=0.3111 BalAcc=0.6130 | 1066s
[2026-02-24 02:25:27] [INFO    ]   [Per-class] MEL=0.476  NV=0.434  BCC=0.768  AK=0.538  BKL=0.498  DF=0.806  VASC=0.974  SCC=0.670
[2026-02-24 02:25:27] [INFO    ] ** New best val acc: 0.3111 **
[2026-02-24 02:43:25] [INFO    ] Epoch  2/25 | Train Loss=0.6422 Acc=0.6183 | Val Loss=2.2011 Acc=0.4729 BalAcc=0.6840 | 1066s
[2026-02-24 02:43:25] [INFO    ]   [Per-class] MEL=0.571  NV=0.552  BCC=0.830  AK=0.600  BKL=0.598  DF=0.833  VASC=1.000  SCC=0.713
[2026-02-24 02:43:25] [INFO    ] ** New best val acc: 0.4729 **
[2026-02-24 03:01:22] [INFO    ] Epoch  3/25 | Train Loss=0.5743 Acc=0.7020 | Val Loss=2.0775 Acc=0.6466 BalAcc=0.7793 | 1066s
[2026-02-24 03:01:22] [INFO    ]   [Per-class] MEL=0.679  NV=0.642  BCC=0.870  AK=0.677  BKL=0.723  DF=0.861  VASC=1.000  SCC=0.755
[2026-02-24 03:01:22] [INFO    ] ** New best val acc: 0.6466 **
[2026-02-24 03:19:18] [INFO    ] Epoch  4/25 | Train Loss=0.5250 Acc=0.7619 | Val Loss=2.1070 Acc=0.5095 BalAcc=0.7510 | 1066s
[2026-02-24 03:19:18] [INFO    ]   [Per-class] MEL=0.643  NV=0.618  BCC=0.858  AK=0.654  BKL=0.697  DF=0.861  VASC=1.000  SCC=0.755
[2026-02-24 03:37:07] [INFO    ] Epoch  5/25 | Train Loss=0.4984 Acc=0.7987 | Val Loss=1.9569 Acc=0.4200 BalAcc=0.7430 | 1066s
[2026-02-24 03:37:07] [INFO    ]   [Per-class] MEL=0.628  NV=0.607  BCC=0.836  AK=0.677  BKL=0.701  DF=0.833  VASC=1.000  SCC=0.713
[2026-02-24 03:54:56] [INFO    ] Epoch  6/25 | Train Loss=0.4787 Acc=0.8195 | Val Loss=1.9755 Acc=0.6968 BalAcc=0.7877 | 1066s
[2026-02-24 03:54:56] [INFO    ]   [Per-class] MEL=0.703  NV=0.671  BCC=0.886  AK=0.723  BKL=0.740  DF=0.861  VASC=1.000  SCC=0.777
[2026-02-24 03:54:56] [INFO    ] ** New best val acc: 0.6968 **
[2026-02-24 04:12:52] [INFO    ] Epoch  7/25 | Train Loss=0.4631 Acc=0.8428 | Val Loss=1.9985 Acc=0.6045 BalAcc=0.7940 | 1066s
[2026-02-24 04:12:52] [INFO    ]   [Per-class] MEL=0.714  NV=0.688  BCC=0.892  AK=0.723  BKL=0.756  DF=0.861  VASC=1.000  SCC=0.798
[2026-02-24 04:12:52] [INFO    ] ** New best val acc: 0.6045 **
[2026-02-24 04:30:46] [INFO    ] Epoch  8/25 | Train Loss=0.4585 Acc=0.8501 | Val Loss=1.9168 Acc=0.6811 BalAcc=0.8255 | 1066s
[2026-02-24 04:30:46] [INFO    ]   [Per-class] MEL=0.779  NV=0.742  BCC=0.922  AK=0.769  BKL=0.804  DF=0.861  VASC=1.000  SCC=0.819
[2026-02-24 04:30:46] [INFO    ] ** New best val acc: 0.6811 **
[2026-02-24 04:48:39] [INFO    ] Epoch  9/25 | Train Loss=0.4447 Acc=0.8681 | Val Loss=1.8917 Acc=0.7421 BalAcc=0.8384 | 1066s
[2026-02-24 04:48:39] [INFO    ]   [Per-class] MEL=0.800  NV=0.763  BCC=0.934  AK=0.785  BKL=0.815  DF=0.861  VASC=1.000  SCC=0.819
[2026-02-24 04:48:39] [INFO    ] ** New best val acc: 0.7421 **
[2026-02-24 05:06:36] [INFO    ] Epoch 10/25 | Train Loss=0.4307 Acc=0.8828 | Val Loss=1.8881 Acc=0.7100 BalAcc=0.8418 | 1066s
[2026-02-24 05:06:36] [INFO    ]   [Per-class] MEL=0.809  NV=0.774  BCC=0.942  AK=0.785  BKL=0.822  DF=0.861  VASC=1.000  SCC=0.830
[2026-02-24 05:06:36] [INFO    ] ** New best val acc: 0.7100 **
[2026-02-24 05:24:30] [INFO    ] Epoch 11/25 | Train Loss=0.4267 Acc=0.8938 | Val Loss=1.8455 Acc=0.8155 BalAcc=0.8675 | 1066s
[2026-02-24 05:24:30] [INFO    ]   [Per-class] MEL=0.857  NV=0.795  BCC=0.958  AK=0.815  BKL=0.849  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 05:24:30] [INFO    ] ** New best val acc: 0.8155 **
[2026-02-24 05:42:28] [INFO    ] Epoch 12/25 | Train Loss=0.4212 Acc=0.9039 | Val Loss=1.8416 Acc=0.7974 BalAcc=0.8706 | 1066s
[2026-02-24 05:42:28] [INFO    ]   [Per-class] MEL=0.865  NV=0.800  BCC=0.960  AK=0.823  BKL=0.851  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 05:42:28] [INFO    ] ** New best val acc: 0.7974 **
[2026-02-24 06:00:20] [INFO    ] Epoch 13/25 | Train Loss=0.4193 Acc=0.9130 | Val Loss=1.8597 Acc=0.7882 BalAcc=0.8580 | 1066s
[2026-02-24 06:00:20] [INFO    ]   [Per-class] MEL=0.850  NV=0.788  BCC=0.954  AK=0.808  BKL=0.839  DF=0.889  VASC=1.000  SCC=0.819
[2026-02-24 06:18:09] [INFO    ] Epoch 14/25 | Train Loss=0.4137 Acc=0.9181 | Val Loss=1.8435 Acc=0.7879 BalAcc=0.8556 | 1066s
[2026-02-24 06:18:09] [INFO    ]   [Per-class] MEL=0.847  NV=0.782  BCC=0.952  AK=0.808  BKL=0.836  DF=0.889  VASC=1.000  SCC=0.819
[2026-02-24 06:35:58] [INFO    ] Epoch 15/25 | Train Loss=0.4081 Acc=0.9261 | Val Loss=1.8598 Acc=0.7818 BalAcc=0.8633 | 1066s
[2026-02-24 06:35:58] [INFO    ]   [Per-class] MEL=0.852  NV=0.797  BCC=0.956  AK=0.823  BKL=0.847  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 06:53:47] [INFO    ] Epoch 16/25 | Train Loss=0.4020 Acc=0.9373 | Val Loss=1.8370 Acc=0.7605 BalAcc=0.8718 | 1066s
[2026-02-24 06:53:47] [INFO    ]   [Per-class] MEL=0.876  NV=0.804  BCC=0.962  AK=0.831  BKL=0.855  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 06:53:47] [INFO    ] ** New best val acc: 0.7605 **
[2026-02-24 07:11:39] [INFO    ] Epoch 17/25 | Train Loss=0.3994 Acc=0.9405 | Val Loss=1.8032 Acc=0.8416 BalAcc=0.8848 | 1066s
[2026-02-24 07:11:39] [INFO    ]   [Per-class] MEL=0.894  NV=0.812  BCC=0.964  AK=0.838  BKL=0.860  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 07:11:39] [INFO    ] ** New best val acc: 0.8416 **
[2026-02-24 07:29:36] [INFO    ] Epoch 18/25 | Train Loss=0.3996 Acc=0.9421 | Val Loss=1.7984 Acc=0.8326 BalAcc=0.8794 | 1066s
[2026-02-24 07:29:36] [INFO    ]   [Per-class] MEL=0.887  NV=0.812  BCC=0.964  AK=0.838  BKL=0.859  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 07:47:26] [INFO    ] Epoch 19/25 | Train Loss=0.3974 Acc=0.9497 | Val Loss=1.8186 Acc=0.8300 BalAcc=0.8742 | 1066s
[2026-02-24 07:47:26] [INFO    ]   [Per-class] MEL=0.879  NV=0.812  BCC=0.964  AK=0.838  BKL=0.855  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 08:05:15] [INFO    ] Epoch 20/25 | Train Loss=0.3961 Acc=0.9529 | Val Loss=1.8036 Acc=0.8366 BalAcc=0.8798 | 1066s
[2026-02-24 08:05:15] [INFO    ]   [Per-class] MEL=0.887  NV=0.812  BCC=0.964  AK=0.838  BKL=0.858  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 08:23:04] [INFO    ] Epoch 21/25 | Train Loss=0.3935 Acc=0.9547 | Val Loss=1.7923 Acc=0.8566 BalAcc=0.8867 | 1066s
[2026-02-24 08:23:04] [INFO    ]   [Per-class] MEL=0.892  NV=0.816  BCC=0.964  AK=0.838  BKL=0.860  DF=0.889  VASC=1.000  SCC=0.830
```

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[2026-02-24 08:23:04] [INFO    ] Epoch 21/25 | Train Loss=0.3935 Acc=0.9547 | Val Loss=1.7923 Acc=0.8566 BalAcc=0.8867 | 1066s
[2026-02-24 08:23:04] [INFO    ]   [Per-class] MEL=0.892  NV=0.816  BCC=0.964  AK=0.838  BKL=0.860  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 08:23:04] [INFO    ] ** New best val acc: 0.8566 **
[2026-02-24 08:41:00] [INFO    ] Epoch 22/25 | Train Loss=0.3943 Acc=0.9578 | Val Loss=1.7885 Acc=0.8576 BalAcc=0.8864 | 1066s
[2026-02-24 08:41:00] [INFO    ]   [Per-class] MEL=0.892  NV=0.818  BCC=0.966  AK=0.838  BKL=0.860  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 08:58:54] [INFO    ] Epoch 23/25 | Train Loss=0.3964 Acc=0.9604 | Val Loss=1.7888 Acc=0.8600 BalAcc=0.8865 | 1066s
[2026-02-24 08:58:54] [INFO    ]   [Per-class] MEL=0.892  NV=0.820  BCC=0.966  AK=0.838  BKL=0.858  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 09:16:47] [INFO    ] Epoch 24/25 | Train Loss=0.3954 Acc=0.9608 | Val Loss=1.7902 Acc=0.8558 BalAcc=0.8859 | 1066s
[2026-02-24 09:16:47] [INFO    ]   [Per-class] MEL=0.892  NV=0.818  BCC=0.966  AK=0.838  BKL=0.858  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 09:34:37] [INFO    ] Epoch 25/25 | Train Loss=0.3893 Acc=0.9617 | Val Loss=1.7912 Acc=0.8603 BalAcc=0.8866 | 1066s
[2026-02-24 09:34:37] [INFO    ]   [Per-class] MEL=0.892  NV=0.822  BCC=0.966  AK=0.838  BKL=0.855  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 09:34:43] [INFO    ] ============================================================
[2026-02-24 09:34:43] [INFO    ]   Training complete!
[2026-02-24 09:34:43] [INFO    ]   Best val accuracy     : 0.8603
[2026-02-24 09:34:43] [INFO    ]   Best balanced accuracy: 0.8867
[2026-02-24 09:34:43] [INFO    ]   Checkpoints: /workspace/layer-wised-personalised/checkpoints
[2026-02-24 09:34:43] [INFO    ] ============================================================
```

Figure 6. Centralized training run terminal output (NVIDIA A40, RunPod): (Top) Epochs 1–13 showing progressive improvement in validation Balanced Accuracy; (Bottom) Epochs 14–25 converging to peak Balanced Accuracy of 0.8867 at Epoch 21, with per-class recall and final training summary.

4.3.2 Moderately Heterogeneous Setting ($\alpha = 0.5$, $K = 5$)

The introduction of significant class imbalance across institutions (K=5 and α=0.5) produces a more realistic clinical scenario. The use of FedSA-Drift to correct for drift provides an improvement over the worst client (lowest accuracy) of **+9.05 percentage points** from 0.4800 to 0.5705 and reduces the inter-client standard deviation of accuracy by **38.7%** from 0.0833 to 0.0511 (observed in a single experimental trial). The results in Figures 9 and 10 indicate that the convergence of the different model types will be slower and noisier than in the near-IID case. Therefore, the improvement to the performance floor is clearly visible in this comparison.

4.3.3 Severely Heterogeneous Setting ($\alpha = 0.1$, $K = 5$)

At α=0.1 the partition becomes genuinely pathological rather than merely imbalanced. Client 1 holds 2687 samples of which 2645 belong to a single class; client 2 holds 11339 samples—53% of the federation—but has no samples at all for melanoma, BCC, AK or VASC; and melanoma itself is

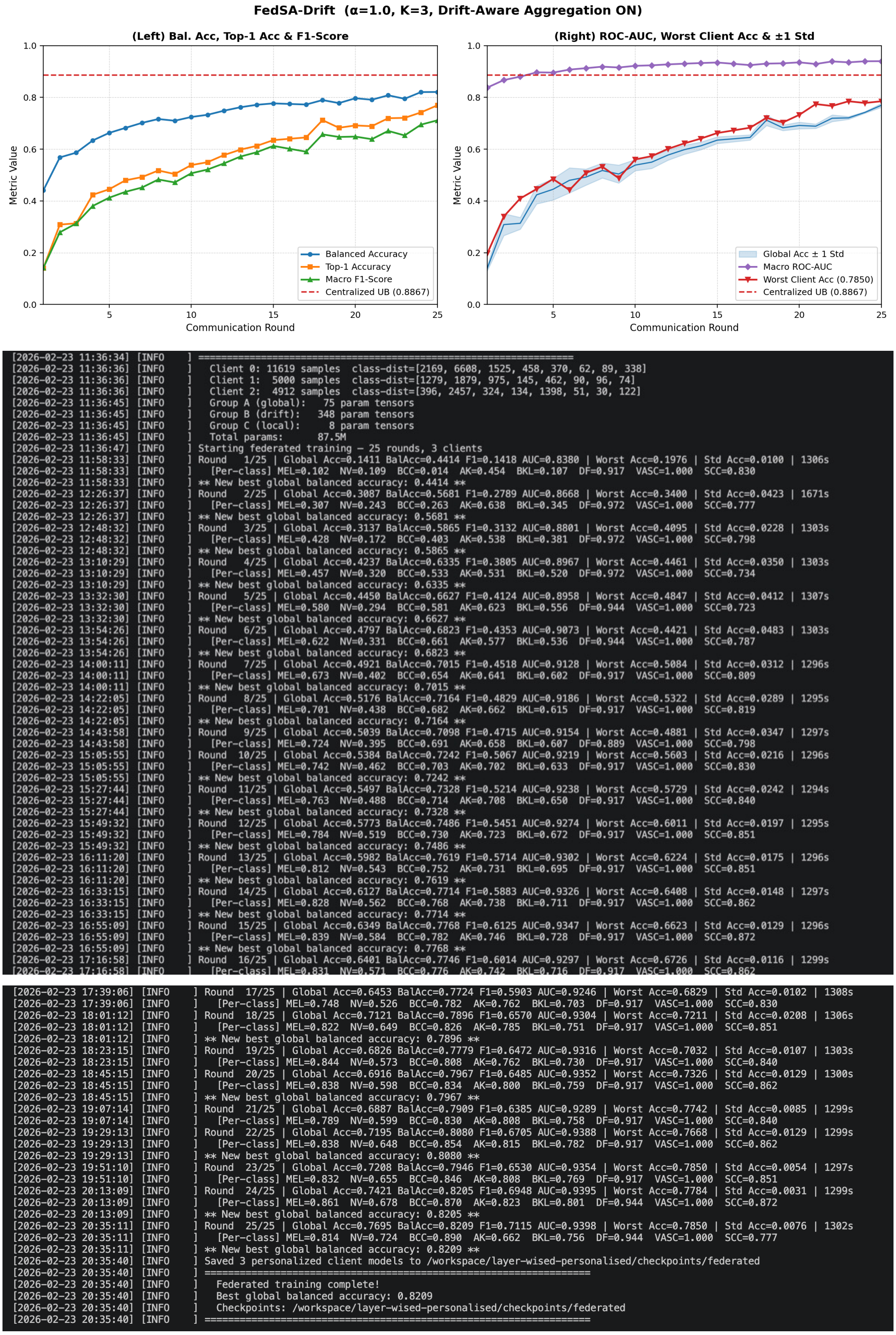
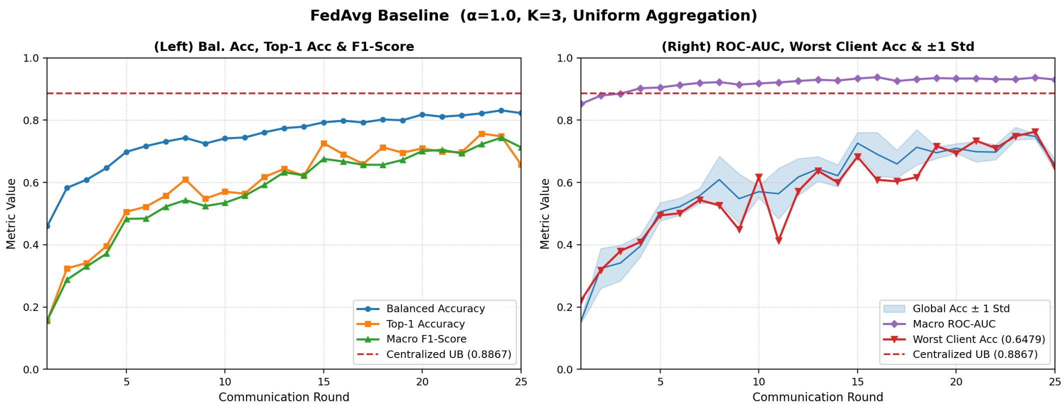


Figure 7. FedSA-Drift near-IID setting (α=1.0, K=3): top panel shows convergence metrics; middle and bottom panels show terminal outputs (Rounds 1–13 and 14–25). Final Balanced Accuracy: 0.8209, Worst Client Accuracy: 0.7850.

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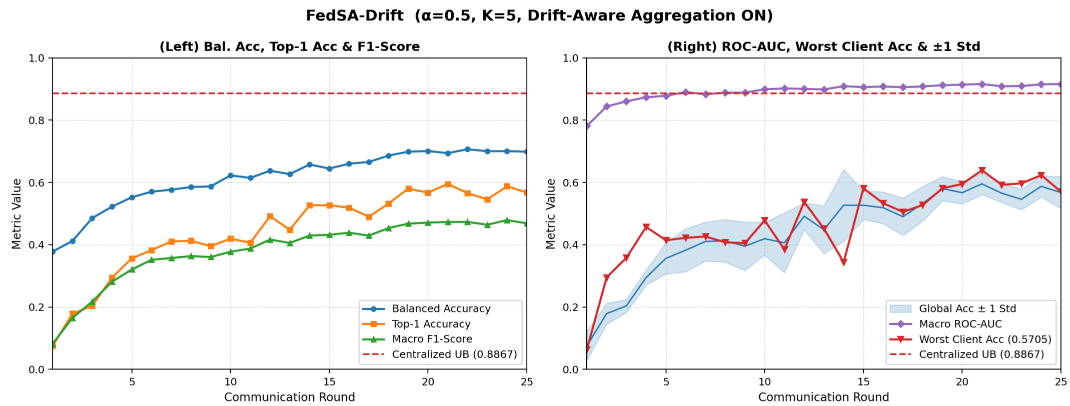
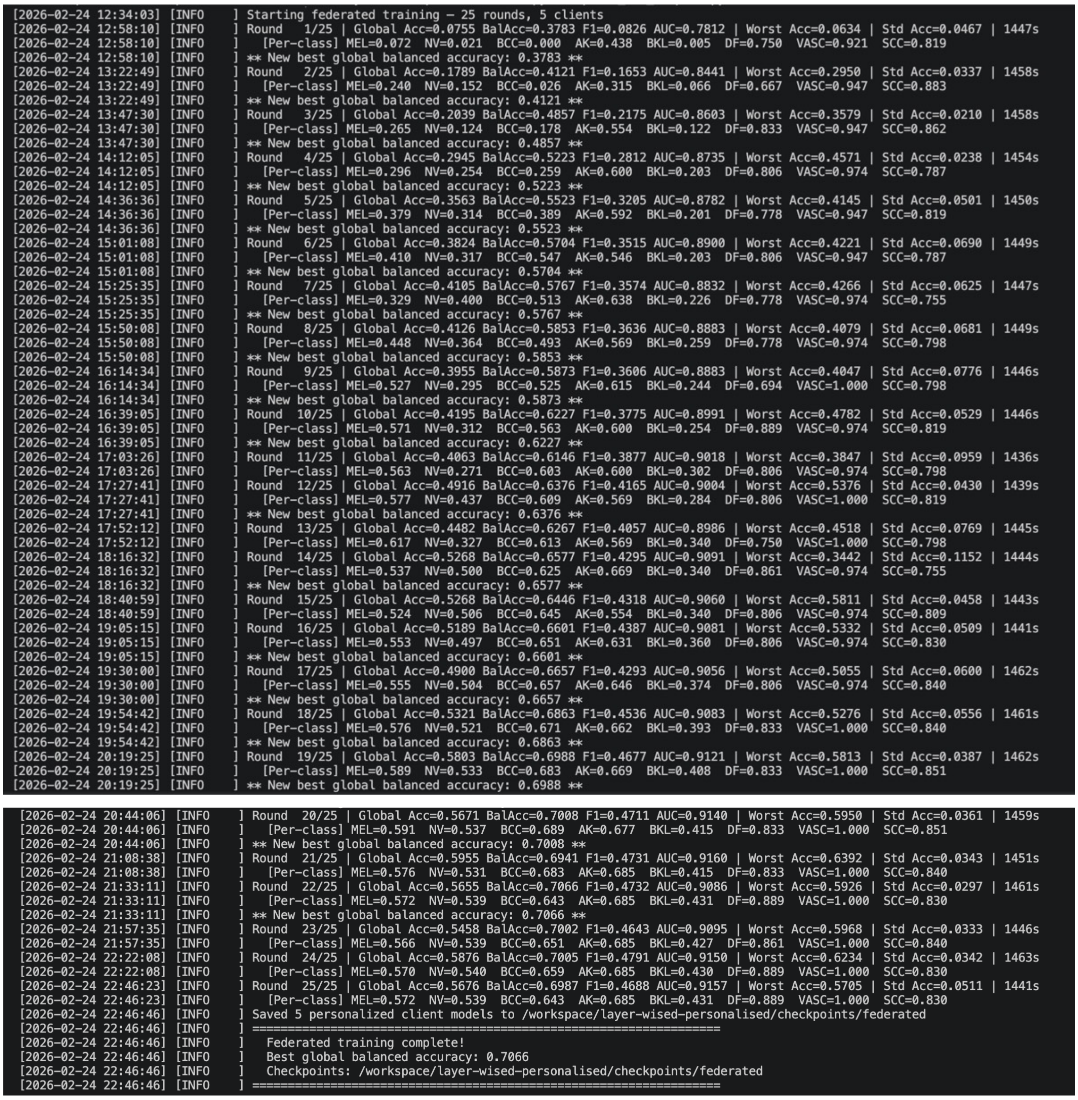
[2026-02-23 11:27:01] [INFO    ] Starting federated training — 25 rounds, 3 clients
[2026-02-23 11:48:41] [INFO    ] Round   1/25 | Global Acc=0.1553 BalAcc=0.4595 F1=0.1570 AUC=0.8521 | Worst Acc=0.2211 | Std Acc=0.0068 | 1300s
[2026-02-23 11:48:41] [INFO    ]   [Per-class] MEL=0.505  NV=0.314  BCC=0.594  AK=0.577  BKL=0.429  DF=0.806  VASC=1.000  SCC=0.713
[2026-02-23 11:48:41] [INFO    ] ** New best global balanced accuracy: 0.4595 **
[2026-02-23 12:10:34] [INFO    ] Round   2/25 | Global Acc=0.3239 BalAcc=0.5827 F1=0.2877 AUC=0.8789 | Worst Acc=0.3187 | Std Acc=0.0641 | 1300s
[2026-02-23 12:10:34] [INFO    ]   [Per-class] MEL=0.590  NV=0.444  BCC=0.727  AK=0.631  BKL=0.556  DF=0.861  VASC=1.000  SCC=0.755
[2026-02-23 12:10:34] [INFO    ] ** New best global balanced accuracy: 0.5827 **
[2026-02-23 12:32:25] [INFO    ] Round   3/25 | Global Acc=0.3416 BalAcc=0.6089 F1=0.3302 AUC=0.8853 | Worst Acc=0.3803 | Std Acc=0.0572 | 1295s
[2026-02-23 12:32:25] [INFO    ]   [Per-class] MEL=0.634  NV=0.459  BCC=0.773  AK=0.662  BKL=0.601  DF=0.889  VASC=1.000  SCC=0.777
[2026-02-23 12:32:25] [INFO    ] ** New best global balanced accuracy: 0.6089 **
[2026-02-23 12:54:22] [INFO    ] Round   4/25 | Global Acc=0.3955 BalAcc=0.6464 F1=0.3713 AUC=0.9023 | Worst Acc=0.4087 | Std Acc=0.0346 | 1299s
[2026-02-23 12:54:22] [INFO    ]   [Per-class] MEL=0.673  NV=0.527  BCC=0.810  AK=0.685  BKL=0.641  DF=0.889  VASC=1.000  SCC=0.798
[2026-02-23 12:54:22] [INFO    ] ** New best global balanced accuracy: 0.6464 **
[2026-02-23 13:16:28] [INFO    ] Round   5/25 | Global Acc=0.5058 BalAcc=0.6979 F1=0.4829 AUC=0.9049 | Worst Acc=0.4942 | Std Acc=0.0291 | 1306s
[2026-02-23 13:16:28] [INFO    ]   [Per-class] MEL=0.728  NV=0.594  BCC=0.856  AK=0.723  BKL=0.686  DF=0.889  VASC=1.000  SCC=0.809
[2026-02-23 13:16:28] [INFO    ] ** New best global balanced accuracy: 0.6979 **
[2026-02-23 13:38:21] [INFO    ] Round   6/25 | Global Acc=0.5221 BalAcc=0.7167 F1=0.4843 AUC=0.9131 | Worst Acc=0.5011 | Std Acc=0.0275 | 1294s
[2026-02-23 13:38:21] [INFO    ]   [Per-class] MEL=0.753  NV=0.612  BCC=0.876  AK=0.738  BKL=0.704  DF=0.889  VASC=1.000  SCC=0.819
[2026-02-23 13:38:21] [INFO    ] ** New best global balanced accuracy: 0.7167 **
[2026-02-23 14:00:11] [INFO    ] Round   7/25 | Global Acc=0.5571 BalAcc=0.7312 F1=0.5221 AUC=0.9195 | Worst Acc=0.5437 | Std Acc=0.0239 | 1295s
[2026-02-23 14:00:11] [INFO    ]   [Per-class] MEL=0.780  NV=0.631  BCC=0.890  AK=0.754  BKL=0.721  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-23 14:00:11] [INFO    ] ** New best global balanced accuracy: 0.7312 **
[2026-02-23 14:22:05] [INFO    ] Round   8/25 | Global Acc=0.6092 BalAcc=0.7431 F1=0.5434 AUC=0.9222 | Worst Acc=0.5263 | Std Acc=0.0750 | 1295s
[2026-02-23 14:22:05] [INFO    ]   [Per-class] MEL=0.806  NV=0.637  BCC=0.902  AK=0.762  BKL=0.733  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-23 14:22:05] [INFO    ] ** New best global balanced accuracy: 0.7431 **
[2026-02-23 14:43:58] [INFO    ] Round   9/25 | Global Acc=0.5479 BalAcc=0.7249 F1=0.5241 AUC=0.9142 | Worst Acc=0.4487 | Std Acc=0.0795 | 1298s
[2026-02-23 14:43:58] [INFO    ]   [Per-class] MEL=0.779  NV=0.617  BCC=0.888  AK=0.746  BKL=0.718  DF=0.889  VASC=1.000  SCC=0.819
[2026-02-23 15:05:55] [INFO    ] Round  10/25 | Global Acc=0.5703 BalAcc=0.7413 F1=0.5345 AUC=0.9178 | Worst Acc=0.6171 | Std Acc=0.0189 | 1298s
[2026-02-23 15:05:55] [INFO    ]   [Per-class] MEL=0.801  NV=0.637  BCC=0.898  AK=0.754  BKL=0.730  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-23 15:27:44] [INFO    ] Round  11/25 | Global Acc=0.5639 BalAcc=0.7443 F1=0.5574 AUC=0.9211 | Worst Acc=0.4132 | Std Acc=0.0813 | 1294s
[2026-02-23 15:27:44] [INFO    ]   [Per-class] MEL=0.821  NV=0.643  BCC=0.906  AK=0.762  BKL=0.742  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-23 15:27:44] [INFO    ] ** New best global balanced accuracy: 0.7443 **
[2026-02-23 15:49:32] [INFO    ] Round  12/25 | Global Acc=0.6174 BalAcc=0.7610 F1=0.5922 AUC=0.9257 | Worst Acc=0.5721 | Std Acc=0.0592 | 1294s
[2026-02-23 15:49:32] [INFO    ]   [Per-class] MEL=0.840  NV=0.660  BCC=0.916  AK=0.777  BKL=0.760  DF=0.889  VASC=1.000  SCC=0.840
[2026-02-23 15:49:32] [INFO    ] ** New best global balanced accuracy: 0.7610 **
[2026-02-23 16:11:20] [INFO    ] Round  13/25 | Global Acc=0.6437 BalAcc=0.7743 F1=0.6329 AUC=0.9297 | Worst Acc=0.6374 | Std Acc=0.0392 | 1292s
[2026-02-23 16:11:20] [INFO    ]   [Per-class] MEL=0.865  NV=0.668  BCC=0.926  AK=0.785  BKL=0.771  DF=0.889  VASC=1.000  SCC=0.851
[2026-02-23 16:11:20] [INFO    ] ** New best global balanced accuracy: 0.7743 **
[2026-02-23 16:33:15] [INFO    ] Round  14/25 | Global Acc=0.6216 BalAcc=0.7791 F1=0.6224 AUC=0.9272 | Worst Acc=0.5997 | Std Acc=0.0351 | 1299s
[2026-02-23 16:33:15] [INFO    ]   [Per-class] MEL=0.876  NV=0.668  BCC=0.930  AK=0.792  BKL=0.779  DF=0.889  VASC=1.000  SCC=0.862
[2026-02-23 16:33:15] [INFO    ] ** New best global balanced accuracy: 0.7791 **
[2026-02-23 16:55:09] [INFO    ] Round  15/25 | Global Acc=0.7261 BalAcc=0.7930 F1=0.6756 AUC=0.9336 | Worst Acc=0.6824 | Std Acc=0.0342 | 1296s
[2026-02-23 16:55:09] [INFO    ]   [Per-class] MEL=0.894  NV=0.680  BCC=0.938  AK=0.800  BKL=0.789  DF=0.889  VASC=1.000  SCC=0.872
[2026-02-23 16:55:09] [INFO    ] ** New best global balanced accuracy: 0.7930 **
[2026-02-23 17:17:05] [INFO    ] Round  16/25 | Global Acc=0.6903 BalAcc=0.7983 F1=0.6673 AUC=0.9379 | Worst Acc=0.6082 | Std Acc=0.0701 | 1298s
[2026-02-23 17:17:05] [INFO    ]   [Per-class] MEL=0.905  NV=0.683  BCC=0.942  AK=0.808  BKL=0.795  DF=0.917  VASC=1.000  SCC=0.872
[2026-02-23 17:17:05] [INFO    ] ** New best global balanced accuracy: 0.7983 **
[2026-02-23 17:39:11] [INFO    ] Round  17/25 | Global Acc=0.6595 BalAcc=0.7926 F1=0.6568 AUC=0.9258 | Worst Acc=0.6037 | Std Acc=0.0443 | 1301s
[2026-02-23 17:39:11] [INFO    ]   [Per-class] MEL=0.896  NV=0.678  BCC=0.938  AK=0.808  BKL=0.791  DF=0.917  VASC=1.000  SCC=0.872
[2026-02-23 18:01:02] [INFO    ] Round  18/25 | Global Acc=0.7129 BalAcc=0.8021 F1=0.6565 AUC=0.9310 | Worst Acc=0.6166 | Std Acc=0.0567 | 1297s
[2026-02-23 18:01:02] [INFO    ]   [Per-class] MEL=0.909  NV=0.687  BCC=0.944  AK=0.815  BKL=0.799  DF=0.917  VASC=1.000  SCC=0.883
[2026-02-23 18:01:02] [INFO    ] ** New best global balanced accuracy: 0.8021 **
[2026-02-23 18:22:54] [INFO    ] Round  19/25 | Global Acc=0.6955 BalAcc=0.7996 F1=0.6724 AUC=0.9352 | Worst Acc=0.7168 | Std Acc=0.0184 | 1294s
[2026-02-23 18:22:54] [INFO    ]   [Per-class] MEL=0.905  NV=0.683  BCC=0.942  AK=0.815  BKL=0.797  DF=0.917  VASC=1.000  SCC=0.883
```

```
[2026-02-23 18:44:50] [INFO    ] Round  20/25 | Global Acc=0.7095 BalAcc=0.8181 F1=0.7006 AUC=0.9336 | Worst Acc=0.6942 | Std Acc=0.0151 | 1304s
[2026-02-23 18:44:50] [INFO    ]   [Per-class] MEL=0.916  NV=0.700  BCC=0.950  AK=0.823  BKL=0.813  DF=0.917  VASC=1.000  SCC=0.883
[2026-02-23 18:44:50] [INFO    ] ** New best global balanced accuracy: 0.8181 **
[2026-02-23 19:06:45] [INFO    ] Round  21/25 | Global Acc=0.6987 BalAcc=0.8111 F1=0.7044 AUC=0.9337 | Worst Acc=0.7342 | Std Acc=0.0330 | 1294s
[2026-02-23 19:06:45] [INFO    ]   [Per-class] MEL=0.909  NV=0.698  BCC=0.948  AK=0.815  BKL=0.805  DF=0.917  VASC=1.000  SCC=0.883
[2026-02-23 19:28:34] [INFO    ] Round  22/25 | Global Acc=0.6974 BalAcc=0.8153 F1=0.6931 AUC=0.9315 | Worst Acc=0.7100 | Std Acc=0.0240 | 1296s
[2026-02-23 19:28:34] [INFO    ]   [Per-class] MEL=0.916  NV=0.701  BCC=0.950  AK=0.823  BKL=0.811  DF=0.917  VASC=1.000  SCC=0.883
[2026-02-23 19:50:24] [INFO    ] Round  23/25 | Global Acc=0.7571 BalAcc=0.8219 F1=0.7227 AUC=0.9311 | Worst Acc=0.7484 | Std Acc=0.0200 | 1297s
[2026-02-23 19:50:24] [INFO    ]   [Per-class] MEL=0.916  NV=0.707  BCC=0.952  AK=0.831  BKL=0.817  DF=0.917  VASC=1.000  SCC=0.872
[2026-02-23 19:50:24] [INFO    ] ** New best global balanced accuracy: 0.8219 **
[2026-02-23 20:12:21] [INFO    ] Round  24/25 | Global Acc=0.7479 BalAcc=0.8314 F1=0.7438 AUC=0.9367 | Worst Acc=0.7629 | Std Acc=0.0083 | 1297s
[2026-02-23 20:12:21] [INFO    ]   [Per-class] MEL=0.916  NV=0.462  BCC=0.900  AK=0.846  BKL=0.701  DF=0.917  VASC=1.000  SCC=0.840
[2026-02-23 20:12:21] [INFO    ] ** New best global balanced accuracy: 0.8314 **
[2026-02-23 20:34:25] [INFO    ] Round  25/25 | Global Acc=0.6576 BalAcc=0.8227 F1=0.7129 AUC=0.9301 | Worst Acc=0.6479 | Std Acc=0.0182 | 1298s
[2026-02-23 20:34:25] [INFO    ]   [Per-class] MEL=0.916  NV=0.462  BCC=0.900  AK=0.846  BKL=0.701  DF=0.917  VASC=1.000  SCC=0.840
[2026-02-23 20:34:54] [INFO    ] Saved 3 personalized client models to /workspace/layer-wised-personalised/checkpoints/federated
[2026-02-23 20:34:54] [INFO    ] ============================================================
[2026-02-23 20:34:54] [INFO    ]   Federated training complete!
[2026-02-23 20:34:54] [INFO    ]   Best global balanced accuracy: 0.8314
[2026-02-23 20:34:54] [INFO    ]   Checkpoints: /workspace/layer-wised-personalised/checkpoints/federated
[2026-02-23 20:34:54] [INFO    ] ============================================================
```

Figure 8. FedAvg near-IID setting (α=1.0, K=3): top panel shows convergence metrics; middle and bottom panels show terminal outputs (Rounds 1–13 and 14–25). Final Balanced Accuracy: 0.8314, Worst Client Accuracy: 0.6479.

concentrated almost entirely on client 3, which holds 3815 of the 3844 cases. Three configurations were trained on this identical partition: the FedAvg baseline with the decomposition removed (Figure 11), decomposition only with drift weighting disabled (Figure 12), and full FedSA-Drift (Figure 13).





```
[2026-02-24 12:34:03] [INFO    ] Starting federated training — 25 rounds, 5 clients
[2026-02-24 12:58:10] [INFO    ] Round   1/25 | Global Acc=0.0755 BalAcc=0.3783 F1=0.0826 AUC=0.7812 | Worst Acc=0.0634 | Std Acc=0.0467 | 1447s
[2026-02-24 12:58:10] [INFO    ]   [Per-class] MEL=0.072  NV=0.021  BCC=0.000  AK=0.438  BKL=0.005  DF=0.750  VASC=0.921  SCC=0.819
[2026-02-24 12:58:10] [INFO    ] ** New best global balanced accuracy: 0.3783 **
[2026-02-24 13:22:49] [INFO    ] Round   2/25 | Global Acc=0.1789 BalAcc=0.4121 F1=0.1653 AUC=0.8441 | Worst Acc=0.2950 | Std Acc=0.0337 | 1458s
[2026-02-24 13:22:49] [INFO    ]   [Per-class] MEL=0.240  NV=0.152  BCC=0.026  AK=0.315  BKL=0.066  DF=0.667  VASC=0.947  SCC=0.883
[2026-02-24 13:22:49] [INFO    ] ** New best global balanced accuracy: 0.4121 **
[2026-02-24 13:47:30] [INFO    ] Round   3/25 | Global Acc=0.2039 BalAcc=0.4857 F1=0.2175 AUC=0.8603 | Worst Acc=0.3579 | Std Acc=0.0210 | 1458s
[2026-02-24 13:47:30] [INFO    ]   [Per-class] MEL=0.265  NV=0.124  BCC=0.178  AK=0.554  BKL=0.122  DF=0.833  VASC=0.947  SCC=0.862
[2026-02-24 13:47:30] [INFO    ] ** New best global balanced accuracy: 0.4857 **
[2026-02-24 14:12:05] [INFO    ] Round   4/25 | Global Acc=0.2945 BalAcc=0.5223 F1=0.2812 AUC=0.8735 | Worst Acc=0.4571 | Std Acc=0.0238 | 1454s
[2026-02-24 14:12:05] [INFO    ]   [Per-class] MEL=0.296  NV=0.254  BCC=0.259  AK=0.600  BKL=0.203  DF=0.806  VASC=0.974  SCC=0.787
[2026-02-24 14:12:05] [INFO    ] ** New best global balanced accuracy: 0.5223 **
[2026-02-24 14:36:36] [INFO    ] Round   5/25 | Global Acc=0.3563 BalAcc=0.5523 F1=0.3205 AUC=0.8782 | Worst Acc=0.4145 | Std Acc=0.0501 | 1450s
[2026-02-24 14:36:36] [INFO    ]   [Per-class] MEL=0.379  NV=0.314  BCC=0.389  AK=0.592  BKL=0.201  DF=0.778  VASC=0.947  SCC=0.819
[2026-02-24 14:36:36] [INFO    ] ** New best global balanced accuracy: 0.5523 **
[2026-02-24 15:01:08] [INFO    ] Round   6/25 | Global Acc=0.3824 BalAcc=0.5704 F1=0.3515 AUC=0.8900 | Worst Acc=0.4221 | Std Acc=0.0690 | 1449s
[2026-02-24 15:01:08] [INFO    ]   [Per-class] MEL=0.410  NV=0.317  BCC=0.547  AK=0.546  BKL=0.203  DF=0.806  VASC=0.947  SCC=0.787
[2026-02-24 15:01:08] [INFO    ] ** New best global balanced accuracy: 0.5704 **
[2026-02-24 15:25:35] [INFO    ] Round   7/25 | Global Acc=0.4105 BalAcc=0.5767 F1=0.3574 AUC=0.8832 | Worst Acc=0.4266 | Std Acc=0.0625 | 1447s
[2026-02-24 15:25:35] [INFO    ]   [Per-class] MEL=0.329  NV=0.400  BCC=0.513  AK=0.638  BKL=0.226  DF=0.778  VASC=0.974  SCC=0.755
[2026-02-24 15:25:35] [INFO    ] ** New best global balanced accuracy: 0.5767 **
[2026-02-24 15:50:08] [INFO    ] Round   8/25 | Global Acc=0.4126 BalAcc=0.5853 F1=0.3636 AUC=0.8883 | Worst Acc=0.4079 | Std Acc=0.0681 | 1449s
[2026-02-24 15:50:08] [INFO    ]   [Per-class] MEL=0.448  NV=0.364  BCC=0.493  AK=0.569  BKL=0.259  DF=0.778  VASC=0.974  SCC=0.798
[2026-02-24 15:50:08] [INFO    ] ** New best global balanced accuracy: 0.5853 **
[2026-02-24 16:14:34] [INFO    ] Round   9/25 | Global Acc=0.3955 BalAcc=0.5873 F1=0.3606 AUC=0.8883 | Worst Acc=0.4047 | Std Acc=0.0776 | 1446s
[2026-02-24 16:14:34] [INFO    ]   [Per-class] MEL=0.527  NV=0.295  BCC=0.525  AK=0.615  BKL=0.244  DF=0.694  VASC=1.000  SCC=0.798
[2026-02-24 16:14:34] [INFO    ] ** New best global balanced accuracy: 0.5873 **
[2026-02-24 16:39:05] [INFO    ] Round  10/25 | Global Acc=0.4195 BalAcc=0.6227 F1=0.3775 AUC=0.8991 | Worst Acc=0.4782 | Std Acc=0.0529 | 1446s
[2026-02-24 16:39:05] [INFO    ]   [Per-class] MEL=0.571  NV=0.312  BCC=0.563  AK=0.600  BKL=0.254  DF=0.889  VASC=0.974  SCC=0.819
[2026-02-24 16:39:05] [INFO    ] ** New best global balanced accuracy: 0.6227 **
[2026-02-24 17:03:26] [INFO    ] Round  11/25 | Global Acc=0.4063 BalAcc=0.6146 F1=0.3877 AUC=0.9018 | Worst Acc=0.3847 | Std Acc=0.0959 | 1436s
[2026-02-24 17:03:26] [INFO    ]   [Per-class] MEL=0.563  NV=0.271  BCC=0.603  AK=0.600  BKL=0.302  DF=0.806  VASC=0.974  SCC=0.798
[2026-02-24 17:27:41] [INFO    ] Round  12/25 | Global Acc=0.4916 BalAcc=0.6376 F1=0.4165 AUC=0.9004 | Worst Acc=0.5376 | Std Acc=0.0430 | 1439s
[2026-02-24 17:27:41] [INFO    ]   [Per-class] MEL=0.577  NV=0.437  BCC=0.609  AK=0.569  BKL=0.284  DF=0.806  VASC=1.000  SCC=0.819
[2026-02-24 17:27:41] [INFO    ] ** New best global balanced accuracy: 0.6376 **
[2026-02-24 17:52:12] [INFO    ] Round  13/25 | Global Acc=0.4482 BalAcc=0.6267 F1=0.4057 AUC=0.8986 | Worst Acc=0.4518 | Std Acc=0.0769 | 1445s
[2026-02-24 17:52:12] [INFO    ]   [Per-class] MEL=0.617  NV=0.327  BCC=0.613  AK=0.569  BKL=0.340  DF=0.750  VASC=1.000  SCC=0.798
[2026-02-24 18:16:32] [INFO    ] Round  14/25 | Global Acc=0.5268 BalAcc=0.6577 F1=0.4295 AUC=0.9091 | Worst Acc=0.3442 | Std Acc=0.1152 | 1444s
[2026-02-24 18:16:32] [INFO    ]   [Per-class] MEL=0.537  NV=0.500  BCC=0.625  AK=0.669  BKL=0.340  DF=0.861  VASC=0.974  SCC=0.755
[2026-02-24 18:16:32] [INFO    ] ** New best global balanced accuracy: 0.6577 **
[2026-02-24 18:40:59] [INFO    ] Round  15/25 | Global Acc=0.5268 BalAcc=0.6446 F1=0.4318 AUC=0.9060 | Worst Acc=0.5811 | Std Acc=0.0458 | 1443s
[2026-02-24 18:40:59] [INFO    ]   [Per-class] MEL=0.524  NV=0.506  BCC=0.645  AK=0.554  BKL=0.340  DF=0.806  VASC=0.974  SCC=0.809
[2026-02-24 19:05:15] [INFO    ] Round  16/25 | Global Acc=0.5189 BalAcc=0.6601 F1=0.4387 AUC=0.9081 | Worst Acc=0.5332 | Std Acc=0.0509 | 1441s
[2026-02-24 19:05:15] [INFO    ]   [Per-class] MEL=0.553  NV=0.497  BCC=0.651  AK=0.631  BKL=0.360  DF=0.806  VASC=0.974  SCC=0.830
[2026-02-24 19:05:15] [INFO    ] ** New best global balanced accuracy: 0.6601 **
[2026-02-24 19:30:00] [INFO    ] Round  17/25 | Global Acc=0.4900 BalAcc=0.6657 F1=0.4293 AUC=0.9056 | Worst Acc=0.5055 | Std Acc=0.0600 | 1462s
[2026-02-24 19:30:00] [INFO    ]   [Per-class] MEL=0.555  NV=0.504  BCC=0.657  AK=0.646  BKL=0.374  DF=0.806  VASC=0.974  SCC=0.840
[2026-02-24 19:30:00] [INFO    ] ** New best global balanced accuracy: 0.6657 **
[2026-02-24 19:54:42] [INFO    ] Round  18/25 | Global Acc=0.5321 BalAcc=0.6863 F1=0.4536 AUC=0.9083 | Worst Acc=0.5276 | Std Acc=0.0556 | 1461s
[2026-02-24 19:54:42] [INFO    ]   [Per-class] MEL=0.576  NV=0.521  BCC=0.671  AK=0.662  BKL=0.393  DF=0.833  VASC=1.000  SCC=0.840
[2026-02-24 19:54:42] [INFO    ] ** New best global balanced accuracy: 0.6863 **
[2026-02-24 20:19:25] [INFO    ] Round  19/25 | Global Acc=0.5803 BalAcc=0.6988 F1=0.4677 AUC=0.9121 | Worst Acc=0.5813 | Std Acc=0.0387 | 1462s
[2026-02-24 20:19:25] [INFO    ]   [Per-class] MEL=0.589  NV=0.533  BCC=0.683  AK=0.669  BKL=0.408  DF=0.833  VASC=1.000  SCC=0.851
[2026-02-24 20:19:25] [INFO    ] ** New best global balanced accuracy: 0.6988 **
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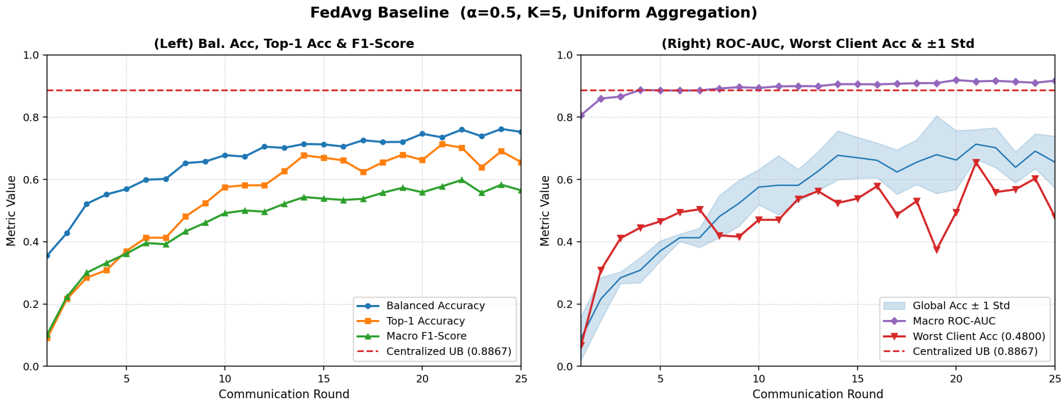
[2026-02-24 20:44:06] [INFO    ] Round  20/25 | Global Acc=0.5671 BalAcc=0.7008 F1=0.4711 AUC=0.9140 | Worst Acc=0.5950 | Std Acc=0.0361 | 1459s
[2026-02-24 20:44:06] [INFO    ]   [Per-class] MEL=0.591  NV=0.537  BCC=0.689  AK=0.677  BKL=0.415  DF=0.833  VASC=1.000  SCC=0.851
[2026-02-24 20:44:06] [INFO    ] ** New best global balanced accuracy: 0.7008 **
[2026-02-24 21:08:38] [INFO    ] Round  21/25 | Global Acc=0.5955 BalAcc=0.6941 F1=0.4731 AUC=0.9160 | Worst Acc=0.6392 | Std Acc=0.0343 | 1451s
[2026-02-24 21:08:38] [INFO    ]   [Per-class] MEL=0.576  NV=0.531  BCC=0.683  AK=0.685  BKL=0.415  DF=0.833  VASC=1.000  SCC=0.840
[2026-02-24 21:33:11] [INFO    ] Round  22/25 | Global Acc=0.5655 BalAcc=0.7066 F1=0.4732 AUC=0.9086 | Worst Acc=0.5926 | Std Acc=0.0297 | 1461s
[2026-02-24 21:33:11] [INFO    ]   [Per-class] MEL=0.572  NV=0.539  BCC=0.643  AK=0.685  BKL=0.431  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 21:33:11] [INFO    ] ** New best global balanced accuracy: 0.7066 **
[2026-02-24 21:57:35] [INFO    ] Round  23/25 | Global Acc=0.5458 BalAcc=0.7002 F1=0.4643 AUC=0.9095 | Worst Acc=0.5968 | Std Acc=0.0333 | 1446s
[2026-02-24 21:57:35] [INFO    ]   [Per-class] MEL=0.566  NV=0.539  BCC=0.651  AK=0.685  BKL=0.427  DF=0.861  VASC=1.000  SCC=0.840
[2026-02-24 22:22:08] [INFO    ] Round  24/25 | Global Acc=0.5876 BalAcc=0.7005 F1=0.4791 AUC=0.9150 | Worst Acc=0.6234 | Std Acc=0.0342 | 1463s
[2026-02-24 22:22:08] [INFO    ]   [Per-class] MEL=0.570  NV=0.540  BCC=0.659  AK=0.685  BKL=0.430  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 22:46:23] [INFO    ] Round  25/25 | Global Acc=0.5676 BalAcc=0.6987 F1=0.4688 AUC=0.9157 | Worst Acc=0.5705 | Std Acc=0.0511 | 1441s
[2026-02-24 22:46:23] [INFO    ]   [Per-class] MEL=0.572  NV=0.539  BCC=0.643  AK=0.685  BKL=0.431  DF=0.889  VASC=1.000  SCC=0.830
[2026-02-24 22:46:46] [INFO    ] Saved 5 personalized client models to /workspace/layer-wised-personalised/checkpoints/federated
[2026-02-24 22:46:46] [INFO    ] ============================================================
[2026-02-24 22:46:46] [INFO    ]   Federated training complete!
[2026-02-24 22:46:46] [INFO    ]   Best global balanced accuracy: 0.7066
[2026-02-24 22:46:46] [INFO    ]   Checkpoints: /workspace/layer-wised-personalised/checkpoints/federated
[2026-02-24 22:46:46] [INFO    ] ============================================================
```

Figure 9. FedSA-Drift non-IID setting (α=0.5, K=5): top panel shows convergence metrics; middle and bottom panels show terminal outputs (Rounds 1–13 and 14–25). Final Balanced Accuracy: 0.7066, Worst Client Accuracy: 0.5705.

Comparing the two personalizing configurations, drift-aware weighting raises worst-client accuracy by **+23.44 percentage points** from 0.1953 to 0.4297 and reduces the inter-client standard deviation of accuracy by **57.4%** from 0.1294 to 0.0551. Both margins exceed those observed at α=0.5 and α=1.0,

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[2026-02-23 11:36:17] [INFO    ] Starting federated training — 25 rounds, 5 clients
[2026-02-23 12:00:18] [INFO    ] Round   1/25 | Global Acc=0.0908 BalAcc=0.3561 F1=0.1025 AUC=0.8064 | Worst Acc=0.0668 | Std Acc=0.0704 | 1437s
[2026-02-23 12:00:18] [INFO    ]   [Per-class] MEL=0.068  NV=0.011  BCC=0.000  AK=0.392  BKL=0.002  DF=0.694  VASC=0.895  SCC=0.798
[2026-02-23 12:00:18] [INFO    ] ** New best global balanced accuracy: 0.3561 **
[2026-02-23 12:24:45] [INFO    ] Round   2/25 | Global Acc=0.2163 BalAcc=0.4285 F1=0.2234 AUC=0.8599 | Worst Acc=0.3089 | Std Acc=0.0690 | 1447s
[2026-02-23 12:24:45] [INFO    ]   [Per-class] MEL=0.232  NV=0.143  BCC=0.032  AK=0.285  BKL=0.059  DF=0.639  VASC=0.921  SCC=0.862
[2026-02-23 12:24:45] [INFO    ] ** New best global balanced accuracy: 0.4285 **
[2026-02-23 12:49:06] [INFO    ] Round   3/25 | Global Acc=0.2845 BalAcc=0.5219 F1=0.3005 AUC=0.8659 | Worst Acc=0.4113 | Std Acc=0.0188 | 1430s
[2026-02-23 12:49:06] [INFO    ]   [Per-class] MEL=0.318  NV=0.184  BCC=0.196  AK=0.523  BKL=0.130  DF=0.806  VASC=0.921  SCC=0.840
[2026-02-23 12:49:06] [INFO    ] ** New best global balanced accuracy: 0.5219 **
[2026-02-23 13:13:31] [INFO    ] Round   4/25 | Global Acc=0.3087 BalAcc=0.5519 F1=0.3320 AUC=0.8873 | Worst Acc=0.4453 | Std Acc=0.0403 | 1436s
[2026-02-23 13:13:31] [INFO    ]   [Per-class] MEL=0.362  NV=0.241  BCC=0.283  AK=0.577  BKL=0.198  DF=0.806  VASC=0.947  SCC=0.766
[2026-02-23 13:13:31] [INFO    ] ** New best global balanced accuracy: 0.5519 **
[2026-02-23 13:37:55] [INFO    ] Round   5/25 | Global Acc=0.3697 BalAcc=0.5689 F1=0.3613 AUC=0.8860 | Worst Acc=0.4650 | Std Acc=0.0328 | 1437s
[2026-02-23 13:37:55] [INFO    ]   [Per-class] MEL=0.447  NV=0.303  BCC=0.373  AK=0.577  BKL=0.197  DF=0.750  VASC=0.921  SCC=0.798
[2026-02-23 13:37:55] [INFO    ] ** New best global balanced accuracy: 0.5689 **
[2026-02-23 14:02:12] [INFO    ] Round   6/25 | Global Acc=0.4129 BalAcc=0.5991 F1=0.3958 AUC=0.8854 | Worst Acc=0.4947 | Std Acc=0.0118 | 1436s
[2026-02-23 14:02:12] [INFO    ]   [Per-class] MEL=0.501  NV=0.371  BCC=0.493  AK=0.554  BKL=0.246  DF=0.778  VASC=0.921  SCC=0.777
[2026-02-23 14:02:12] [INFO    ] ** New best global balanced accuracy: 0.5991 **
[2026-02-23 14:26:30] [INFO    ] Round   7/25 | Global Acc=0.4129 BalAcc=0.6017 F1=0.3920 AUC=0.8858 | Worst Acc=0.5042 | Std Acc=0.0307 | 1435s
[2026-02-23 14:26:30] [INFO    ]   [Per-class] MEL=0.506  NV=0.380  BCC=0.505  AK=0.554  BKL=0.256  DF=0.750  VASC=0.947  SCC=0.766
[2026-02-23 14:26:30] [INFO    ] ** New best global balanced accuracy: 0.6017 **
[2026-02-23 14:50:57] [INFO    ] Round   8/25 | Global Acc=0.4808 BalAcc=0.6525 F1=0.4328 AUC=0.8919 | Worst Acc=0.4203 | Std Acc=0.0678 | 1446s
[2026-02-23 14:50:57] [INFO    ]   [Per-class] MEL=0.561  NV=0.446  BCC=0.567  AK=0.608  BKL=0.323  DF=0.806  VASC=0.974  SCC=0.787
[2026-02-23 14:50:57] [INFO    ] ** New best global balanced accuracy: 0.6525 **
[2026-02-23 15:15:29] [INFO    ] Round   9/25 | Global Acc=0.5237 BalAcc=0.6573 F1=0.4608 AUC=0.8963 | Worst Acc=0.4163 | Std Acc=0.0737 | 1440s
[2026-02-23 15:15:29] [INFO    ]   [Per-class] MEL=0.600  NV=0.452  BCC=0.595  AK=0.615  BKL=0.345  DF=0.778  VASC=0.974  SCC=0.798
[2026-02-23 15:15:29] [INFO    ] ** New best global balanced accuracy: 0.6573 **
[2026-02-23 15:39:53] [INFO    ] Round  10/25 | Global Acc=0.5755 BalAcc=0.6779 F1=0.4918 AUC=0.8939 | Worst Acc=0.4705 | Std Acc=0.0565 | 1437s
[2026-02-23 15:39:53] [INFO    ]   [Per-class] MEL=0.635  NV=0.491  BCC=0.625  AK=0.631  BKL=0.372  DF=0.806  VASC=0.974  SCC=0.808
[2026-02-23 15:39:53] [INFO    ] ** New best global balanced accuracy: 0.6779 **
[2026-02-23 16:04:21] [INFO    ] Round  11/25 | Global Acc=0.5813 BalAcc=0.6735 F1=0.5004 AUC=0.8988 | Worst Acc=0.4703 | Std Acc=0.0952 | 1440s
[2026-02-23 16:04:21] [INFO    ]   [Per-class] MEL=0.641  NV=0.478  BCC=0.635  AK=0.638  BKL=0.366  DF=0.806  VASC=0.974  SCC=0.808
[2026-02-23 16:28:37] [INFO    ] Round  12/25 | Global Acc=0.5811 BalAcc=0.7051 F1=0.4965 AUC=0.8996 | Worst Acc=0.5379 | Std Acc=0.0512 | 1438s
[2026-02-23 16:28:37] [INFO    ]   [Per-class] MEL=0.685  NV=0.497  BCC=0.667  AK=0.654  BKL=0.411  DF=0.806  VASC=0.974  SCC=0.819
[2026-02-23 16:28:37] [INFO    ] ** New best global balanced accuracy: 0.7051 **
[2026-02-23 16:52:56] [INFO    ] Round  13/25 | Global Acc=0.6263 BalAcc=0.7014 F1=0.5218 AUC=0.8994 | Worst Acc=0.5624 | Std Acc=0.0622 | 1437s
[2026-02-23 16:52:56] [INFO    ]   [Per-class] MEL=0.699  NV=0.486  BCC=0.659  AK=0.662  BKL=0.404  DF=0.806  VASC=0.974  SCC=0.808
[2026-02-23 17:17:14] [INFO    ] Round  14/25 | Global Acc=0.6774 BalAcc=0.7139 F1=0.5434 AUC=0.9060 | Worst Acc=0.5245 | Std Acc=0.0788 | 1439s
[2026-02-23 17:17:14] [INFO    ]   [Per-class] MEL=0.730  NV=0.504  BCC=0.681  AK=0.662  BKL=0.432  DF=0.833  VASC=0.974  SCC=0.819
[2026-02-23 17:17:14] [INFO    ] ** New best global balanced accuracy: 0.7139 **
[2026-02-23 17:41:45] [INFO    ] Round  15/25 | Global Acc=0.6697 BalAcc=0.7126 F1=0.5386 AUC=0.9059 | Worst Acc=0.5387 | Std Acc=0.0660 | 1441s
[2026-02-23 17:41:45] [INFO    ]   [Per-class] MEL=0.726  NV=0.503  BCC=0.677  AK=0.662  BKL=0.428  DF=0.833  VASC=0.974  SCC=0.819
[2026-02-23 18:05:58] [INFO    ] Round  16/25 | Global Acc=0.6613 BalAcc=0.7057 F1=0.5341 AUC=0.9053 | Worst Acc=0.5787 | Std Acc=0.0556 | 1438s
[2026-02-23 18:05:58] [INFO    ]   [Per-class] MEL=0.718  NV=0.497  BCC=0.671  AK=0.654  BKL=0.419  DF=0.833  VASC=0.974  SCC=0.808
[2026-02-23 18:30:11] [INFO    ] Round  17/25 | Global Acc=0.6239 BalAcc=0.7259 F1=0.5378 AUC=0.9073 | Worst Acc=0.4871 | Std Acc=0.0713 | 1436s
[2026-02-23 18:30:11] [INFO    ]   [Per-class] MEL=0.738  NV=0.509  BCC=0.695  AK=0.669  BKL=0.445  DF=0.833  VASC=0.974  SCC=0.830
[2026-02-23 18:30:11] [INFO    ] ** New best global balanced accuracy: 0.7259 **
[2026-02-23 18:54:38] [INFO    ] Round  18/25 | Global Acc=0.6555 BalAcc=0.7200 F1=0.5575 AUC=0.9091 | Worst Acc=0.5305 | Std Acc=0.0718 | 1440s
[2026-02-23 18:54:38] [INFO    ]   [Per-class] MEL=0.731  NV=0.507  BCC=0.685  AK=0.662  BKL=0.435  DF=0.833  VASC=0.974  SCC=0.830
[2026-02-23 19:18:57] [INFO    ] Round  19/25 | Global Acc=0.6795 BalAcc=0.7206 F1=0.5732 AUC=0.9094 | Worst Acc=0.3739 | Std Acc=0.1246 | 1440s
[2026-02-23 19:18:57] [INFO    ]   [Per-class] MEL=0.739  NV=0.507  BCC=0.689  AK=0.662  BKL=0.437  DF=0.833  VASC=0.974  SCC=0.830
[2026-02-23 19:43:09] [INFO    ] Round  20/25 | Global Acc=0.6624 BalAcc=0.7465 F1=0.5587 AUC=0.9191 | Worst Acc=0.4947 | Std Acc=0.0939 | 1435s
[2026-02-23 19:43:09] [INFO    ]   [Per-class] MEL=0.762  NV=0.527  BCC=0.713  AK=0.685  BKL=0.459  DF=0.861  VASC=0.974  SCC=0.840
[2026-02-23 19:43:09] [INFO    ] ** New best global balanced accuracy: 0.7465 **
```

```
[2026-02-23 20:07:35] [INFO    ] Round  21/25 | Global Acc=0.7132 BalAcc=0.7352 F1=0.5780 AUC=0.9147 | Worst Acc=0.6547 | Std Acc=0.0472 | 1435s
[2026-02-23 20:07:35] [INFO    ]   [Per-class] MEL=0.748  NV=0.520  BCC=0.703  AK=0.677  BKL=0.449  DF=0.833  VASC=0.974  SCC=0.830
[2026-02-23 20:31:54] [INFO    ] Round  22/25 | Global Acc=0.7018 BalAcc=0.7600 F1=0.5984 AUC=0.9166 | Worst Acc=0.5595 | Std Acc=0.0640 | 1441s
[2026-02-23 20:31:54] [INFO    ]   [Per-class] MEL=0.776  NV=0.541  BCC=0.725  AK=0.692  BKL=0.473  DF=0.861  VASC=0.974  SCC=0.851
[2026-02-23 20:31:54] [INFO    ] ** New best global balanced accuracy: 0.7600 **
[2026-02-23 20:56:21] [INFO    ] Round  23/25 | Global Acc=0.6389 BalAcc=0.7386 F1=0.5569 AUC=0.9135 | Worst Acc=0.5679 | Std Acc=0.0498 | 1435s
[2026-02-23 20:56:21] [INFO    ]   [Per-class] MEL=0.752  NV=0.523  BCC=0.707  AK=0.677  BKL=0.453  DF=0.861  VASC=0.974  SCC=0.830
[2026-02-23 21:20:42] [INFO    ] Round  24/25 | Global Acc=0.6908 BalAcc=0.7622 F1=0.5832 AUC=0.9106 | Worst Acc=0.6032 | Std Acc=0.0558 | 1439s
[2026-02-23 21:20:42] [INFO    ]   [Per-class] MEL=0.779  NV=0.543  BCC=0.727  AK=0.692  BKL=0.477  DF=0.861  VASC=0.974  SCC=0.851
[2026-02-23 21:20:42] [INFO    ] ** New best global balanced accuracy: 0.7622 **
[2026-02-23 21:45:05] [INFO    ] Round  25/25 | Global Acc=0.6555 BalAcc=0.7526 F1=0.5649 AUC=0.9171 | Worst Acc=0.4800 | Std Acc=0.0833 | 1437s
[2026-02-23 21:45:05] [INFO    ]   [Per-class] MEL=0.752  NV=0.569  BCC=0.820  AK=0.792  BKL=0.609  DF=0.861  VASC=0.947  SCC=0.670
[2026-02-23 21:45:43] [INFO    ] Saved 5 personalized client models to /workspace/layer-wised-personalised/checkpoints/federated
[2026-02-23 21:45:43] [INFO    ] ==================================================================
[2026-02-23 21:45:43] [INFO    ]   Federated training complete!
[2026-02-23 21:45:43] [INFO    ]   Best global balanced accuracy: 0.7622
[2026-02-23 21:45:43] [INFO    ]   Checkpoints: /workspace/layer-wised-personalised/checkpoints/federated
[2026-02-23 21:45:43] [INFO    ] ==================================================================
```

Figure 10. FedAvg non-IID setting (α=0.5, K=5): top panel shows convergence metrics; middle and bottom panels show terminal outputs (Rounds 1–13 and 14–25). Final Balanced Accuracy: 0.7622, Worst Client Accuracy: 0.4800.

indicating that the benefit of drift correction grows as heterogeneity worsens. On the held-out test split the same comparison gives $+21.53$ points of worst-client accuracy and a 67.1% reduction in dispersion, so the effect is not specific to the validation partition. Averaging over the final five communication rounds rather than reading the final round alone gives $+23.04$ points and -56.7%, confirming these are not single-round artefacts.

The FedAvg baseline in Figure 11 deserves separate attention. Its convergence curve is unremarkable and its overall accuracy is the highest of the three, yet the per-class lines of its terminal output record `MEL=0.000` at every one of the 25 rounds. The model reaches competitive aggregate accuracy by learning four classes well and never predicting the other four at all. Section 4.5.1 examines this behaviour in detail.

4.3.4 Interpreting the Global Accuracy Trade-off

FedSA-Drift achieves a lower global balanced accuracy than the uniform-aggregation baseline at the milder settings (0.8209 vs. 0.8314 at α=1.0; 0.7066 vs. 0.7622 at α=0.5). The more pronounced gap at α=0.5 (5.6 percentage points, vs. 1.1 at α=1.0) reflects a genuine cost of the fairness–utility trade-off that is well-recognized in the federated learning literature (Li et al., 2021; Mohri et al., 2019). We do not claim this cost is negligible; a global accuracy reduction of this magnitude in medical imaging warrants careful clinical evaluation before deployment.

The relevant comparison, however, is not global accuracy in isolation. Under FedAvg at α=0.5, the worst-performing client reaches only 48.0% balanced accuracy—little better than chance for an eight-class problem. FedSA-Drift raises this floor to 57.1%. The minimax fairness objective (Mohri et al., 2019; Li et al., 2021), which optimizes the worst-case client rather than the population average, has been explicitly motivated in federated healthcare AI contexts where a model that fails disproportionately at one institution creates inequities in clinical outcomes (Darzi et al., 2024; Rieke et al., 2020). Improving the minimum performance guarantee at the cost of some global average is the stated design goal of FedSA-Drift, and the observed trade-off is consistent with that goal. A comprehensive clinical validation comparing deployment outcomes across institutions is an important direction for future work.

The direction of this trade-off is not fixed. At α=0.1 it reverses within the decomposed family: FedSA-Drift attains *higher* global balanced accuracy than decomposition alone (0.5196 vs. 0.4477), so drift-aware weighting costs nothing in aggregate utility once heterogeneity is severe, while still raising the worst-client floor by 23.4 points. The utility cost seen at $\alpha \in \{0.5, 1.0\}$ therefore appears to be a feature of mild heterogeneity rather than an inherent price of the method.

A separate comparison applies once the decomposition itself is removed. The no-decomposition FedAvg baseline at α=0.1 records the highest mean per-client accuracy of the three configurations (0.4702 on the held-out test split, against 0.4264 for FedSA-Drift), and marginally higher balanced accuracy (0.3778 vs. 0.3681). We do not present FedSA-Drift as superior on aggregate accuracy in this regime. As Section 4.5.1 shows, that baseline reaches its figure by abandoning four of the eight diagnostic classes outright, so the aggregate comparison and the clinical comparison point in opposite directions.

4.4 Contextual Reference Against Related Methods

Table 3 places FedSA-Drift alongside related published methods for contextual reference. **This is not a controlled head-to-head comparison.** The methods listed use different backbone architectures, training datasets, class distributions, federation setups, and evaluation protocols; direct numerical comparison is therefore not appropriate. The table is intended only to situate the design choices and claims of FedSA-Drift within the broader literature.

4.4.1 Comparison with Centralized Baselines

We do not benchmark our centralized figure against published ISIC 2019 results. Ours is measured on an internal validation split that shares lesions with the training partition, whereas challenge results

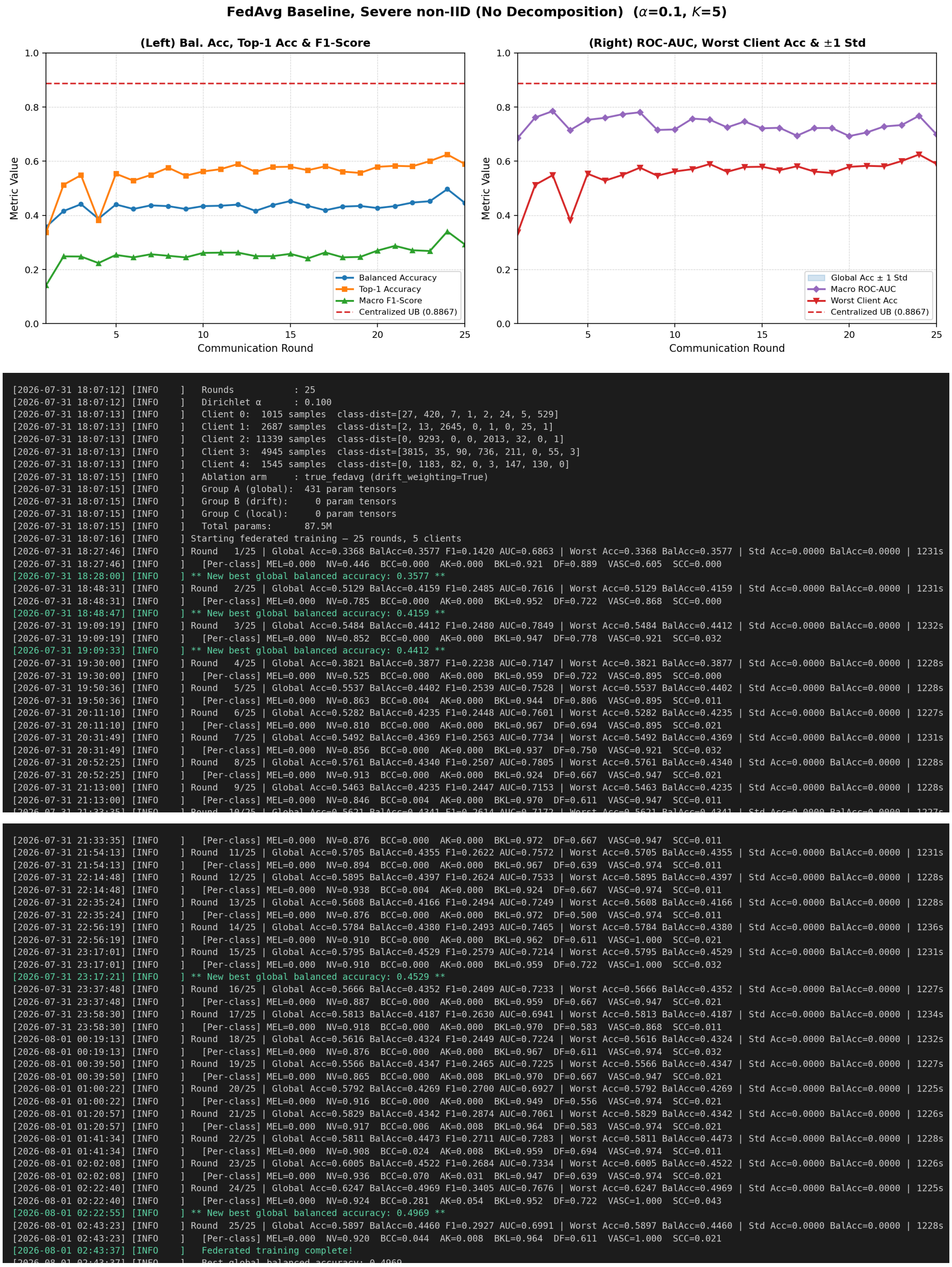
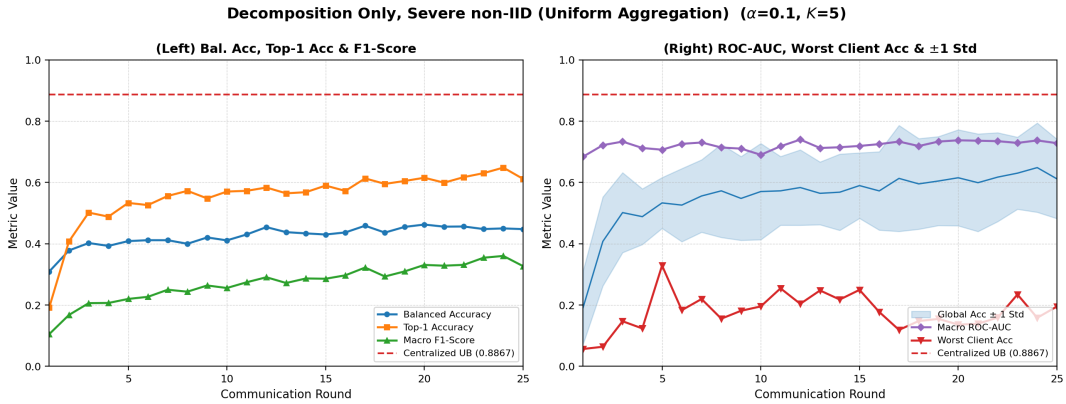
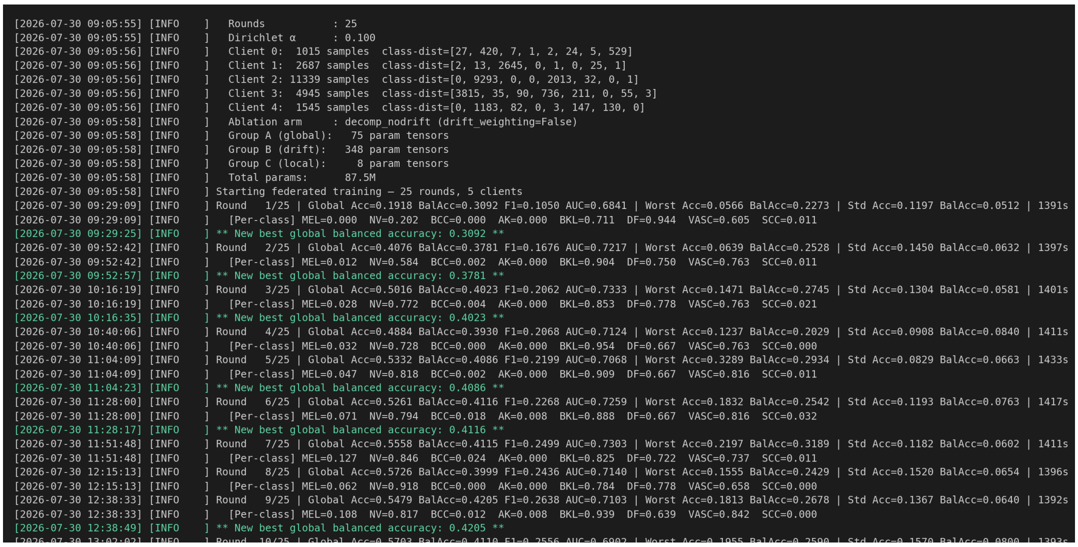


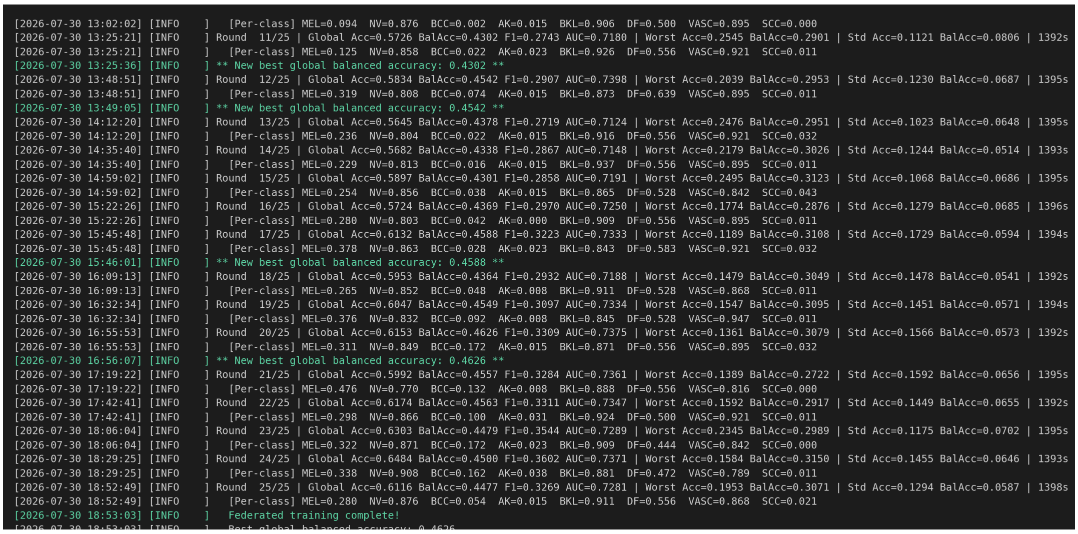
Figure 11. FedAvg baseline with the decomposition removed (α=0.1, K=5): top panel shows convergence metrics; middle and bottom panels show terminal outputs. The per-class lines record `MEL=0.000` at every one of the 25 rounds, with `BCC`, `AK` and `SCC` at or near zero: the run reaches the highest overall accuracy of the three configurations while never predicting melanoma. Inter-client standard deviation is identically zero because no client-specific head exists. The `Ablation arm` line reports `drift_weighting=True` because the command-line flag retains its default; with Group B empty (0 tensors, shown two lines below) the drift rule has no parameters to act on and is inert, so this configuration is standard FedAvg.

are computed on the official held-out test collection; the two are not comparable, and the comparison is omitted rather than qualified.





```
[2026-07-30 09:05:55] [INFO    ]   Rounds          : 25
[2026-07-30 09:05:55] [INFO    ]   Dirichlet α     : 0.100
[2026-07-30 09:05:56] [INFO    ]   Client 0:  1015 samples  class-dist=[27, 420, 7, 1, 2, 24, 5, 529]
[2026-07-30 09:05:56] [INFO    ]   Client 1:  2687 samples  class-dist=[2, 13, 2645, 0, 1, 0, 25, 1]
[2026-07-30 09:05:56] [INFO    ]   Client 2: 11339 samples  class-dist=[0, 9293, 0, 0, 2013, 32, 0, 1]
[2026-07-30 09:05:56] [INFO    ]   Client 3:  4945 samples  class-dist=[3815, 35, 90, 736, 211, 0, 55, 3]
[2026-07-30 09:05:56] [INFO    ]   Client 4:  1545 samples  class-dist=[0, 1183, 82, 0, 3, 147, 130, 0]
[2026-07-30 09:05:58] [INFO    ]   Ablation arm    : decomp_nodrift (drift_weighting=False)
[2026-07-30 09:05:58] [INFO    ]   Group A (global):   75 param tensors
[2026-07-30 09:05:58] [INFO    ]   Group B (drift):   348 param tensors
[2026-07-30 09:05:58] [INFO    ]   Group C (local):     8 param tensors
[2026-07-30 09:05:58] [INFO    ]   Total params:      87.5M
[2026-07-30 09:05:58] [INFO    ] Starting federated training — 25 rounds, 5 clients
[2026-07-30 09:29:09] [INFO    ] Round   1/25 | Global Acc=0.1918 BalAcc=0.3092 F1=0.1050 AUC=0.6841 | Worst Acc=0.0566 BalAcc=0.2273 | Std Acc=0.1197 BalAcc=0.0512 | 1391s
[2026-07-30 09:29:09] [INFO    ]   [Per-class] MEL=0.000  NV=0.202  BCC=0.000  AK=0.000  BKL=0.711  DF=0.944  VASC=0.605  SCC=0.011
[2026-07-30 09:29:25] [INFO    ] ** New best global balanced accuracy: 0.3092 **
[2026-07-30 09:52:42] [INFO    ] Round   2/25 | Global Acc=0.4076 BalAcc=0.3781 F1=0.1676 AUC=0.7217 | Worst Acc=0.0639 BalAcc=0.2528 | Std Acc=0.1450 BalAcc=0.0632 | 1397s
[2026-07-30 09:52:42] [INFO    ]   [Per-class] MEL=0.012  NV=0.584  BCC=0.002  AK=0.000  BKL=0.904  DF=0.750  VASC=0.763  SCC=0.011
[2026-07-30 09:52:57] [INFO    ] ** New best global balanced accuracy: 0.3781 **
[2026-07-30 10:16:19] [INFO    ] Round   3/25 | Global Acc=0.5016 BalAcc=0.4023 F1=0.2062 AUC=0.7333 | Worst Acc=0.1471 BalAcc=0.2745 | Std Acc=0.1304 BalAcc=0.0581 | 1401s
[2026-07-30 10:16:19] [INFO    ]   [Per-class] MEL=0.028  NV=0.772  BCC=0.004  AK=0.000  BKL=0.853  DF=0.778  VASC=0.763  SCC=0.021
[2026-07-30 10:16:35] [INFO    ] ** New best global balanced accuracy: 0.4023 **
[2026-07-30 10:40:06] [INFO    ] Round   4/25 | Global Acc=0.4884 BalAcc=0.3930 F1=0.2068 AUC=0.7124 | Worst Acc=0.1237 BalAcc=0.2029 | Std Acc=0.0908 BalAcc=0.0840 | 1411s
[2026-07-30 10:40:06] [INFO    ]   [Per-class] MEL=0.032  NV=0.728  BCC=0.000  AK=0.000  BKL=0.954  DF=0.667  VASC=0.763  SCC=0.000
[2026-07-30 11:04:09] [INFO    ] Round   5/25 | Global Acc=0.5332 BalAcc=0.4086 F1=0.2199 AUC=0.7068 | Worst Acc=0.3289 BalAcc=0.2934 | Std Acc=0.0829 BalAcc=0.0663 | 1433s
[2026-07-30 11:04:09] [INFO    ]   [Per-class] MEL=0.047  NV=0.818  BCC=0.002  AK=0.000  BKL=0.909  DF=0.667  VASC=0.816  SCC=0.011
[2026-07-30 11:04:23] [INFO    ] ** New best global balanced accuracy: 0.4086 **
[2026-07-30 11:28:00] [INFO    ] Round   6/25 | Global Acc=0.5261 BalAcc=0.4116 F1=0.2268 AUC=0.7259 | Worst Acc=0.1832 BalAcc=0.2542 | Std Acc=0.1193 BalAcc=0.0763 | 1417s
[2026-07-30 11:28:00] [INFO    ]   [Per-class] MEL=0.071  NV=0.794  BCC=0.018  AK=0.008  BKL=0.888  DF=0.667  VASC=0.816  SCC=0.032
[2026-07-30 11:28:17] [INFO    ] ** New best global balanced accuracy: 0.4116 **
[2026-07-30 11:51:48] [INFO    ] Round   7/25 | Global Acc=0.5558 BalAcc=0.4115 F1=0.2499 AUC=0.7303 | Worst Acc=0.2197 BalAcc=0.3189 | Std Acc=0.1182 BalAcc=0.0602 | 1411s
[2026-07-30 11:51:48] [INFO    ]   [Per-class] MEL=0.127  NV=0.846  BCC=0.024  AK=0.000  BKL=0.825  DF=0.722  VASC=0.737  SCC=0.011
[2026-07-30 12:15:13] [INFO    ] Round   8/25 | Global Acc=0.5726 BalAcc=0.3999 F1=0.2436 AUC=0.7140 | Worst Acc=0.1555 BalAcc=0.2429 | Std Acc=0.1520 BalAcc=0.0654 | 1396s
[2026-07-30 12:15:13] [INFO    ]   [Per-class] MEL=0.062  NV=0.918  BCC=0.000  AK=0.000  BKL=0.784  DF=0.778  VASC=0.658  SCC=0.000
[2026-07-30 12:38:33] [INFO    ] Round   9/25 | Global Acc=0.5479 BalAcc=0.4205 F1=0.2638 AUC=0.7103 | Worst Acc=0.1813 BalAcc=0.2678 | Std Acc=0.1367 BalAcc=0.0640 | 1392s
[2026-07-30 12:38:33] [INFO    ]   [Per-class] MEL=0.108  NV=0.817  BCC=0.012  AK=0.008  BKL=0.939  DF=0.639  VASC=0.842  SCC=0.000
[2026-07-30 12:38:49] [INFO    ] ** New best global balanced accuracy: 0.4205 **
```

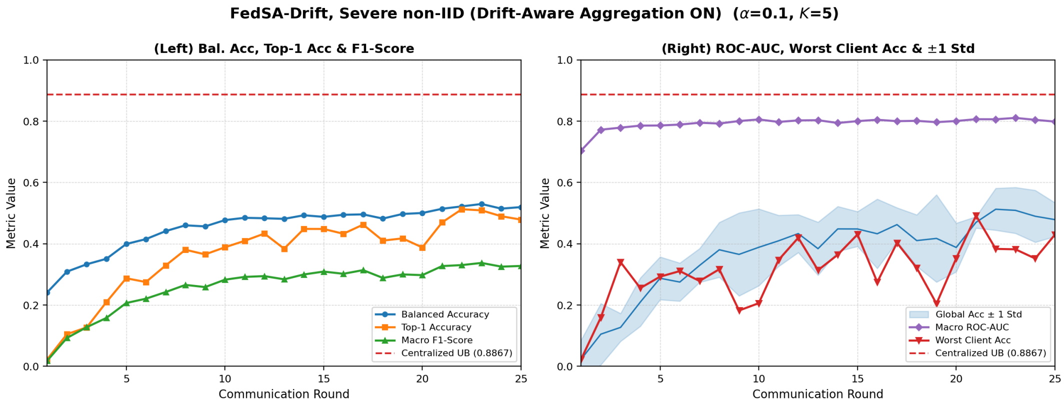


```
[2026-07-30 13:02:02] [INFO    ]   [Per-class] MEL=0.094  NV=0.876  BCC=0.002  AK=0.015  BKL=0.906  DF=0.500  VASC=0.895  SCC=0.000
[2026-07-30 13:25:21] [INFO    ] Round  11/25 | Global Acc=0.5726 BalAcc=0.4302 F1=0.2743 AUC=0.7180 | Worst Acc=0.2545 BalAcc=0.2901 | Std Acc=0.1121 BalAcc=0.0806 | 1392s
[2026-07-30 13:25:21] [INFO    ]   [Per-class] MEL=0.125  NV=0.858  BCC=0.022  AK=0.023  BKL=0.926  DF=0.556  VASC=0.921  SCC=0.011
[2026-07-30 13:25:36] [INFO    ] ** New best global balanced accuracy: 0.4302 **
[2026-07-30 13:48:51] [INFO    ] Round  12/25 | Global Acc=0.5834 BalAcc=0.4542 F1=0.2907 AUC=0.7398 | Worst Acc=0.2039 BalAcc=0.2953 | Std Acc=0.1230 BalAcc=0.0687 | 1395s
[2026-07-30 13:48:51] [INFO    ]   [Per-class] MEL=0.319  NV=0.808  BCC=0.074  AK=0.015  BKL=0.873  DF=0.639  VASC=0.895  SCC=0.011
[2026-07-30 13:49:05] [INFO    ] ** New best global balanced accuracy: 0.4542 **
[2026-07-30 14:12:20] [INFO    ] Round  13/25 | Global Acc=0.5645 BalAcc=0.4378 F1=0.2719 AUC=0.7124 | Worst Acc=0.2476 BalAcc=0.2951 | Std Acc=0.1023 BalAcc=0.0648 | 1395s
[2026-07-30 14:12:20] [INFO    ]   [Per-class] MEL=0.236  NV=0.804  BCC=0.022  AK=0.015  BKL=0.916  DF=0.556  VASC=0.921  SCC=0.032
[2026-07-30 14:35:40] [INFO    ] Round  14/25 | Global Acc=0.5682 BalAcc=0.4338 F1=0.2867 AUC=0.7148 | Worst Acc=0.2179 BalAcc=0.3026 | Std Acc=0.1244 BalAcc=0.0514 | 1393s
[2026-07-30 14:35:40] [INFO    ]   [Per-class] MEL=0.229  NV=0.813  BCC=0.016  AK=0.015  BKL=0.937  DF=0.556  VASC=0.895  SCC=0.011
[2026-07-30 14:59:02] [INFO    ] Round  15/25 | Global Acc=0.5897 BalAcc=0.4301 F1=0.2858 AUC=0.7191 | Worst Acc=0.2495 BalAcc=0.3123 | Std Acc=0.1068 BalAcc=0.0686 | 1395s
[2026-07-30 14:59:02] [INFO    ]   [Per-class] MEL=0.254  NV=0.856  BCC=0.038  AK=0.015  BKL=0.865  DF=0.528  VASC=0.842  SCC=0.043
[2026-07-30 15:22:26] [INFO    ] Round  16/25 | Global Acc=0.5724 BalAcc=0.4369 F1=0.2970 AUC=0.7250 | Worst Acc=0.1774 BalAcc=0.2876 | Std Acc=0.1279 BalAcc=0.0685 | 1396s
[2026-07-30 15:22:26] [INFO    ]   [Per-class] MEL=0.280  NV=0.803  BCC=0.042  AK=0.000  BKL=0.909  DF=0.556  VASC=0.895  SCC=0.011
[2026-07-30 15:45:48] [INFO    ] Round  17/25 | Global Acc=0.6132 BalAcc=0.4588 F1=0.3223 AUC=0.7333 | Worst Acc=0.1189 BalAcc=0.3108 | Std Acc=0.1729 BalAcc=0.0594 | 1394s
[2026-07-30 15:45:48] [INFO    ]   [Per-class] MEL=0.378  NV=0.863  BCC=0.028  AK=0.023  BKL=0.843  DF=0.583  VASC=0.921  SCC=0.032
[2026-07-30 15:46:01] [INFO    ] ** New best global balanced accuracy: 0.4588 **
[2026-07-30 16:09:13] [INFO    ] Round  18/25 | Global Acc=0.5953 BalAcc=0.4364 F1=0.2932 AUC=0.7188 | Worst Acc=0.1479 BalAcc=0.3049 | Std Acc=0.1478 BalAcc=0.0541 | 1392s
[2026-07-30 16:09:13] [INFO    ]   [Per-class] MEL=0.265  NV=0.852  BCC=0.048  AK=0.008  BKL=0.911  DF=0.528  VASC=0.868  SCC=0.011
[2026-07-30 16:32:34] [INFO    ] Round  19/25 | Global Acc=0.6047 BalAcc=0.4549 F1=0.3097 AUC=0.7334 | Worst Acc=0.1547 BalAcc=0.3095 | Std Acc=0.1451 BalAcc=0.0571 | 1394s
[2026-07-30 16:32:34] [INFO    ]   [Per-class] MEL=0.376  NV=0.832  BCC=0.092  AK=0.008  BKL=0.845  DF=0.528  VASC=0.947  SCC=0.011
[2026-07-30 16:55:53] [INFO    ] Round  20/25 | Global Acc=0.6153 BalAcc=0.4626 F1=0.3309 AUC=0.7375 | Worst Acc=0.1361 BalAcc=0.3079 | Std Acc=0.1566 BalAcc=0.0573 | 1392s
[2026-07-30 16:55:53] [INFO    ]   [Per-class] MEL=0.311  NV=0.849  BCC=0.172  AK=0.015  BKL=0.871  DF=0.556  VASC=0.895  SCC=0.032
[2026-07-30 16:56:07] [INFO    ] ** New best global balanced accuracy: 0.4626 **
[2026-07-30 17:19:22] [INFO    ] Round  21/25 | Global Acc=0.5992 BalAcc=0.4557 F1=0.3284 AUC=0.7361 | Worst Acc=0.1389 BalAcc=0.2722 | Std Acc=0.1592 BalAcc=0.0656 | 1395s
[2026-07-30 17:19:22] [INFO    ]   [Per-class] MEL=0.476  NV=0.770  BCC=0.132  AK=0.008  BKL=0.888  DF=0.556  VASC=0.816  SCC=0.000
[2026-07-30 17:42:41] [INFO    ] Round  22/25 | Global Acc=0.6174 BalAcc=0.4563 F1=0.3311 AUC=0.7347 | Worst Acc=0.1592 BalAcc=0.2917 | Std Acc=0.1449 BalAcc=0.0655 | 1392s
[2026-07-30 17:42:41] [INFO    ]   [Per-class] MEL=0.298  NV=0.866  BCC=0.100  AK=0.031  BKL=0.924  DF=0.500  VASC=0.921  SCC=0.011
[2026-07-30 18:06:04] [INFO    ] Round  23/25 | Global Acc=0.6303 BalAcc=0.4479 F1=0.3544 AUC=0.7289 | Worst Acc=0.2345 BalAcc=0.2989 | Std Acc=0.1175 BalAcc=0.0702 | 1395s
[2026-07-30 18:06:04] [INFO    ]   [Per-class] MEL=0.322  NV=0.871  BCC=0.172  AK=0.023  BKL=0.909  DF=0.444  VASC=0.842  SCC=0.000
[2026-07-30 18:29:25] [INFO    ] Round  24/25 | Global Acc=0.6484 BalAcc=0.4500 F1=0.3602 AUC=0.7371 | Worst Acc=0.1584 BalAcc=0.3150 | Std Acc=0.1455 BalAcc=0.0646 | 1393s
[2026-07-30 18:29:25] [INFO    ]   [Per-class] MEL=0.338  NV=0.908  BCC=0.162  AK=0.038  BKL=0.881  DF=0.472  VASC=0.789  SCC=0.011
[2026-07-30 18:52:49] [INFO    ] Round  25/25 | Global Acc=0.6116 BalAcc=0.4477 F1=0.3269 AUC=0.7281 | Worst Acc=0.1953 BalAcc=0.3071 | Std Acc=0.1294 BalAcc=0.0587 | 1398s
[2026-07-30 18:52:49] [INFO    ]   [Per-class] MEL=0.280  NV=0.876  BCC=0.054  AK=0.015  BKL=0.911  DF=0.556  VASC=0.868  SCC=0.021
[2026-07-30 18:53:03] [INFO    ]   Federated training complete!
```

Figure 12. Decomposition-only setting (α=0.1, K=5, drift weighting disabled): top panel shows convergence metrics; middle and bottom panels show terminal outputs. All eight classes retain non-zero recall, but the worst-client floor is low and inter-client dispersion is wide. Final Balanced Accuracy: 0.4477, Worst Client Accuracy: 0.1953, standard deviation 0.1294.

4.4.2 Comparison with Federated Vision Transformers

Under non-IID data, standard FedAvg with self-attention results in feature collapse due to misaligned attention matrices.




```
[2026-07-30 09:05:55] [INFO    ]   Rounds          : 25
[2026-07-30 09:05:55] [INFO    ]   Dirichlet α     : 0.100
[2026-07-30 09:05:56] [INFO    ]   Client 0:  1015 samples  class-dist=[27, 420, 7, 1, 2, 24, 5, 529]
[2026-07-30 09:05:56] [INFO    ]   Client 1:  2687 samples  class-dist=[2, 13, 2645, 0, 1, 0, 25, 1]
[2026-07-30 09:05:56] [INFO    ]   Client 2: 11339 samples  class-dist=[0, 9293, 0, 0, 2013, 32, 0, 1]
[2026-07-30 09:05:56] [INFO    ]   Client 3:  4945 samples  class-dist=[3815, 35, 90, 736, 211, 0, 55, 3]
[2026-07-30 09:05:56] [INFO    ]   Client 4:  1545 samples  class-dist=[0, 1183, 82, 0, 3, 147, 130, 0]
[2026-07-30 09:05:58] [INFO    ]   Ablation arm    : full (drift_weighting=True)
[2026-07-30 09:05:58] [INFO    ]   Group A (global):   75 param tensors
[2026-07-30 09:05:58] [INFO    ]   Group B (drift):   348 param tensors
[2026-07-30 09:05:58] [INFO    ]   Group C (local):     8 param tensors
[2026-07-30 09:05:58] [INFO    ]   Total params:      87.5M
[2026-07-30 09:05:58] [INFO    ] Starting federated training — 25 rounds, 5 clients
[2026-07-30 09:29:10] [INFO    ] Round   1/25 | Global Acc=0.0229 BalAcc=0.2414 F1=0.0184 AUC=0.7039 | Worst Acc=0.0229 BalAcc=0.2130 | Std Acc=0.0629 BalAcc=0.0644 | 1391s
[2026-07-30 09:29:10] [INFO    ]   [Per-class] MEL=0.000  NV=0.005  BCC=0.004  AK=0.000  BKL=0.005  DF=0.917  VASC=0.947  SCC=0.053
[2026-07-30 09:29:26] [INFO    ] ** New best global balanced accuracy: 0.2414 **
[2026-07-30 09:52:44] [INFO    ] Round   2/25 | Global Acc=0.1050 BalAcc=0.3090 F1=0.0926 AUC=0.7721 | Worst Acc=0.1595 BalAcc=0.2824 | Std Acc=0.1001 BalAcc=0.0663 | 1398s
[2026-07-30 09:52:44] [INFO    ]   [Per-class] MEL=0.040  NV=0.096  BCC=0.156  AK=0.008  BKL=0.033  DF=0.889  VASC=0.974  SCC=0.277
[2026-07-30 09:53:00] [INFO    ] ** New best global balanced accuracy: 0.3090 **
[2026-07-30 10:16:23] [INFO    ] Round   3/25 | Global Acc=0.1271 BalAcc=0.3330 F1=0.1277 AUC=0.7788 | Worst Acc=0.3400 BalAcc=0.3050 | Std Acc=0.0454 BalAcc=0.0772 | 1403s
[2026-07-30 10:16:23] [INFO    ]   [Per-class] MEL=0.109  NV=0.079  BCC=0.271  AK=0.008  BKL=0.074  DF=0.917  VASC=0.974  SCC=0.234
[2026-07-30 10:16:38] [INFO    ] ** New best global balanced accuracy: 0.3330 **
[2026-07-30 10:40:08] [INFO    ] Round   4/25 | Global Acc=0.2097 BalAcc=0.3509 F1=0.1580 AUC=0.7854 | Worst Acc=0.2553 BalAcc=0.2804 | Std Acc=0.0789 BalAcc=0.0843 | 1410s
[2026-07-30 10:40:08] [INFO    ]   [Per-class] MEL=0.105  NV=0.253  BCC=0.210  AK=0.015  BKL=0.104  DF=0.944  VASC=0.974  SCC=0.202
[2026-07-30 10:40:28] [INFO    ] ** New best global balanced accuracy: 0.3509 **
[2026-07-30 11:04:27] [INFO    ] Round   5/25 | Global Acc=0.2874 BalAcc=0.3991 F1=0.2066 AUC=0.7858 | Worst Acc=0.2926 BalAcc=0.2678 | Std Acc=0.0698 BalAcc=0.0946 | 1440s
[2026-07-30 11:04:27] [INFO    ]   [Per-class] MEL=0.087  NV=0.366  BCC=0.337  AK=0.031  BKL=0.127  DF=0.889  VASC=0.974  SCC=0.383
[2026-07-30 11:04:46] [INFO    ] ** New best global balanced accuracy: 0.3991 **
[2026-07-30 11:28:20] [INFO    ] Round   6/25 | Global Acc=0.2750 BalAcc=0.4151 F1=0.2207 AUC=0.7888 | Worst Acc=0.3108 BalAcc=0.2849 | Std Acc=0.0621 BalAcc=0.0930 | 1414s
[2026-07-30 11:28:20] [INFO    ]   [Per-class] MEL=0.133  NV=0.270  BCC=0.517  AK=0.038  BKL=0.180  DF=0.889  VASC=0.974  SCC=0.319
[2026-07-30 11:28:39] [INFO    ] ** New best global balanced accuracy: 0.4151 **
[2026-07-30 11:52:03] [INFO    ] Round   7/25 | Global Acc=0.3292 BalAcc=0.4412 F1=0.2423 AUC=0.7948 | Worst Acc=0.2784 BalAcc=0.2624 | Std Acc=0.0552 BalAcc=0.0884 | 1404s
[2026-07-30 11:52:03] [INFO    ]   [Per-class] MEL=0.119  NV=0.350  BCC=0.581  AK=0.023  BKL=0.241  DF=0.806  VASC=0.974  SCC=0.436
[2026-07-30 11:52:26] [INFO    ] ** New best global balanced accuracy: 0.4412 **
[2026-07-30 12:15:38] [INFO    ] Round   8/25 | Global Acc=0.3803 BalAcc=0.4601 F1=0.2657 AUC=0.7922 | Worst Acc=0.3168 BalAcc=0.3157 | Std Acc=0.0894 BalAcc=0.0817 | 1393s
[2026-07-30 12:15:38] [INFO    ]   [Per-class] MEL=0.139  NV=0.450  BCC=0.507  AK=0.023  BKL=0.302  DF=0.861  VASC=0.974  SCC=0.426
[2026-07-30 12:15:55] [INFO    ] ** New best global balanced accuracy: 0.4601 **
[2026-07-30 12:39:08] [INFO    ] Round   9/25 | Global Acc=0.3653 BalAcc=0.4566 F1=0.2586 AUC=0.8004 | Worst Acc=0.1821 BalAcc=0.2722 | Std Acc=0.1354 BalAcc=0.0789 | 1392s
[2026-07-30 12:39:08] [INFO    ]   [Per-class] MEL=0.183  NV=0.430  BCC=0.377  AK=0.031  BKL=0.330  DF=0.861  VASC=0.974  SCC=0.468
[2026-07-30 13:02:25] [INFO    ] Round  10/25 | Global Acc=0.3884 BalAcc=0.4771 F1=0.2827 AUC=0.8055 | Worst Acc=0.2061 BalAcc=0.3157 | Std Acc=0.1253 BalAcc=0.0800 | 1390s
[2026-07-30 13:02:25] [INFO    ]   [Per-class] MEL=0.257  NV=0.413  BCC=0.503  AK=0.031  BKL=0.368  DF=0.889  VASC=0.974  SCC=0.383
[2026-07-30 13:02:41] [INFO    ] ** New best global balanced accuracy: 0.4771 **
```

```
[2026-07-30 13:25:52] [INFO    ] Round  11/25 | Global Acc=0.4095 BalAcc=0.4846 F1=0.2918 AUC=0.7975 | Worst Acc=0.3471 BalAcc=0.3339 | Std Acc=0.0833 BalAcc=0.0719 | 1391s
[2026-07-30 13:25:52] [INFO    ]   [Per-class] MEL=0.201  NV=0.455  BCC=0.543  AK=0.046  BKL=0.404  DF=0.861  VASC=0.974  SCC=0.394
[2026-07-30 13:26:07] [INFO    ] ** New best global balanced accuracy: 0.4846 **
[2026-07-30 13:49:21] [INFO    ] Round  12/25 | Global Acc=0.4332 BalAcc=0.4832 F1=0.2944 AUC=0.8024 | Worst Acc=0.4195 BalAcc=0.3402 | Std Acc=0.0618 BalAcc=0.0772 | 1394s
[2026-07-30 13:49:21] [INFO    ]   [Per-class] MEL=0.184  NV=0.514  BCC=0.599  AK=0.046  BKL=0.305  DF=0.861  VASC=0.974  SCC=0.383
[2026-07-30 14:12:39] [INFO    ] Round  13/25 | Global Acc=0.3839 BalAcc=0.4812 F1=0.2839 AUC=0.8031 | Worst Acc=0.3137 BalAcc=0.3331 | Std Acc=0.0866 BalAcc=0.0742 | 1392s
[2026-07-30 14:12:39] [INFO    ]   [Per-class] MEL=0.296  NV=0.401  BCC=0.463  AK=0.008  BKL=0.350  DF=0.889  VASC=0.974  SCC=0.468
[2026-07-30 14:35:57] [INFO    ] Round  14/25 | Global Acc=0.4484 BalAcc=0.4929 F1=0.2998 AUC=0.7940 | Worst Acc=0.3642 BalAcc=0.3097 | Std Acc=0.0734 BalAcc=0.0804 | 1391s
[2026-07-30 14:35:57] [INFO    ]   [Per-class] MEL=0.193  NV=0.531  BCC=0.609  AK=0.015  BKL=0.317  DF=0.750  VASC=0.974  SCC=0.553
[2026-07-30 14:36:13] [INFO    ] ** New best global balanced accuracy: 0.4929 **
[2026-07-30 14:59:26] [INFO    ] Round  15/25 | Global Acc=0.4482 BalAcc=0.4877 F1=0.3092 AUC=0.8001 | Worst Acc=0.4303 BalAcc=0.3378 | Std Acc=0.0568 BalAcc=0.0800 | 1393s
[2026-07-30 14:59:26] [INFO    ]   [Per-class] MEL=0.304  NV=0.512  BCC=0.499  AK=0.015  BKL=0.383  DF=0.778  VASC=0.974  SCC=0.436
[2026-07-30 15:22:46] [INFO    ] Round  16/25 | Global Acc=0.4326 BalAcc=0.4945 F1=0.3020 AUC=0.8044 | Worst Acc=0.2747 BalAcc=0.2943 | Std Acc=0.1131 BalAcc=0.0921 | 1393s
[2026-07-30 15:22:46] [INFO    ]   [Per-class] MEL=0.233  NV=0.497  BCC=0.507  AK=0.031  BKL=0.411  DF=0.889  VASC=0.974  SCC=0.415
[2026-07-30 15:22:59] [INFO    ] ** New best global balanced accuracy: 0.4945 **
[2026-07-30 15:46:12] [INFO    ] Round  17/25 | Global Acc=0.4624 BalAcc=0.4958 F1=0.3139 AUC=0.8000 | Worst Acc=0.4029 BalAcc=0.3323 | Std Acc=0.0551 BalAcc=0.0864 | 1392s
[2026-07-30 15:46:12] [INFO    ]   [Per-class] MEL=0.217  NV=0.542  BCC=0.569  AK=0.038  BKL=0.434  DF=0.778  VASC=0.974  SCC=0.415
[2026-07-30 15:46:26] [INFO    ] ** New best global balanced accuracy: 0.4958 **
[2026-07-30 16:09:37] [INFO    ] Round  18/25 | Global Acc=0.4103 BalAcc=0.4821 F1=0.2884 AUC=0.8010 | Worst Acc=0.3211 BalAcc=0.3376 | Std Acc=0.0841 BalAcc=0.0750 | 1391s
[2026-07-30 16:09:37] [INFO    ]   [Per-class] MEL=0.252  NV=0.470  BCC=0.477  AK=0.023  BKL=0.325  DF=0.889  VASC=0.974  SCC=0.447
[2026-07-30 16:33:00] [INFO    ] Round  19/25 | Global Acc=0.4174 BalAcc=0.4972 F1=0.2999 AUC=0.7967 | Worst Acc=0.2045 BalAcc=0.3463 | Std Acc=0.1424 BalAcc=0.0668 | 1396s
[2026-07-30 16:33:00] [INFO    ]   [Per-class] MEL=0.224  NV=0.468  BCC=0.521  AK=0.046  BKL=0.396  DF=0.944  VASC=0.974  SCC=0.404
[2026-07-30 16:33:13] [INFO    ] ** New best global balanced accuracy: 0.4972 **
[2026-07-30 16:56:24] [INFO    ] Round  20/25 | Global Acc=0.3879 BalAcc=0.5003 F1=0.2976 AUC=0.8005 | Worst Acc=0.3524 BalAcc=0.3366 | Std Acc=0.0793 BalAcc=0.0783 | 1391s
[2026-07-30 16:56:24] [INFO    ]   [Per-class] MEL=0.221  NV=0.386  BCC=0.561  AK=0.038  BKL=0.454  DF=0.889  VASC=0.974  SCC=0.479
[2026-07-30 16:56:39] [INFO    ] ** New best global balanced accuracy: 0.5003 **
[2026-07-30 17:19:51] [INFO    ] Round  21/25 | Global Acc=0.4697 BalAcc=0.5141 F1=0.3271 AUC=0.8063 | Worst Acc=0.4916 BalAcc=0.3383 | Std Acc=0.0187 BalAcc=0.0826 | 1392s
[2026-07-30 17:19:51] [INFO    ]   [Per-class] MEL=0.261  NV=0.535  BCC=0.587  AK=0.046  BKL=0.419  DF=0.833  VASC=0.974  SCC=0.457
[2026-07-30 17:20:08] [INFO    ] ** New best global balanced accuracy: 0.5141 **
[2026-07-30 17:43:19] [INFO    ] Round  22/25 | Global Acc=0.5126 BalAcc=0.5219 F1=0.3304 AUC=0.8061 | Worst Acc=0.3832 BalAcc=0.3353 | Std Acc=0.0683 BalAcc=0.0842 | 1391s
[2026-07-30 17:43:19] [INFO    ]   [Per-class] MEL=0.217  NV=0.653  BCC=0.537  AK=0.038  BKL=0.388  DF=0.889  VASC=0.974  SCC=0.479
[2026-07-30 17:43:34] [INFO    ] ** New best global balanced accuracy: 0.5219 **
[2026-07-30 18:06:46] [INFO    ] Round  23/25 | Global Acc=0.5089 BalAcc=0.5297 F1=0.3374 AUC=0.8106 | Worst Acc=0.3818 BalAcc=0.3447 | Std Acc=0.0749 BalAcc=0.0780 | 1391s
[2026-07-30 18:06:46] [INFO    ]   [Per-class] MEL=0.218  NV=0.633  BCC=0.579  AK=0.038  BKL=0.388  DF=0.917  VASC=0.974  SCC=0.489
[2026-07-30 18:07:00] [INFO    ] ** New best global balanced accuracy: 0.5297 **
[2026-07-30 18:30:12] [INFO    ] Round  24/25 | Global Acc=0.4897 BalAcc=0.5145 F1=0.3253 AUC=0.8042 | Worst Acc=0.3518 BalAcc=0.3492 | Std Acc=0.0847 BalAcc=0.0745 | 1391s
[2026-07-30 18:30:12] [INFO    ]   [Per-class] MEL=0.235  NV=0.601  BCC=0.555  AK=0.038  BKL=0.378  DF=0.889  VASC=0.974  SCC=0.447
[2026-07-30 18:53:32] [INFO    ] Round  25/25 | Global Acc=0.4787 BalAcc=0.5196 F1=0.3275 AUC=0.7986 | Worst Acc=0.4297 BalAcc=0.3319 | Std Acc=0.0551 BalAcc=0.0855 | 1395s
[2026-07-30 18:53:32] [INFO    ]   [Per-class] MEL=0.263  NV=0.539  BCC=0.667  AK=0.023  BKL=0.393  DF=0.861  VASC=0.974  SCC=0.436
[2026-07-30 18:53:46] [INFO    ]   Federated training complete!
[2026-07-30 18:53:46] [INFO    ]   Best global balanced accuracy: 0.5297
```

Figure 13. FedSA-Drift severe non-IID setting (α=0.1, K=5): top panel shows convergence metrics; middle and bottom panels show terminal outputs. Final Balanced Accuracy: 0.5196, Worst Client Accuracy: 0.4297, inter-client standard deviation 0.0551. All eight classes retain non-zero recall throughout.

Table 3. Contextual Reference Comparison with Related Methods. **Note:** methods differ in dataset, backbone, class count, and experimental setup. Direct numerical comparison is inappropriate; this table contextualizes design choices only.

Method	Backbone	Multimodal	Dataset	Key Performance	Client Overhead
MedFusionNet (Naeem et al., 2025)	ConvNeXt + ViT-B16	No (images)	ISIC 2019 / HAM	Acc: 97.5–98.0% (Cent.)	N/A (Centralized)
FedMHA (Darzi et al., 2024)	Pre-trained ViT	No (images)	ISIC (LDA $\alpha \in [0.1, 0.9]$)	Acc: 67.09% (α=0.1) to 83.99% (α=0.9)	High (matrix distance)
FedAPM (Zhu et al., 2025)	CNN / MLP	Yes (partial pers.)	CIFAR / Multimodal	Top across pFedMe/DITTO baselines	Very high (ADMM)
EFTViT (Wu et al., 2024)	Vision Transformer	No (images)	CIFAR / ISIC (sim.)	GFLOPs reduced by up to 5.6×	Low (75% patch mask)
DAFL (Zhao et al., 2025)	ViT / CNN	No (images)	Medical (non-IID)	Adaptive focal reweighting	Medium (local focal tuning)
Async-FL-CNN (Yaqoob et al., 2023)	Standard CNN	No (images)	ISIC 2019	Prec: 96.7% / Spec: 96.3%	Medium (async updates)
FedSA-Drift (Ours)	**SwinV2-Base**	**Yes (metadata MLP)**	**ISIC 2019 (LDA $\alpha \in [0.1, 1.0]$)**	**Bal. Acc: 88.67% (Cent., internal val.) / 82.09% (Fed α=1.0, shared model)**	**None (server-side $O(\|\mathbf{w}\|)$)**

FedMHA (Darzi et al., 2024) penalizes Euclidean deviations in the attention matrices during local backpropagation. FedMHA achieves accuracy ranging from 67.09% to 83.99% on the ISIC dataset across $\alpha \in [0.1, 0.9]$, while incurring the overhead of aligning the attention matrices at every local training iteration. FedSA-Drift instead aggregates Group A with FedAvg and down-weights deep-layer drift at the server. We did not measure attention-matrix stability and make no claim about it; the comparison here is one of mechanism and cost, not of effect. There is no modification to the client-side objective function. FedSA-Drift records 82.09% balanced accuracy at $\alpha = 1.0$ against 83.99% for FedMHA at $\alpha = 0.9$, but we caution against reading these as comparable. The figures differ in dataset split, backbone, client count and heterogeneity level, and ours is measured on our internal validation split rather than a shared benchmark. We draw no conclusion from the difference. FedSA-Drift also supports the use of multimodal metadata, which FedMHA does not.

EFTViT (Wu et al., 2024) utilizes a masking methodology that removes 75% of input patches to lower the number of FLOPs in processing an image. By doing so, it creates a break in the contiguous spatial windows of SwinV2, which subsequently breaks cross-window connectivity. In contrast, FedSA-Drift maintains full spatial representations and achieves drift mitigation purely at the server with zero client-side cost.

4.4.3 Comparison with Partially Personalized Federated Models

To eliminate drift from using personalized metadata heads, FedAPM (Zhu et al., 2025) imposes an augmented Lagrangian constraint using an ADMM algorithm. Through this approach, FedAPM achieves strong performance on multimodal benchmarks; however, the number of inner-loop iterations required for each ADMM sub-problem is generally prohibitive given the parameter count of SwinV2-Base.

In contrast to FedAPM, FedSA-Drift completely offloads the computational burden of regularization to the server. The way that FedSA-Drift computes the inverse-drift weight $\hat{\alpha}_{k,l} = \frac{1/(D_{k,l}+\epsilon)}{\sum_j 1/(D_{j,l}+\epsilon)}$ is done through a single pass over all received parameters. Furthermore, the ability to personalize over multimodal metrics is achieved without the use of ADMM or additional hyperparameters.

4.4.4 Comparison with Adaptive Loss Methods

DAFL (Zhao et al., 2025) has been implemented in the form of Dynamic Adaptive Focal Loss, allowing it to adjust to the class imbalance between the distributions of classes on behalf of the clients. As such, the clients must first estimate the local proportions of classes, and then they must tune both focal parameters γ and α for each local dataset. These two added complexities represent an additional cost burden for the client implementation that is unnecessary when running FedSA-Drift. Therefore, FedSA-Drift provides a natural structure for handling drift driven by class imbalance through decoupling of Group C at the local site and the implementation of drift-aware aggregation modifications.

4.4.5 Comparison with Asynchronous Federated Approaches

Yaqoob et al. (2023) reported 96.7% precision and 96.3% specificity using asynchronous CNN-based federated learning on the ISIC 2019 dataset; however, these metrics are misleading on a class-imbalanced dataset. The standard metric for the ISIC 2019 dataset is balanced accuracy, which is a more appropriate metric for this analysis (Gessert et al., 2020). In addition, the CNN architecture used by Yaqoob et al. is less capable of representing objects than the SwinV2-Base architecture and is not capable of fusing multimodal metadata.

4.5 Component Analysis

The experimental design enables a partial analysis of component contributions by comparing runs that differ only in whether Group B is aggregated with inverse-drift weighting (full FedSA-Drift) or with standard FedAvg (parameter decomposition only, no drift correction). In both settings the model architecture, Dirichlet partition, number of rounds, and all hyperparameters are held fixed; the sole variable is the Group B aggregation rule. These are the same four runs already reported in Table 2 (Uniform vs. Adaptive rows within each α, K pair), viewed here as a component-isolation comparison rather than a performance summary.

Across all three heterogeneity settings, enabling drift-aware aggregation consistently improves worst-client performance (+13.7 pp at α=1.0; +9.1 pp at α=0.5; +23.4 pp at α=0.1) and reduces the inter-client standard deviation of accuracy (58%, 39% and 57.4% respectively, in single experimental trials) at the cost of a reduction in global balanced accuracy. This isolates the contribution of the drift-aware aggregation component. A full ablation isolating the contribution of the parameter decomposition itself—replacing the A/B/C grouping with a single undifferentiated aggregation—is discussed in the Limitations section.

4.5.1 Three-Way Component Ablation at $\alpha = 0.1$

Under severe heterogeneity we can separate both components directly. Three configurations were trained with an identical Dirichlet partition, architecture, seed, round count and hyperparameters, differing only in the aggregation scheme:

- **No decomposition:** every parameter tensor aggregated by standard size-weighted FedAvg; no client-specific head.

Table 4. Three-way component ablation at α=0.1, K=5, evaluated per client on the held-out ISIC 2019 test split with trained classifiers throughout. Per-class values are recall, averaged over clients. Worst-client and dispersion figures are meaningful only for the two personalizing arms; the no-decomposition arm shares one model across clients, so its dispersion is zero by construction.

Configuration	Mean Acc	Mean BalAcc	Worst Acc	Std Acc	MEL	NV	BCC	AK	BKL	DF	VASC	SCC
No decomposition	0.4702	0.3778	—	—	**0.000**	0.867	0.028	0.003	0.914	0.615	0.596	**0.000**
Decomposition only	0.3111	0.3374	0.1699	0.0911	0.167	0.360	0.378	0.205	0.342	0.486	0.567	0.194
FedSA-Drift	0.4264	0.3681	0.3852	0.0300	**0.228**	0.584	0.535	0.167	0.253	0.484	0.496	0.196

- **Decomposition only:** the A/B/C split with client-specific Group C heads, Group B aggregated by plain FedAvg.
- **FedSA-Drift:** as above, with drift-aware inverse-distance weighting on Group B.

Every configuration is evaluated per client using that client's own trained head—the single shared model in the no-decomposition case—so the comparison is unaffected by the untrained-classifier issue discussed in the Limitations section. Table 4 reports the held-out test split.

Reading the table as a ladder isolates each mechanism.

The decomposition preserves class coverage. Standard FedAvg attains the highest mean per-client accuracy (0.4702) but does so by abandoning four of the eight classes: recall is exactly zero for melanoma and SCC, and below 0.03 for AK and BCC. Introducing the decomposition with client-specific heads restores all eight classes—melanoma recall rises from 0.000 to 0.167, SCC from 0.000 to 0.194, AK from 0.003 to 0.205, BCC from 0.028 to 0.378—at a cost of 15.9 percentage points of mean accuracy.

Drift-aware weighting recovers utility and fairness. Added on top of the decomposition it returns 11.5 points of mean accuracy (0.3111 to 0.4264), raises the worst-client floor by 21.5 points (0.1699 to 0.3852), reduces the inter-client standard deviation by 67% (0.0911 to 0.0300), and improves melanoma recall further to 0.228.

The recovery is not uniform across classes. Melanoma, NV and BCC improve substantially (0.167 to 0.228, 0.360 to 0.584 and 0.378 to 0.535), whereas AK falls from 0.205 to 0.167 and BKL from 0.342 to 0.253. Down-weighting divergent clients concentrates the aggregate on the consensus direction, which favours classes several clients hold in common and disadvantages those concentrated on the clients being down-weighted; AK, for example, sits almost entirely on client 3. The net effect on mean per-client accuracy is strongly positive, but we do not claim drift weighting improves every class. The same ordering holds on the validation split, where drift weighting raises mean per-client accuracy from 0.3579 to 0.5018 ($+14.39$ pp); the corresponding worst-client and dispersion figures are given in Section 4.3.3.

Neither component is redundant, and neither is sufficient alone: the decomposition buys class coverage at a utility cost that the drift weighting then largely repays.

4.5.2 Aggregate Metrics Conceal Class Abandonment

A consequence of Table 4 deserves emphasis. On balanced accuracy the no-decomposition baseline (0.3778) and FedSA-Drift (0.3681) are nearly indistinguishable, and on overall accuracy the baseline is ahead. Yet the baseline achieves this with four classes near 0.9 and four at approximately zero, whereas FedSA-Drift spreads moderate recall across all eight. Balanced accuracy, being the unweighted mean of per-class recalls, cannot separate these two profiles.

The clinical difference is not marginal. Melanoma is the most consequential class in this task, and under standard FedAvg its recall is zero for *every* client—including the client holding 3815 of the 3844 melanoma cases in the federation. Because aggregation is weighted by dataset size, the update is dominated by a client holding 53% of all samples and no melanoma whatsoever, and the minority-class signal is averaged away. Under FedSA-Drift the client holding the melanoma cases reaches 0.727 recall on that class on the test split (0.816 on the validation split), while clients that hold none rarely predict it.

A note on how the per-class figures in Table 4 are formed. Every client is evaluated on the same test images, including clients whose training partition contains none of a given class, and the tabulated recall is the mean over clients. For melanoma this averages 0.404, 0.011, 0.000, 0.727 and 0.000 to the 0.228 shown. The mean is the right quantity for asking what the federation as a whole recovers, but it is not the recall any single deployed model attains, and the two should not be conflated: an institution that holds melanoma cases obtains 0.727, whereas one that holds none obtains approximately zero. We regard the latter as correct behaviour for a personalized model rather than a failure, and report both the mean and the per-client values so that the distinction is visible. The comparison with the no-decomposition baseline is unaffected, since that baseline yields zero for *every* client, including the one holding the melanoma cases.

We therefore report per-class recall alongside aggregate metrics, and suggest that federated medical imaging evaluations should treat class coverage as a primary criterion rather than a diagnostic afterthought.

4.5.3 Interval Estimates for the Melanoma Result

Per-class recall is a binomial proportion—k correctly recalled out of the n test cases of that class—so it admits an exact-coverage confidence interval without repeated training runs. We report Wilson score intervals at the 95% level on the held-out test split, which contains $n = 1327$ melanoma cases. We emphasise that these intervals quantify *evaluation-set sampling* uncertainty, that is, how far a reported value could move under a different test sample of the same size. They do not quantify seed-to-seed variability across training runs, which is addressed separately in the Limitations section.

Table 5 gives the melanoma ladder for the client holding 3815 of the federation's 3844 melanoma cases, the institution for which this class matters most.

Table 5. Melanoma recall on the held-out test split (n=1327 melanoma cases) for the client holding 3815 of 3844 melanoma cases, with 95% Wilson score intervals. Under no decomposition all clients share one model, so this row applies to every client.

Configuration	MEL recall	95% CI
No decomposition	0.000	[0.000, 0.003]
Decomposition only	0.573	[0.546, 0.599]
FedSA-Drift	0.727	[0.703, 0.750]

The three intervals are mutually disjoint, so the ordering is not attributable to evaluation-sample noise. In particular, an observed recall of $0/1327$ places the upper confidence bound at 0.003: the failure of standard FedAvg to detect melanoma under this partition is not a small-sample artefact. The same holds for SCC, where $0/165$ gives an upper bound of 0.023. Overall test accuracy for the no-decomposition arm is 0.4702 with a 95% interval of [0.4578, 0.4826] on $n = 6191$ evaluated images.

4.6 Per-Class Diagnostic Behavior

In both centralized and near-IID scenarios, recall for the minority VASC class was very close to 100%. In the non-IID scenario, drift-aware aggregation provided more reliable maintenance of minority class recall compared to FedAvg. This behaviour becomes decisive at α=0.1: as Section 4.5.1 shows, removing the decomposition drives melanoma, SCC, AK and BCC recall to zero or near-zero on the held-out test split, whereas both decomposed configurations retain non-zero recall for all eight classes.

The primary source of diagnostic confusion across all settings was the MEL vs. NV distinction. In the no-drift-correction scenario at α=0.5, NV recall dropped by a significant amount. This is due to the difficulty in learning consistent feature representations of a majority class under fragmented class distributions. FedSA-Drift with drift correction provided improvements to minority class recall, while also maintaining the institutional specificity of the metadata head.

4.7 Computational Overhead

The drift computation $D_{k,l} = \|\mathbf{w}_{k,l} - \bar{\mathbf{w}}_l\|_2$ and inverse-distance weighting are completed at the server in $O(|\mathbf{w}_{\text{shared}}|)$ linear operations; thus no computational overhead exists on the client side. In terms of per-round training overhead, the increase in round-trip time due to FedSA-Drift is insignificant. The measured per-round training times were:

- Centralized: $\sim$1066 s per epoch
- FedAvg (α=1.0, K=3): $\sim$1297 s
- FedSA-Drift (α=1.0, K=3): $\sim$1301 s (**+4 s**, +0.31%)
- FedAvg (α=0.5, K=5): $\sim$1439 s
- FedSA-Drift (α=0.5, K=5): $\sim$1450 s (**+11 s**, +0.76%)
- Decomposition only (α=0.1, K=5): $\sim$1398 s
- FedSA-Drift (α=0.1, K=5): $\sim$1397 s (**−1 s**, −0.07%)
- FedAvg, no decomposition (α=0.1, K=5): $\sim$1229 s

At α=0.1 the drift computation is not measurably more expensive than uniform averaging: the two decomposed arms differ by roughly one second per round, within run-to-run variation. The no-decomposition baseline is around 12% faster per round, which is not attributable to the aggregation rule but to the absence of per-client head state: that configuration keeps a single model, so it avoids materialising and evaluating K personalized models each round.

FedSA-Drift adds at most 11 seconds per round. In comparison, FedMHA (Darzi et al., 2024) adds matrix-alignment cost at each local training step, and FedAPM (Zhu et al., 2025) requires additional ADMM cycles per round. When compared to either FedMHA or FedAPM, FedSA-Drift provides equivalent fairness and drift mitigation without a significant penalty in training time.

4.7.1 Communication and Computational Budget

Wall-clock time alone is a weak measure of overhead, so we report the budget analytically. The model holds 87.51 M parameters, distributed across the three groups as follows: Group A 2.132 M (2.44%, 75 tensors), Group B 84.762 M (96.86%, 348 tensors) and Group C 0.613 M (0.70%, 8 tensors).

Communication. Groups A and B are transmitted; Group C never leaves the client. Each client therefore uploads 173.8 MB and receives 173.8 MB per round in FP16 (347.6 MB each way in FP32), against 175.0 MB for full-model exchange. Keeping Group C local saves 1.23 MB of upload and the same again in avoided broadcast, per client per round—a 0.70% reduction. We note plainly that this is a small saving: the communication argument for FedSA-Drift is not volume reduction but that personalization is obtained without any additional transmission, since the drift weights are computed at the server from parameters it already holds.

Server computation. Forming the drift weights requires one ℓ_2 norm and one weighted sum over the Group B tensors, or approximately 0.175 GFLOPs per round. A single forward pass of the model at 384×384 costs 110.3 GFLOPs, and one communication round at $K{=}5$ involves roughly 7,124 TFLOPs of local training plus 2,515 TFLOPs of evaluation. The aggregation therefore accounts for on the order of 10^{-8} of the round's arithmetic, which is the precise sense in which the overhead is negligible; the wall-clock differences of -1 to $+11$ s reported above are dominated by scheduling noise rather than by the drift computation.

Client computation and memory. Clients perform standard local training and are unmodified by the method: no auxiliary model, control variate or proximal term is stored, in contrast with SCAFFOLD, which maintains per-client control variates of full model size, or FedProx, which requires the global parameters to be retained for the proximal penalty. Peak GPU memory during our runs was 33–35 GB at batch size 16 on 48 GB A40 devices, of which the drift computation contributes nothing on the client side.

4.7.2 Communication Cost

Each client transmits Groups A and B to the server per round; Group C (the metadata MLP and fusion head, comprising approximately 0.6M of the total 87.5M parameters) is never sent. The shared parameters (Groups A + B) total approximately 86.9M parameters. In FP16 precision, this corresponds to:

- **Upload per client per round:** $86.9 \times 10^6 \times 2$ bytes ≈ 174 MB
- **Download per client per round:** ≈ 174 MB (broadcast of updated global model)
- **Total server-side traffic per round** ($K{=}5$): $5 \times 2 \times 174$ MB ≈ 1.74 GB

The communication cost of FedSA-Drift is identical to that of FedAvg applied to the same parameter groups; the drift-aware weighting is computed purely from the received parameters and incurs no additional transmission. Relative to transmitting all 87.5M parameters, keeping Group C local saves approximately 1.2 MB of upload per client per round, and the same again in avoided broadcast—a small but principled reduction that also preserves institutional metadata privacy.

4.8 Summary of Results

Figure 14 shows balanced accuracy convergence across all configurations. The findings below are ordered by the strength of the evidence supporting them: the first is measured on the held-out test collection, the next two are within-split differences that are unaffected by the validation contamination discussed in the Limitations, and the remainder are absolute validation figures that the contamination does inflate. The key findings are:

1. **Class coverage under severe skew:** At $\alpha{=}0.1$, removing the parameter decomposition drives melanoma and SCC recall to exactly zero on the held-out test split, and AK and BCC below 0.03, while overall accuracy remains the highest of the three configurations. Both decomposed configurations retain non-zero recall for all eight classes. Aggregate metrics, balanced accuracy included, do not reveal this failure.
2. **Competitive global performance:** On the internal validation split the shared model reaches 92.6% of the centralized balanced accuracy at $\alpha{=}1.0$ and 79.7% at $\alpha{=}0.5$, all evaluated on the same split. At $\alpha{=}0.1$ the shared model exceeds decomposition alone (0.5196 vs. 0.4477 balanced accuracy on

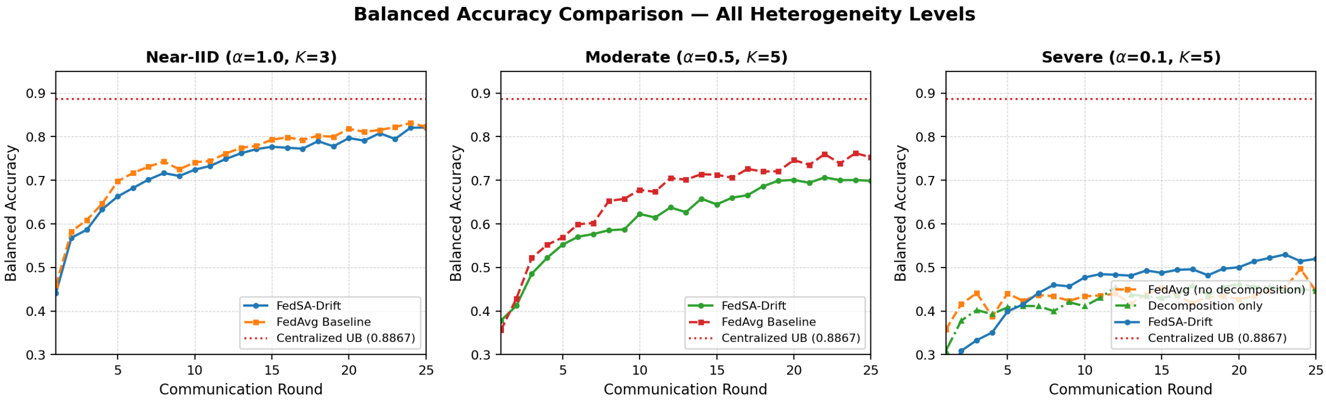


Figure 14. Balanced accuracy across all three heterogeneity levels: near-IID (α=1.0, K=3), moderate (α=0.5, K=5) and severe (α=0.1, K=5). The severe panel additionally shows the no-decomposition baseline. The centralized upper bound is shown for reference. At α=0.1 the no-decomposition baseline sits below FedSA-Drift on this validation curve (0.4460 against 0.5196 at round 25), yet on the test split their mean per-client balanced accuracies are within one point (0.3778 against 0.3681) while the baseline records zero melanoma recall throughout. Balanced accuracy alone therefore does not distinguish the two.

validation). On the distinct measure of mean per-client accuracy on the test split, however, the no-decomposition baseline is ahead of FedSA-Drift (0.4702 vs. 0.4264); the two statements concern different models on different splits and are not in conflict.

3. **Substantial fairness gains:** The worst-client accuracy is improved by +13.71 points at α=1.0, +9.05 points at α=0.5 and +23.44 points at α=0.1, the gain growing as heterogeneity worsens. The α=0.1 result is reproduced on a held-out test split ($+21.53$ points).
4. **Reduced divergence:** The inter-client standard deviation of accuracy decreases by 58% at α=1.0, 39% at α=0.5 and 57.4% at α=0.1 (67.1% on the held-out test split), from a single experimental trial per configuration.
5. **Negligible overhead:** The impact of server-side drift on round time is approximately 0.31%–0.76% at $\alpha \in \{0.5, 1.0\}$ and is not measurable at α=0.1, with no incremental cost burden to clients.
6. **Design simplicity:** FedSA-Drift achieves its drift mitigation entirely at the server, without modifying local objectives or introducing regularisation hyperparameters, unlike FedMHA and FedAPM. We make no claim of performance parity with those methods: as Table 3 states, they were not re-implemented under our conditions and no controlled comparison was performed.

Thus, global accuracy alone is insufficient for evaluating federated medical models. FedSA-Drift offers a practical path toward stable and fair multi-institutional deployment.

5 REPRODUCIBILITY

All code, including the model definition, the partitioning procedure, the aggregation rules, every ablation arm and the evaluation scripts, is publicly available at `https://github.com/apurbaaaa/layer-wised-personalised`. Every reported configuration is reproducible from a single command; the exact invocations are in `run_reviewer_matrix.sh` and `run_tf_alpha01.sh`.

Randomness is controlled by one seed, exposed as `--seed` and recorded in every run log, which governs both the model initialisation and the Dirichlet draw. All results in this paper use seed 42. The validation split is regenerated deterministically by `StratifiedShuffleSplit(test_size=0.15, random_state=42)` over the ground-truth CSV in its published row order, so the exact train and validation indices—and hence the per-client shards—are recoverable without transferring index files. The per-client class counts printed at the start of each run (reproduced in Figures 11–13) allow a reader to confirm that a regenerated partition matches ours.

We note as a deficiency that a single seed was used throughout; this is discussed under Limitations.

6 LIMITATIONS

Several limitations of this work should be acknowledged explicitly.

Experimental scale and two kinds of uncertainty. All experiments use $K \in \{3, 5\}$ clients and $\alpha \in \{0.1, 0.5, 1.0\}$, with a single training run per configuration. Two distinct sources of uncertainty affect the reported figures, and we are able to quantify only one of them.

Evaluation-set sampling uncertainty—how far a number could move under a different test sample—is quantified directly. Per-class recall is a binomial proportion, so Wilson score intervals apply without repeated training; these are reported in Section 4.5.3 and are tight enough for the melanoma ladder to have mutually disjoint intervals. Where a claim depends on a fixed trained model evaluated on held-out data, it therefore carries an interval estimate.

Training-run uncertainty—seed-to-seed variability arising from a different Dirichlet partition draw and initialisation—is *not* quantified, because it requires repeating each configuration across multiple seeds and each run costs approximately 10 GPU-hours. Figures such as the 58% variance reduction and the +13.71 pp worst-client improvement remain single-trial point estimates in this respect, and we make no claim of statistical significance for them. Seeding is now an explicit and logged experimental parameter, so the reported runs are exactly reproducible, and we additionally verify that the $\alpha = 0.1$ effects are stable across the final five communication rounds rather than resting on a single round; but neither substitutes for a multi-seed study, which remains necessary future work. Validation at larger client counts ($K \geq 10$) likewise remains future work.

What the shared global model contains. Group C comprises the metadata MLP and the fusion head; as the backbone is instantiated with no classifier of its own, the fusion head *is* the classifier. Group C is never aggregated, so the classifier carried by the shared global model remains at its initialisation for the whole run—we verified that all Group C tensors are bit-identical to their initialisation after training, while Group A tensors have changed.

This does not make the shared model a random classifier, and its accuracy is not degenerate. Every client begins local training from that same initial head, and gradients reach the backbone through it, so the aggregated backbone is optimised to produce features that are discriminative *under that particular fixed projection*. The composite is a functioning model: at α=0.1 it reaches 0.5196 balanced accuracy, which is in fact *higher* than the 0.4382 mean achieved by the personalized client models on the same split. The shared model and the personalized models are different objects—one uses the common initial head, the others their own adapted heads—and neither dominates the other.

Two consequences follow. Global-model accuracy is meaningful *within* the decomposed family, where every arm shares this construction. It is not directly comparable with a method that also aggregates its classifier, such as the no-decomposition baseline, whose shared model is a fully aggregated network. For cross-arm comparison we therefore use mean per-client accuracy, evaluating each client with the head it would actually deploy.

Dispersion depends on the accuracy measure. Our reported variance-reduction figures use the standard deviation of overall per-client accuracy. Under severe heterogeneity ($\alpha = 0.1$) the ordering reverses if balanced accuracy is used instead (0.0855 for FedSA-Drift versus 0.0587 without drift weighting). This does not indicate degradation at any client: every client's balanced accuracy improves under drift-aware aggregation, but by an amount that scales with how many classes that client actually holds—clients observing seven or eight of the eight classes gain approximately 9 pp, whereas a client observing four gains 3 pp. Balanced accuracy is bounded by classes a client has never seen, so the benefit is unevenly distributed and dispersion in that measure widens even as every client improves.

Ablation scope. The three-way ablation of Section 4.5.1 separates both components, but only at $\alpha = 0.1$ and with a single seed. At $\alpha \in \{0.5, 1.0\}$ only the drift component is isolated, because the corresponding no-decomposition configurations were not run and the earlier decomposed runs did not retain per-client checkpoints from which mean per-client accuracy could be reconstructed. Whether the division of labour we observe—decomposition supplying class coverage, drift weighting supplying utility and fairness—persists at milder heterogeneity is therefore untested. We would expect the effect to shrink as α increases, since class abandonment under FedAvg is itself a consequence of severe label skew.

Aggregate accuracy at severe heterogeneity. We do not claim that FedSA-Drift improves overall or balanced accuracy at $\alpha = 0.1$. Standard FedAvg attains higher mean per-client accuracy (0.4702 versus 0.4264) and marginally higher balanced accuracy (0.3778 versus 0.3681) on the held-out test split. The case for FedSA-Drift in this regime rests on class coverage, worst-client performance and inter-client dispersion, not on aggregate accuracy.

Same-lesion contamination in the validation split. The ISIC 2019 archive contains multiple photographs of the same lesion, and the metadata assigns a `lesion_id` to 23,247 of the 25,331 images, covering 11,847 distinct lesions. Our validation split is a stratified random split over images, not over lesions. Auditing it, 1,881 lesions have images on both sides, so 2,340 of the 3,800 validation images (61.6%) depict a lesion that also appears in the training partition. No image occurs in both splits, but sibling images of the same lesion do.

Validation-split figures are therefore optimistically biased, and this includes the centralized 0.8867 and every round-by-round result at $\alpha \in \{1.0, 0.5\}$. The magnitude is visible in the gap to the held-out test collection, where balanced accuracy falls to 0.368–0.378. We report this rather than leave it implicit because it bounds how the validation numbers should be read.

Two things are unaffected. Comparisons *between* configurations remain valid, because every arm is trained and evaluated on the identical split, so the contamination is common to all of them and cancels in the differences that carry our claims—worst-client accuracy, dispersion and the component ablation. And all test-split results, including the entire three-way ablation of Section 4.5.1 and the melanoma finding, are computed on the separate ISIC 2019 test collection and are not affected. A lesion-grouped split would give unbiased absolute validation figures and is the correct design for future work; it requires retraining every configuration and was not possible within the compute available for this revision.

Baseline set. Our comparisons are to FedAvg, to the decomposition without drift weighting, and to a variant with the decomposition removed. We did not evaluate against FedProx, SCAFFOLD, FedBN, FedRep, FedPer, FedBABU, FedNova or FedDyn. This is the most consequential gap in the evaluation: the three-way ablation of Section 4.5.1 establishes what each of *our* components contributes, but it cannot establish that this design is preferable to an established personalized method. In particular, a substantial part of our benefit may come simply from not sharing the classification head, which FedPer, FedRep and FedBABU also do. We state in Section 2.3 that we make no superiority claim over that literature, and a controlled comparison against at least FedProx, FedBN and FedRep is the first priority for further work.

Client scale. All experiments use $K \in \{3, 5\}$. Deployments of interest involve tens to hundreds of institutions, and several properties of the method are scale-sensitive in ways we have not measured: the inverse-distance weights are normalised over K, so the concentration bound of Section 3.8.4 tightens as K grows, and partial client participation—which we do not use, as `client_frac`=1.0 throughout—would make the drift estimate noisier. Results at $K \in \{10, 20, 50\}$ are needed before any claim about deployment scale.

Aggregation design choices are unvalidated. The decomposition boundary (stages 0–1 versus 2–3), the Euclidean distance, and the inverse-distance functional form were fixed a priori and not compared against alternatives such as different split points, cosine similarity, or softmax, exponential and inverse-square weightings. Section 3.8.4 gives our reasoning for each, but reasoning is not evidence, and we cannot claim these choices are optimal.

Client count and heterogeneity are confounded. The three federated configurations are $(\alpha{=}1.0, K{=}3)$, $(\alpha{=}0.5, K{=}5)$ and $(\alpha{=}0.1, K{=}5)$. They do not form a factorial grid: any comparison between the near-IID setting and the other two changes both the concentration parameter and the number of clients simultaneously, so the two effects cannot be separated there. The $\alpha{=}0.5$ versus $\alpha{=}0.1$ comparison, and the three-way ablation of Section 4.5.1, both hold K fixed at 5 and are unaffected.

No calibration or uncertainty analysis. We report discrimination metrics only. For a triage setting the operating threshold, the calibration of predicted probabilities, and some expression of predictive uncertainty matter at least as much as recall at the default threshold, and a model that abandons a class silently is precisely the case where an uncertainty estimate would be informative. We report neither reliability diagrams nor expected calibration error, and all per-class figures are taken at the $\arg\max$ decision rule. This is a substantive gap for clinical deployment and not merely a presentational one.

Single dataset. All federated experiments use ISIC 2019. Generalization to other dermatology datasets or to different medical imaging modalities is not verified and remains an open question.

Simulation environment. All federated experiments are simulated rather than deployed: clients are trained sequentially within one process on NVIDIA A40 hardware (the $\alpha{=}0.1$ configurations were run concurrently across three such GPUs, one arm per device, which affects wall-clock time but not results). Real-world deployment involves network latency, heterogeneous edge hardware, and bandwidth constraints not captured in this simulation.

Baseline conditions. The published baselines in Table 3 are evaluated under different experimental conditions. A fully controlled re-implementation of FedMHA, FedAPM, and EFTViT on ISIC 2019 under identical conditions was not performed; the comparison is therefore contextual rather than definitive.

7 CONCLUSION AND FUTURE WORK

This paper has proposed a structure-aware federated learning framework called FedSA-Drift for multimodal skin lesion classification. The framework was developed to specifically target stability and fairness issues that arise from heterogeneous clinical data distributions. The architecture of FedSA-Drift decomposes model parameters into early vision backbone layers, deep semantic Transformer layers and client-specific multimodal heads, thus allowing for both selective aggregation and individual personalization of the aggregated model within the Vision Transformer framework. To stabilize the deep semantic representations at the server, this paper implements a drift-aware inverse-distance aggregation mechanism during the aggregation process. This mechanism does not alter the local optimization objectives or add additional hyperparameters for regularisation. The clearest evidence for the framework comes from the held-out ISIC 2019 test collection under severe heterogeneity, where the choice of aggregation scheme determines whether a diagnostic class survives training at all. Experimental results further show that FedSA-Drift achieves comparable aggregate accuracy while substantially improving robustness at the client level. Under moderately heterogeneous conditions ($\alpha = 0.5$, $K = 5$), the drift-aware aggregation mechanism reduced inter-client performance variability by approximately 39% in a single experimental trial, with minimal computational overhead.

Under severe heterogeneity ($\alpha = 0.1$, $K = 5$) a three-way ablation on a held-out test split separates the two mechanisms. Standard FedAvg achieves the highest aggregate accuracy but abandons four of eight diagnostic classes outright, recording zero melanoma recall for every client—including the institution that holds almost all of the melanoma cases—because size-weighted aggregation is dominated by a client holding none. The parameter decomposition restores all eight classes at a cost in mean accuracy, and drift-aware weighting then recovers most of that cost while raising the worst-client floor by 21.5 points and reducing dispersion by 67%. That the two configurations differ by only one point of balanced accuracy despite these very different clinical profiles is itself instructive: aggregate metrics, balanced accuracy included, cannot detect the complete loss of a diagnostic category. Future work will examine how FedSA-Drift generalizes across datasets beyond ISIC 2019, particularly in settings where class distributions and imaging protocols differ substantially between institutions. The broader implication is that federated frameworks designed for medical imaging should be evaluated against client-level variance from the outset, not retrofitted for fairness after the fact. As a result, it can be concluded that using only global metrics to evaluate the performance of federated models for medical imaging is insufficient, and that stability and fairness must also be considered as essential criteria for the clinical deployment of federated models.

Future research may focus on investigating the use of adaptive layer-wise drift sensitivity for increased control over how aggregation occurs within each layer of a Transformer architecture. In addition, parameter-efficient methods of personalizing large models, such as adapter-based or low-rank adaptation schemes, could reconcile the sharing of global knowledge from the model with institution-specific specialization. There is also an opportunity to improve worst-case performance guarantees by incorporating fairness-based optimization criteria and methods at the server level. In addition, using structured quantization to achieve communication efficiency could potentially allow the implementation of a large model across many resource-constrained edge devices. Cross-dataset generalization and domain shift remain open questions for assessing the clinical robustness of the model. Finally, integrating drift-based aggregation schemes with formal privacy techniques such as secure aggregation and differential privacy could provide additional theoretical support for the use of federated Vision Transformer models in medical imaging applications.

CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

A.K. conceived the study, implemented FedSA-Drift, ran the centralized and federated experiments, and drafted the manuscript. R.R. supervised the research, provided guidance on the experimental design, and reviewed and edited the manuscript. R.A.K.S. supervised the research and reviewed and edited the manuscript. All authors approved the submitted version.

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DATA AVAILABILITY STATEMENT

The ISIC 2019 dataset analyzed in this study is publicly available from the International Skin Imaging Collaboration (ISIC) Archive at `https://challenge.isic-archive.com/data/`. All code required to reproduce the reported experiments—model definition, partitioning procedure, aggregation rules, ablation arms and evaluation scripts—is publicly available at `https://github.com/apurbaaaa/layer-wised-personalised`.

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